

Section 2 Alterations in Body Temperature

18 Fever

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Body temperature is controlled by the hypothalamus. Neurons in both the preoptic anterior hypothalamus and the posterior hypothalamus receive two kinds of signals: one from peripheral nerves that transmit information from warmth/cold receptors in the skin and the other from the temperature of the blood bathing the region. These two types of signals are integrated by the thermoregulatory center of the hypothalamus to maintain normal temperature. In a neutral temperature environment, the human metabolic rate produces more heat than is necessary to maintain the core body temperature in the range of $36.5\text{--}37.5^{\circ}\text{C}$ ($97.7\text{--}99.5^{\circ}\text{F}$).

A normal body temperature is ordinarily maintained despite environmental variations because the hypothalamic thermoregulatory center balances the excess heat production derived from metabolic activity in muscle and the liver with heat dissipation from the skin and lungs. According to a study of $>35,000$ individuals ≥ 18 years of age seen in routine medical visits, the mean oral temperature is 36.6°C (95% confidence interval, $35.7\text{--}37.3^{\circ}\text{C}$). In light of this study, a temperature of $>37.7^{\circ}\text{C}$ ($>99.9^{\circ}\text{F}$), which represents the 99th percentile for healthy individuals, defines a fever. Importantly, higher ambient temperatures are linked to higher baseline body temperatures. Additionally, body temperatures have diurnal and seasonal variation, with low levels at 8 A.M. and during summer and higher levels at 4 P.M. and during winter. Baseline temperatures are also affected by age (lower by 0.02°C for every 10-year increase in age), demographics (African-American women have temperatures 0.052°C higher than white men), and comorbid conditions (cancer is associated with 0.02°C higher temperatures; hypothyroidism is linked to temperatures lower by 0.01°C). After controlling for age, sex, race, vital signs, and comorbidities, an increase in baseline temperature of 0.15°C (or 1 standard deviation) intriguingly translates into a 0.52% absolute increase in 1-year mortality.

Rectal temperatures are generally 0.4°C (0.7°F) higher than oral readings. The lower oral readings are probably attributable to mouth breathing, which is a factor in patients with respiratory infections and rapid breathing. Lower-esophageal temperatures closely reflect core temperature. Tympanic membrane thermometers measure radiant heat from the tympanic membrane and nearby ear canal and display that absolute value (*unadjusted mode*) or a value automatically calculated from the absolute reading on the basis of nomograms relating the radiant temperature measured to actual core temperatures obtained in clinical studies (*adjusted mode*). These measurements, although convenient, may be more variable than directly determined oral or rectal values. Studies in adults show that readings are lower with unadjusted-mode than with adjusted-mode tympanic membrane thermometers and that unadjusted-mode tympanic membrane values are 0.8°C (1.6°F) lower than rectal temperatures.

In women who menstruate, the A.M. temperature is generally lower during the 2 weeks before ovulation; it then rises by $\sim 0.6^{\circ}\text{C}$ (1°F) with ovulation and stays at that level until menses occur. During the luteal phase, the amplitude of the circadian rhythm remains the same.

FEVER VERSUS HYPERTHERMIA

Fever is an elevation of body temperature that exceeds the normal daily variation and occurs in conjunction with an increase in the hypothalamic set point (e.g., from 37°C to 39°C). This shift of the set point from “normothermic” to febrile levels very much resembles the resetting of

the home thermostat to a higher level in order to raise the ambient temperature in a room. Once the hypothalamic set point is raised, neurons in the vasomotor center are activated and vasoconstriction commences. The individual first notices vasoconstriction in the hands and feet. Shunting of blood away from the periphery to the internal organs essentially decreases heat loss from the skin, and the person feels cold. For most fevers, body temperature increases by $1\text{--}2^{\circ}\text{C}$. Shivering, which increases heat production from the muscles, may begin at this time; however, shivering is not required if mechanisms of heat conservation raise blood temperature sufficiently. Nonshivering heat production from the liver also contributes to increasing core temperature. Behavioral adjustments (e.g., putting on more clothing or bedding) help raise body temperature by decreasing heat loss.

The processes of heat conservation (vasoconstriction) and heat production (shivering and increased nonshivering thermogenesis) continue until the temperature of the blood bathing the hypothalamic neurons matches the new “thermostat setting”. Once that point is reached, the hypothalamus maintains the temperature at the febrile level by the same mechanisms of heat balance that function in the afebrile state. When the hypothalamic set point is again reset downward (in response to either a reduction in the concentration of pyrogens or the use of antipyretics), the processes of heat loss through vasodilation and sweating are initiated. Loss of heat by sweating and vasodilation continues until the blood temperature at the hypothalamic level matches the lower setting. Behavioral changes (e.g., removal of clothing) facilitate heat loss.

A fever of $>41.5^{\circ}\text{C}$ ($>106.7^{\circ}\text{F}$) is called *hyperpyrexia*. This extraordinarily high fever can develop in patients with severe infections but most commonly occurs in patients with central nervous system (CNS) hemorrhages. In the preantibiotic era, fever due to a variety of infectious diseases rarely exceeded 106°F , and there has been speculation that this natural “thermal ceiling” is mediated by neuropeptides functioning as central antipyretics.

In rare cases, the hypothalamic set point is elevated as a result of local trauma, hemorrhage, tumor, or intrinsic hypothalamic malfunction. The term *hypothalamic fever* is sometimes used to describe elevated temperature caused by abnormal hypothalamic function. However, most patients with hypothalamic damage have *subnormal*, not *supranormal*, body temperatures.

Although most patients with elevated body temperature have fever, there are circumstances in which elevated temperature represents not fever but hyperthermia (heat stroke). Hyperthermia is characterized by an uncontrolled increase in body temperature that exceeds the body’s ability to lose heat. The setting of the hypothalamic thermoregulatory center is unchanged. In contrast to fever in infections, hyperthermia does not involve pyrogenic molecules. Exogenous heat exposure and endogenous heat production are two mechanisms by which hyperthermia can result in dangerously high internal temperatures. Excessive heat production can easily cause hyperthermia despite physiologic and behavioral control of body temperature. For example, work or exercise in hot environments can produce heat faster than peripheral mechanisms can lose it. **For a detailed discussion of hyperthermia, see Chap. 465.**

It is important to distinguish between fever and hyperthermia since hyperthermia can be rapidly fatal and characteristically does not respond to antipyretics. In an emergency situation, however, making this distinction can be difficult. For example, in systemic sepsis, fever (hyperpyrexia) can be rapid in onset, and temperatures can exceed 40.5°C (104.9°F). Hyperthermia is often diagnosed on the basis of the events immediately preceding the elevation of core temperature—e.g., heat exposure or treatment with drugs that interfere with thermoregulation. In patients with heat stroke syndromes and in those taking drugs that block sweating, the skin is hot but dry, whereas in fever, the skin can be cold as a consequence of vasoconstriction. Antipyretics do not reduce the elevated temperature in hyperthermia, whereas in fever—and even in hyperpyrexia—adequate doses of either aspirin or acetaminophen usually result in some decrease in body temperature.

PATHOGENESIS OF FEVER

■ PYROGENS

The term *pyrogen* (Greek *pyro*, “fire”) is used to describe any substance that causes fever. **Exogenous** pyrogens are derived from outside the patient; most are microbial products, microbial toxins, or whole microorganisms (including viruses). The classic example of an exogenous pyrogen is the **lipopolysaccharide (endotoxin)** produced by all gram-negative bacteria. Pyrogenic products of gram-positive organisms include the **enterotoxins of *Staphylococcus aureus*** and the **groups A and B streptococcal toxins, also called *superantigens***. One staphylococcal toxin of clinical importance is that associated with isolates of *S. aureus* from patients with toxic shock syndrome. These products of staphylococci and streptococci cause fever in experimental animals when injected intravenously at concentrations of 1–10 µg/kg. **Endotoxin** is a highly pyrogenic molecule in humans: when injected intravenously into volunteers, a dose of 2–3 ng/kg produces fever, leukocytosis, acute-phase proteins, and generalized symptoms of malaise.

■ PYROGENIC CYTOKINES

Cytokines are small proteins (molecular mass, 10,000–20,000 Da) that regulate immune, inflammatory, and hematopoietic processes. For example, the elevated leukocytosis seen in several infections with an **absolute neutrophilia is attributable to the cytokines interleukin (IL) 1 and IL-6**. Some cytokines also cause fever; formerly referred to as **endogenous pyrogens**, they are now called **pyrogenic cytokines**. The pyrogenic cytokines include IL-1, IL-6, tumor necrosis factor (TNF), and ciliary neurotropic factor, a member of the IL-6 family. Fever is a prominent side effect of **interferon α therapy**. Each pyrogenic cytokine is encoded by a separate gene, and each has been shown to cause fever in laboratory animals and in humans. When injected into humans at low doses (10–100 ng/kg), IL-1 and TNF produce fever; in contrast, for IL-6, a dose of 1–10 µg/kg is required for fever production.

A wide spectrum of **bacterial and fungal products induce the synthesis and release of pyrogenic cytokines**. However, fever can be a manifestation of disease in the absence of microbial infection. For example, **inflammatory processes such as pericarditis, trauma, stroke, and routine immunizations** induce the production of IL-1, TNF, and/or IL-6; individually or in combination, these cytokines trigger the hypothalamus to raise the set point to febrile levels.

■ ELEVATION OF THE HYPOTHALAMIC SET POINT BY CYTOKINES

During fever, levels of prostaglandin E_2 (PGE₂) are elevated in hypothalamic tissue and the third cerebral ventricle. The concentrations of PGE₂ are highest near the circumventricular vascular organs (**organum vasculosum of lamina terminalis**)—networks of enlarged capillaries surrounding the hypothalamic regulatory centers. **Destruction of these organs reduces the ability of pyrogens to produce fever**. Most studies in animals have failed to show, however, that **pyrogenic cytokines pass from the circulation into the brain itself**. Thus, it appears that both exogenous pyrogens and pyrogenic cytokines interact with the endothelium of these capillaries and that this interaction is the first step in initiating fever—i.e., in raising the set point to febrile levels.

The key events in the production of fever are illustrated in **Fig. 18-1**. Myeloid and endothelial cells are the primary cell types that produce pyrogenic cytokines. Pyrogenic cytokines such as IL-1, IL-6, and TNF are released from these cells and enter the systemic circulation. Although these circulating cytokines lead to fever by inducing the synthesis of PGE₂, they also induce PGE₂ in peripheral tissues. The increase in PGE₂ in the periphery accounts for the nonspecific myalgias and arthralgias that often accompany fever. It is thought that some systemic PGE₂ escapes destruction by the lung and gains access to the hypothalamus via the internal carotid. However, it is the elevation of PGE₂ in the brain that starts the process of raising the hypothalamic set point for core temperature.

There are four receptors for PGE₂, and each signals the cell in different ways. Of the four receptors, the third (EP-3) is essential for fever: when the gene for this receptor is deleted in mice, no fever follows the

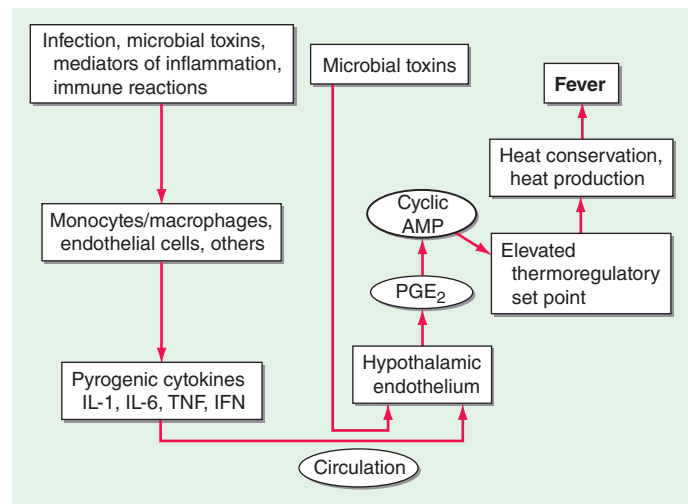


FIGURE 18-1 Chronology of events required for the induction of fever. AMP, adenosine 5'-monophosphate; IFN, interferon; IL, interleukin; PGE₂, prostaglandin E₂; TNF, tumor necrosis factor.

injection of IL-1 or endotoxin. Deletion of the other PGE₂ receptor genes leaves the fever mechanism intact. Although PGE₂ is essential for fever, it is not a neurotransmitter. Rather, the release of PGE₂ from the brain side of the hypothalamic endothelium triggers the PGE₂ receptor on glial cells, and this stimulation results in the rapid release of cyclic adenosine 5'-monophosphate (cAMP), which is a neurotransmitter. As shown in Fig. 18-1, the release of cAMP from glial cells activates neuronal endings from the thermoregulatory center that extend into the area. The elevation of cAMP is thought to account for changes in the hypothalamic set point either directly or indirectly (by inducing the release of neurotransmitters). Distinct receptors for microbial products are located on the hypothalamic endothelium. These receptors are called **Toll-like receptors** and are similar in many ways to IL-1 receptors. IL-1 receptors and Toll-like receptors share the same signal-transducing mechanism. Thus, the **direct activation of Toll-like receptors or IL-1 receptors results in PGE₂ production and fever**.

■ PRODUCTION OF CYTOKINES IN THE CNS

Cytokines produced in the brain may account for the hyperpyrexia of CNS hemorrhage, trauma, or infection. Viral infections of the CNS induce microglial and possibly neuronal production of IL-1, TNF, and IL-6. In experimental animals, the concentration of a cytokine required to cause fever is several orders of magnitude lower with direct injection into the brain substance or brain ventricles than with systemic injection. Therefore, cytokines produced in the CNS can raise the hypothalamic set point, bypassing the circumventricular organs. CNS cytokines likely account for the hyperpyrexia of CNS hemorrhage, trauma, or infection.

APPROACH TO THE PATIENT

Fever

HISTORY AND PHYSICAL EXAMINATION

There are a range of disease processes that present with fever as a cardinal manifestation, and a thorough history can help distinguish between these broad categories (**Table 18-1**). The chronology of events preceding fever, including exposure to other symptomatic individuals or to vectors of disease, should be ascertained. Electronic devices for measuring oral, tympanic membrane, or rectal temperatures are reliable, but the same site should be used consistently to monitor a febrile disease. Moreover, physicians should be aware that newborns, elderly patients, patients with chronic hepatic or renal failure, and patients taking glucocorticoids or being treated with an anticytokine may have active disease in the absence of fever because of a blunted febrile response.

TABLE 18-1 Disease Categories That Present with Fever as a Cardinal Sign

Infectious diseases

Autoimmune and noninfectious inflammatory disorders

Cancer

Medication related (e.g., vaccines, drug fever)

Endocrine disorders (e.g., hyperthyroidism)

Intrinsic hypothalamic malfunction

LABORATORY TESTS

The workup should include a complete blood count; a differential count should be performed manually or with an instrument sensitive to the identification of juvenile or band forms, toxic granulations, and Döhle bodies, which are suggestive of bacterial infection. Neutropenia may be present with some viral infections.

Measurement of circulating cytokines in patients with fever is not helpful since levels of cytokines such as IL-1 and TNF in the circulation often are below the detection limit of the assay or do not coincide with fever. However, in patients with low-grade fevers or with suspected occult disease, the most valuable measurements are the C-reactive protein (CRP) level and the erythrocyte sedimentation rate. These markers of inflammatory processes are particularly helpful in detecting occult disease. Measurement of circulating IL-6, which induces CRP, can be useful. However, whereas IL-6 levels may vary during a febrile disease, CRP levels remain elevated. Acute-phase reactants are discussed in Chap. 304.

FEVER IN PATIENTS RECEIVING ANTICYTOKINE THERAPY

Patients receiving long-term treatment with anticytokine-based regimens are at increased risk of infection because of lowered host defenses. For example, latent *Mycobacterium tuberculosis* infection can disseminate in patients receiving anti-TNF therapy. With the increasing use of anticytokines to reduce the activity of IL-1, IL-6, IL-12, IL-17, or TNF in patients with Crohn's disease, rheumatoid arthritis, or psoriasis, the possibility that these therapies blunt the febrile response should be kept in mind.

The blocking of cytokine activity has the distinct clinical drawback of lowering the level of host defenses against both routine bacterial and opportunistic infections such as *M. tuberculosis* and fungal infections. The use of monoclonal antibodies to reduce IL-17 in psoriasis increases the risk of systemic candidiasis.

In nearly all reported cases of infection associated with anticytokine therapy, fever is among the presenting signs. However, the extent to which the febrile response is blunted in these patients remains unknown. Therefore, low-grade fever in patients receiving anticytokine therapies is of considerable concern. The physician should conduct an early and rigorous diagnostic evaluation in these cases. The febrile response is also blunted in patients receiving chronic glucocorticoid therapy or anti-inflammatory agents such as nonsteroidal anti-inflammatory drugs (NSAIDs).

TREATMENT**Fever****THE DECISION TO TREAT FEVER**

In deciding whether to treat fever, it is important to remember that fever itself is not an illness; it is an ordinary response to a perturbation of normal host physiology. Most fevers are associated with self-limited infections, such as common viral diseases. The use of antipyretics is not contraindicated in these infections; no significant clinical evidence indicates either that antipyretics delay the resolution of viral or bacterial infections or that fever facilitates recovery from infection or acts as an adjuvant to the immune system. In short, treatment of fever and its symptoms with routine antipyretics

does no harm and does not slow the resolution of common viral and bacterial infections.

However, in bacterial infections, the withholding of antipyretic therapy can be helpful in evaluating the effectiveness of a particular antibiotic, especially in the absence of positive cultures of the infecting organism, and the routine use of antipyretics can mask an inadequately treated bacterial infection. Withholding antipyretics in some cases may facilitate the diagnosis of an unusual febrile disease. Temperature-pulse dissociation (*relative bradycardia*) occurs in typhoid fever, brucellosis, leptospirosis, some drug-induced fevers, and factitious fever. As stated earlier, in newborns, elderly patients, patients with chronic liver or kidney failure, and patients taking glucocorticoids, fever may not be present despite infection. Hypothermia can develop in patients with septic shock.

Some infections have characteristic patterns in which febrile episodes are separated by intervals of normal temperature. For example, *Plasmodium vivax* causes fever every third day, whereas fever occurs every fourth day with *Plasmodium malariae*. Another relapsing fever is related to *Borrelia* infection, with days of fever followed by a several-day afebrile period and then a relapse into additional days of fever. In the Pel-Ebstein pattern, fever lasting 3–10 days is followed by afebrile periods of 3–10 days; this pattern can be classic for Hodgkin's disease and other lymphomas. In cyclic neutropenia, fevers occur every 21 days and accompany the neutropenia. There are also a number of periodic fever syndromes (e.g., familial Mediterranean fever, TNF receptor-associated periodic syndrome [TRAPS]) that differ in their periodicity, duration of attack, constellation of clinical features, genetic causes, and therapies (Chap. 369). Understanding these clinical differences can help tailor diagnostic testing to confirm the diagnosis and guide therapy.

ANTICYTOKINE THERAPY TO REDUCE FEVER IN AUTOIMMUNE AND AUTOINFLAMMATORY DISEASES

Recurrent fever is documented at some point in most autoimmune diseases and many autoinflammatory diseases, which include the periodic fever syndromes as well as disorders of inflammasomes (e.g., NLRP3, pyrin) and other components of the innate immune system (Chap. 349). Although fever can be a manifestation of autoimmune diseases, recurrent fevers are characteristic of autoinflammatory diseases, including uncommon diseases such as adult and juvenile Still's disease, familial Mediterranean fever, and hyper-IgD syndrome but also common diseases such as idiopathic pericarditis and gout. In addition to recurrent fevers, neutrophilia and serosal inflammation characterize autoinflammatory diseases. The fevers associated with many of these illnesses are dramatically reduced by blocking of IL-1 activity with anakinra or canakinumab. Anticytokines therefore reduce fever in autoimmune and autoinflammatory diseases. Although fevers in autoinflammatory diseases are mediated by IL-1 β , patients also respond to antipyretics.

MECHANISMS OF ANTIPYRETIC AGENTS

The reduction of fever by lowering of the elevated hypothalamic set point is a direct function of reduction of the PGE₂ level in the thermoregulatory center. The synthesis of PGE₂ depends on the constitutively expressed enzyme cyclooxygenase. The substrate for cyclooxygenase is arachidonic acid released from the cell membrane, and this release is the rate-limiting step in the synthesis of PGE₂. Therefore, inhibitors of cyclooxygenase are potent antipyretics. The antipyretic potency of various drugs is directly correlated with the inhibition of brain cyclooxygenase. Acetaminophen is a poor cyclooxygenase inhibitor in peripheral tissue and lacks noteworthy anti-inflammatory activity; in the brain, however, acetaminophen is oxidized by the P450 cytochrome system, and the oxidized form inhibits cyclooxygenase activity. Moreover, in the brain, the inhibition of another enzyme, COX-3, by acetaminophen may account for the antipyretic effect of this agent. However, COX-3 is not found outside the CNS.

Oral aspirin and acetaminophen are equally effective in reducing fever in humans. NSAIDs such as ibuprofen and specific inhibitors of COX-2 also are excellent antipyretics. Chronic, high-dose

therapy with antipyretics such as aspirin or any NSAID does not reduce normal core body temperature. Thus, PGE₂ appears to play no role in normal thermoregulation.

As effective antipyretics, glucocorticoids act at two levels. First, similar to the cyclooxygenase inhibitors, glucocorticoids reduce PGE₂ synthesis by inhibiting the activity of phospholipase A₂, which is needed to release arachidonic acid from the cell membrane. Second, glucocorticoids block the transcription of the mRNA for the pyrogenic cytokines. Limited experimental evidence indicates that ibuprofen and COX-2 inhibitors reduce IL-1-induced IL-6 production and may contribute to the antipyretic activity of NSAIDs.

REGIMENS FOR THE TREATMENT OF FEVER

The objectives in treating fever are first to reduce the elevated hypothalamic set point and second to facilitate heat loss. Reducing fever with antipyretics also reduces systemic symptoms of headache, myalgias, and arthralgias.

Oral aspirin and NSAIDs effectively reduce fever but can adversely affect platelets and the gastrointestinal tract. Therefore, acetaminophen is preferred as an antipyretic. In children, acetaminophen or oral ibuprofen must be used because aspirin increases the risk of Reye's syndrome. If the patient cannot take oral antipyretics, parenteral preparations of NSAIDs and rectal suppositories of various antipyretics can be used.

Treatment of fever in some patients is highly recommended. Fever increases the demand for oxygen (i.e., for every increase of 1°C over 37°C, there is a 13% increase in oxygen consumption) and can aggravate the condition of patients with preexisting impairment of cardiac, pulmonary, or CNS function. Children with a history of febrile or nonfebrile seizure should be aggressively treated to reduce fever. However, it is unclear what triggers the febrile seizure, and there is no correlation between absolute temperature elevation and onset of a febrile seizure in susceptible children.

In hyperpyrexia, the use of cooling blankets facilitates the reduction of temperature; however, cooling blankets should not be used without oral antipyretics. In hyperpyretic patients with CNS disease or trauma (CNS bleeding), reducing core temperature mitigates the detrimental effects of high temperature on the brain.

For a discussion of treatment for hyperthermia, see Chap. 465.

FURTHER READING

- DINARELLO CA et al: Treating inflammation by blocking interleukin-1 in a broad spectrum of diseases. *Nature Rev* 11:633, 2012.
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19

Fever and Rash

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The acutely ill patient with fever and rash often presents a diagnostic challenge for physicians, yet the distinctive appearance of an eruption in concert with a clinical syndrome can facilitate a prompt diagnosis and the institution of life-saving therapy or critical infection-control interventions. **Representative images of many of the rashes discussed in this chapter are included in Chap. A1.**

APPROACH TO THE PATIENT

Fever and Rash

A thorough history of patients with fever and rash includes the following relevant information: immune status, medications taken within the previous month, specific travel history, immunization status, exposure to domestic pets and other animals, history of animal (including arthropod) bites, recent dietary exposures, existence of cardiac abnormalities, presence of prosthetic material, recent exposure to ill individuals, and sexual exposures. The history should also include the site of onset of the rash and its direction and rate of spread.

PHYSICAL EXAMINATION

A thorough physical examination entails close attention to the rash, with an assessment and precise definition of its salient features. First, it is critical to determine what *type* of lesions make up the eruption. *Macules* are flat lesions defined by an area of changed color (i.e., a blanchable erythema). *Papules* are raised, solid lesions <5 mm in diameter; *plaques* are lesions >5 mm in diameter with a flat, plateau-like surface; and *nodules* are lesions >5 mm in diameter with a more rounded configuration. *Wheals* (urticaria, hives) are papules or plaques that are pale pink and may appear annular (ringlike) as they enlarge; classic (nonvasculitic) wheals are transient, lasting only 24 h in any defined area. *Vesicles* (<5 mm) and *bullae* (>5 mm) are circumscribed, elevated lesions containing fluid. *Pustules* are raised lesions containing purulent exudate; vesicular processes such as varicella or herpes simplex may evolve to pustules. *Nonpalpable purpura* is a flat lesion that is due to bleeding into the skin. If <3 mm in diameter, the purpuric lesions are termed *petechiae*; if >3 mm, they are termed *ecchymoses*. *Palpable purpura* is a raised lesion that is due to inflammation of the vessel wall (vasculitis) with subsequent hemorrhage. An *ulcer* is a defect in the skin extending at least into the upper layer of the dermis, and an *eschar* (tâche noire) is a necrotic lesion covered with a black crust.

Other pertinent features of rashes include their *configuration* (i.e., annular or target), the *arrangement* of their lesions, and their *distribution* (i.e., central or peripheral).

For further discussion, see Chaps. 56, 58, 122, and 129.

CLASSIFICATION OF RASH

This chapter reviews rashes that reflect systemic disease, but it does not include localized skin eruptions (i.e., cellulitis, impetigo) that may also be associated with fever (**Chap. 129**). The chapter is not intended to be all-inclusive, but it covers the most important and most common diseases associated with fever and rash. Rashes are classified herein on the basis of lesion morphology and distribution. For practical purposes, this classification system is based on the most typical disease presentations. However, morphology may vary as rashes evolve, and the presentation of diseases with rashes is subject to many variations (**Chap. 58**). For instance, the classic petechial rash of Rocky Mountain spotted fever (**Chap. 187**) may initially consist of blanchable erythematous macules distributed peripherally; at times, however, the rash associated with this disease may not be predominantly acral, or no rash may develop at all.

Diseases with fever and rash may be classified by type of eruption: centrally distributed maculopapular, peripheral, confluent desquamative erythematous, vesicubullous, urticaria-like, nodular, purpuric, ulcerated, or with eschars. Diseases are listed by these categories in **Table 19-1**, and many are highlighted in the text. However, for a more detailed discussion of each disease associated with a rash, the reader is referred to the chapter dealing with that specific disease. (**Reference chapters are cited in the text and listed in Table 19-1.**)

CENTRALLY DISTRIBUTED MACULOPAPULAR ERUPTIONS

Centrally distributed rashes, in which lesions are primarily truncal, are the most common type of eruption. The rash of *rubeola* (measles) starts

Section 1 Basic Considerations in Infectious Diseases

119 Approach to the Patient with an Infectious Disease

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HISTORICAL PERSPECTIVE

The origins of the field of infectious diseases are humble. The notion that communicable diseases were due to a *miasma* (“bad air”) can be traced back to at least the mid-sixteenth century. Not until the work of Louis Pasteur and Robert Koch in the late nineteenth century was there credible evidence supporting the germ theory of disease—i.e., that microorganisms are the direct cause of infections. In contrast to this relatively slow start, the twentieth century saw remarkable advances in the field of infectious diseases, and the etiologic agents of numerous infectious diseases were soon identified. Furthermore, the discovery of antibiotics and the advent of vaccines against some of the most deadly and debilitating infections greatly altered the landscape of human health. Indeed, the twentieth century saw the elimination of smallpox, one of the great scourges in the history of humanity. These remarkable successes prompted Sir Frank MacFarlane Burnet, a noted immunologist and Nobel laureate, to write in a 1962 publication entitled *Natural History of Infectious Diseases*: “In many ways one can think of the middle of the twentieth century as the end of one of the most important social revolutions in history, the virtual elimination of infectious disease.” Professor Burnet was not alone in this view. Robert Petersdorf, a renowned infectious disease expert and former editor of this textbook, wrote in 1978 that “even with my great personal loyalties to infectious diseases, I cannot conceive a need for 309 more [graduating trainees in infectious diseases] unless they spend their time culturing each other.” Given the enormous growth of interest in the microbiome in the past 15 years, Dr. Petersdorf’s statement might have been ironically clairvoyant, although he could have had no idea what was in store for humanity, with an onslaught of new, emerging, and reemerging infectious diseases.

Clearly, even with all the advances of the twentieth century, infectious diseases continue to represent a formidable challenge for patients and physicians alike. Furthermore, during the latter half of the century, several chronic diseases were demonstrated to be directly or indirectly caused by infectious microbes; perhaps the most notable examples are the associations of *Helicobacter pylori* with peptic ulcer disease and gastric carcinoma, human papillomavirus with cervical cancer, and hepatitis B and C viruses with liver cancer. In fact, ~16% of all malignancies are now known to be associated with an infectious cause. In addition, numerous emerging and reemerging infectious diseases continue to have a dire impact on global health: HIV/AIDS, SARS-CoV-2, Ebola, and Zika are but a few examples. The fear of weaponizing pathogens for bioterrorism is ever present and poses a potentially enormous threat to public health. Moreover, escalating antimicrobial resistance in clinically relevant microbes (e.g., carbapenem-resistant Enterobacteriaceae and *Acinetobacter* spp., *Candida auris*, drug-resistant *Mycobacterium tuberculosis*, and vancomycin-resistant enterococci) signifies that the administration of antimicrobial agents—once thought to be a panacea—requires appropriate stewardship. For all these reasons, infectious diseases continue to exert grim effects on individual patients as well as on international public health. Even with all the successes of the past century, physicians must be as thoughtful about infectious diseases now as they were at the beginning of the twentieth century.

GLOBAL CONSIDERATIONS

Infectious diseases remain the second leading cause of death worldwide. Although the rate of infectious disease–related deaths has decreased dramatically over the past 25 years, there were still 10.3 million such deaths in 2017 (Fig. 119-1A). These deaths disproportionately affect children <1 year of age, adults older than 70 years, and persons living in low- and middle-income countries (Fig. 119-1B and 119-1C; Chap. 474); in 2017, ~18% of all deaths worldwide were related to infectious diseases, with a rate as high as ~58% in sub-Saharan Africa.

Given that infectious diseases are still a major cause of global mortality, understanding the local epidemiology of disease is critically important in evaluating patients. Diseases such as HIV/AIDS have decimated sub-Saharan Africa, with HIV-infected adults representing 20–23% of the total population in countries like South Africa, Botswana, and Lesotho. Moreover, drug-resistant tuberculosis is rampant throughout the former Soviet-bloc countries, India, China, and South Africa. The ready availability of this type of information allows physicians to develop appropriate differential diagnoses and treatment plans for individual patients. Programs such as the Global Burden of Disease seek to quantify human losses (e.g., deaths, disability-adjusted life-years) due to diseases by age, sex, and country over time; these data not only help inform local, national, and international health policy but can also help guide local medical decision-making.

Even though some diseases (e.g., pandemic influenza, Middle East respiratory syndrome) are seemingly geographically restricted, the increasing ease of rapid worldwide travel has raised concern about their swift spread around the globe. Indeed, human migration has historically been the source of epidemics: *Yersinia pestis* spread along trade routes in the fourteenth century, Native American populations were devastated by diseases such as smallpox and measles that were imported by European explorers in the fifteenth and sixteenth centuries, military maneuvers helped facilitate the spread of the 1918 influenza pandemic, and religious pilgrimages (e.g., the Hajj) provide the means for worldwide dissemination of diseases. The continued effects of global travel on the spread of infectious diseases are perhaps best highlighted by the SARS-CoV-2 pandemic (Chap. 199). Although this virus was first identified in Wuhan, China, it quickly spread across the globe and brought an abrupt end to virtually all travel and commerce throughout the world, plunging economies into a deep recession, and resulting at one point in over half the world’s population living under stay-at-home orders. Not only can travelers carry person-to-person transmitted infections (e.g., influenza, HIV) anywhere in the world, but they can also introduce vector-borne infections to new geographic areas (e.g., chikungunya and Zika viruses) and contribute to the worldwide spread of multidrug-resistant organisms. The world’s increasing interconnectedness has profound implications not only for the global economy but also for medicine and the spread of infectious diseases.

UNDERSTANDING THE MICROBIOTA

Normal, healthy humans are colonized with ~50 trillion bacteria as well as countless viruses, fungi, and archaea; taken together, these microorganisms outnumber human cells by ~10 times in the human body (Chap. 471). The major reservoir of these microbes is the gastrointestinal tract, but substantial numbers of microbes live in the female genital tract, the oral cavity, and the nasopharynx. There is increasing interest in the skin and lungs as sites where microbial colonization might be highly relevant to the biology and disease susceptibility of the host. These commensal organisms provide the host with myriad benefits, from aiding in metabolism to shaping the immune system. With regard to infectious diseases, the vast majority of infections are caused by organisms that are part of the normal microbiota (e.g., *Staphylococcus aureus*, *Streptococcus pneumoniae*, *Pseudomonas aeruginosa*), with relatively few infections due to organisms that are strictly pathogens (e.g., *Neisseria gonorrhoeae*, rabies virus). Perhaps it is not surprising that a general understanding of the microbiota is essential in the evaluation of infectious diseases.

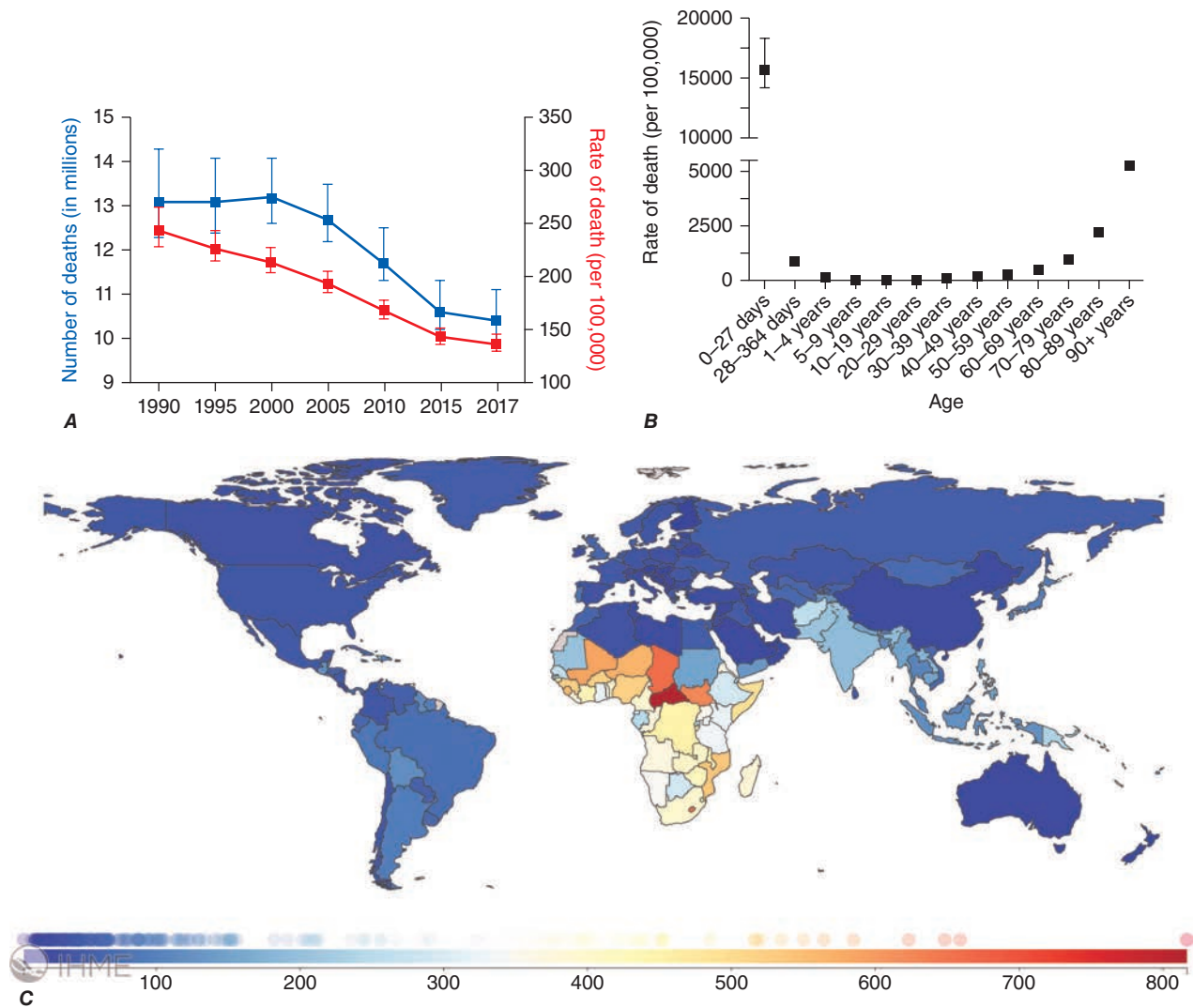


FIGURE 119-1 Magnitude of infectious disease-related deaths globally. **A.** The absolute number (blue line; left axis) and rate (red line; right axis) of infectious disease-related deaths throughout the world since 1990. **B.** Age-specific rates of infectious disease-related deaths in 2017. In both **A** and **B**, the charts depict the mean estimate and 95% uncertainty intervals. **C.** A map depicting country-specific data for the rate of total deaths that were attributable to communicable, maternal, neonatal, and nutritional disorders in 2017. (Source: Global Burden of Disease Study, Institute for Health Metrics and Evaluation.)

Individuals' microbiotas have a major impact on their susceptibility to infectious diseases and even their responses to vaccines. Site-specific knowledge of the indigenous microbiota may facilitate appropriate interpretation of culture results, aid in selection of empirical antimicrobial therapy based on the likely causative agents, and provide additional impetus for rational antibiotic use to minimize the untoward effects of these drugs on the "beneficial" microbes that inhabit the body.

■ WHEN TO CONSIDER AN INFECTIOUS ETIOLOGY

The title of this chapter may appear to presuppose that the physician knows when a patient has an infectious disease. In reality, this chapter can serve only as a guide to the evaluation of a patient in whom an infectious disease is a possibility. Once a specific diagnosis is made, the reader should consult the subsequent chapters that deal with specific microorganisms in detail. The challenge for the physician is to recognize which patients may have an infectious disease as opposed to some other underlying disorder. This task is greatly complicated by the fact that infections have an infinite range of presentations, from acute life-threatening conditions (e.g., meningococemia) to chronic diseases of varying severity (e.g., *H. pylori*-associated peptic ulcer disease) to no symptoms at all (e.g., latent *M. tuberculosis* infection). While it is impossible to generalize about a presentation that encompasses all infections, common findings in the history, physical examination, and

basic laboratory testing often suggest that the patient either has an infectious disease or should be more closely evaluated for one. This chapter focuses on these common findings and how they may direct the ongoing evaluation of the patient.

APPROACH TO THE PATIENT

Infectious Disease

See also Chap. 122.

HISTORY

As in all of medicine, a complete and thorough history is paramount in the evaluation of a patient with a possible infectious disease. The history is critical for developing a focused differential diagnosis and for guiding the physical exam and initial diagnostic testing. Although a detailing of all the elements of a history is beyond the scope of this chapter, specific components relevant to infectious diseases require particular attention. In general, these aspects focus on two areas: (1) an exposure history that may identify microorganisms with which the patient may have come into contact and (2) host-specific factors that may predispose to the development of an infection.

Exposure History • History of infections or exposure to drug-resistant microbes

Information about a patient's previous infections, with the associated microbial susceptibility profiles, is very helpful in determining possible etiologic agents. Specifically, knowing whether a patient has a history of infection with drug-resistant organisms (e.g., methicillin-resistant *S. aureus*, vancomycin-resistant *Enterococcus* species, enteric organisms that produce an extended-spectrum β -lactamase or carbapenemase) or may have been exposed to drug-resistant microbes (e.g., during a recent stay in a hospital, nursing home, or long-term acute-care facility) may alter the choice of empirical antibiotics. For example, a patient presenting with sepsis who is known to have a history of invasive infection with a multidrug-resistant isolate of *P. aeruginosa* should be treated empirically with an antimicrobial regimen that will cover this strain.

Social history Although the social history taken by physicians is often limited to inquiries about a patient's alcohol and tobacco use, a complete social history can offer a number of clues to the underlying diagnosis. Knowing whether the patient has any high-risk behaviors (e.g., unsafe sexual behaviors, intravenous [IV] drug use), potential hobby-associated exposures (e.g., avid gardening, with possible *Sporothrix schenckii* exposure), or occupational exposures (e.g., increased risk for *M. tuberculosis* exposure in funeral service workers) can facilitate diagnosis. The importance of the social history is exemplified by a case in 2009 in which a laboratory researcher died of a *Y. pestis* infection acquired during his work; although this patient had visited both an outpatient clinic and an emergency department, his records at both sites failed to include his occupation—information that potentially could have led quickly to appropriate treatment and infection control measures.

Dietary habits Because certain pathogens are associated with specific dietary habits, inquiring about a patient's diet can provide insight into possible exposures. For example, Shiga toxin-producing strains of *Escherichia coli*, and *Toxoplasma gondii* are associated with the consumption of raw or undercooked meat; *Salmonella typhimurium*, *Listeria monocytogenes*, and *Mycobacterium bovis* with unpasteurized milk; *Leptospira* species, parasites, and enteric bacteria with unpurified water; and *Vibrio* species, norovirus, helminths, and protozoa with raw seafood.

Animal exposures Because animals are often important vectors of infectious diseases, patients should be asked about exposures to any animals, including contact with their own pets, visits to petting zoos, or random encounters (e.g., home rodent infestation). For example, dogs can carry ticks that serve as agents for the transmission of several infectious diseases, including Lyme disease, Rocky Mountain spotted fever, and ehrlichiosis. Cats are associated with *Bartonella henselae* infection, reptiles with *Salmonella* infection, rodents with leptospirosis, and rabbits with tularemia (Chap. 141).

Travel history Attention should be paid to both international and domestic travel. Fever in a patient who has recently returned from abroad significantly broadens the differential diagnosis (Chap. 124); even a remote history of international travel may reflect patients' exposure to infections with pathogens such as *M. tuberculosis* or *Strongyloides stercoralis*. Similarly, domestic travel may have exposed patients to pathogens that are not normally found in their local environment and therefore may not routinely be considered in the differential diagnosis. For example, a patient who has recently visited California or Martha's Vineyard may have been exposed to *Coccidioides immitis* or *Francisella tularensis*, respectively. Beyond simply identifying locations that a patient may have visited, the physician needs to delve deeper to learn what kinds of activities and behaviors the patient engaged in during travel (e.g., the types of food and sources of water consumed, freshwater swimming, animal exposures) and whether the patient had the necessary immunizations and/or took the necessary prophylactic medications prior to travel; these additional exposures, which the patient may

not think to report without specific prompting, are as important as exposures during a patient's routine daily living.

Host-Specific Factors Because many opportunistic infections (e.g., with *Pneumocystis jirovecii*, *Aspergillus* species, or JC virus) affect primarily immunocompromised patients, it is of vital importance to determine the immune status of the patient. Defects in the immune system may be due to an underlying disease (e.g., malignancy, HIV infection, malnutrition), a medication (e.g., chemotherapy, glucocorticoids, monoclonal antibodies to components of the immune system), a treatment modality (e.g., total body irradiation, splenectomy), or a primary immunodeficiency. The type of infection for which the patient is at increased risk varies with the specific type of immune defect. In concert with determining whether a patient is immunocompromised for any reason, the physician should review the immunization record to ensure that the patient is adequately protected against vaccine-preventable diseases (Chap. 123).

PHYSICAL EXAMINATION

Like the history, a thorough physical examination is crucial in evaluating patients with an infectious disease. Some elements of the physical exam (e.g., skin, lymphatics) that are often performed in a cursory manner as a result of the ever-increasing pace of medical practice may help identify the underlying diagnosis. Moreover, serial exams are critical since new findings may appear as the illness progresses. A description of all the elements of a physical exam is beyond the scope of this chapter, but the following components have particular relevance to infectious diseases.

Vital Signs Given that elevations in temperature are often a hallmark of infection, paying close attention to the temperature may be of value in diagnosing an infectious disease. The idea that 37°C (98.6°F) is the normal human body temperature dates back to the nineteenth century and was initially based on axillary measurements. Rectal temperatures more accurately reflect the core body temperature and are 0.4°C (0.7°F) and 0.8°C (1.4°F) higher than oral and axillary temperatures, respectively. This idea of a "normal" body temperature does not take into account the fact that temperatures tend to be higher later in the day, in women, and in younger people. Moreover, the average body temperature seems to have dropped ~0.03°C every decade since the early 1800s to a new normal of ~36.7°C. Although the definition of fever varies greatly throughout the medical literature, the most common definition, which is based on studies defining fever of unknown origin (Chap. 20), uses a core temperature $\geq 38.3^{\circ}\text{C}$ ($\geq 101^{\circ}\text{F}$). Although fever is very commonly associated with infection, it is also documented in many other diseases (Chap. 18). For every 1°C (1.8°F) increase in core temperature, the heart rate typically rises by 15–20 beats/min. Table 119-1 lists infections that are associated with relative bradycardia (*Faget's sign*), where patients have a lower heart rate than might be expected for a given body temperature. Although this pulse-temperature dissociation is not highly sensitive or specific for establishing a diagnosis, it is potentially useful in low-resource settings given its ready availability and simplicity.

Lymphatics There are ~600 lymph nodes throughout the body, and infections are an important cause of lymphadenopathy. A physical examination should include evaluation of lymph nodes in multiple regions (e.g., popliteal, inguinal, epitrochlear, axillary, multiple cervical regions), with notation of the location, size (normal, <1 cm), presence or absence of tenderness, and consistency (soft, firm, or rubbery) and of whether the nodes are matted (i.e., connected and moving together). Nodes that are small and firm can also be described as "shotty," referring to the size and consistency of buckshot pellets. Of note, palpable epitrochlear nodes are always pathologic. Of patients presenting with lymphadenopathy, 75% have localized findings, and the remaining 25% have generalized lymphadenopathy (i.e., that involving more than one anatomic region). Localized lymphadenopathy in the head and neck region is

TABLE 119-1 Causes of Relative Bradycardia

Infectious Causes	
Intracellular organisms	
Gram-negative bacteria	<i>Salmonella typhi</i> <i>Francisella tularensis</i> <i>Brucella</i> spp. <i>Coxiella burnetii</i> (Q fever) <i>Leptospira interrogans</i> <i>Legionella pneumophila</i> <i>Mycoplasma pneumoniae</i>
Tick-borne organisms	<i>Rickettsia</i> spp. <i>Orientia tsutsugamushi</i> (scrub typhus) <i>Babesia</i> spp.
Other	<i>Corynebacterium diphtheriae</i> <i>Plasmodium</i> spp. (malaria)
Viruses/viral infections	Yellow fever virus Dengue virus Viral hemorrhagic fevers ^a Viral myocarditis
Noninfectious Causes	
	Drug fever Beta blocker use Central nervous system lesions Malignant lymphoma Factitious fever

^aPrimarily early in the course of infection with Marburg or Ebola virus.

found in 55% of patients, inguinal lymphadenopathy in 14%, and axillary lymphadenopathy in 5%. Determining whether the patient has generalized versus localized lymphadenopathy can help narrow the differential diagnosis, as various infections present differently.

Skin The fact that many infections have cutaneous manifestations gives the skin examination particular importance in the evaluation of patients (Chaps. 19, 58, 129, and A1). It is important to perform a complete skin exam, with attention to both front and back. Specific rashes are often extremely helpful in narrowing the differential diagnosis of an infection (Chaps. 19 and A1). In numerous anecdotal instances, patients in the intensive care unit have had “fever of unknown origin” that was actually due to unrecognized pressure ulcers. Moreover, close examination of the distal extremities for splinter hemorrhages, Janeway lesions, or Osler’s nodes may yield evidence of endocarditis or other causes of septic emboli.

Foreign Bodies As previously mentioned, many infections are caused by members of the indigenous microbiota. These infections typically occur when these microbes escape their normal habitat and enter a new one. Thus, maintenance of epithelial barriers is one of the most important mechanisms in protection against infection. However, hospitalization of patients is often associated with breaches of these barriers—e.g., due to placement of IV lines, surgical drains, or tubes (e.g., endotracheal tubes and Foley catheters) that allow microorganisms to localize in sites to which they normally would not have access (Chap. 142). Accordingly, knowing what lines, tubes, and drains are in place is helpful in ascertaining what body sites might be infected.

DIAGNOSTIC TESTING

Laboratory and radiologic testing has advanced greatly over the past few decades and has become an important component in the evaluation of patients. The dramatic increase in the number of serologic diagnostics, antigen tests, and molecular diagnostics available to the physician has, in fact, revolutionized medical care. However, all of these tests should be viewed as adjuncts to the history and physical examination—not a replacement for them. The selection

of initial tests should be based directly on the patient’s history and physical exam findings. Moreover, diagnostic testing should generally be limited to those conditions that are reasonably likely and treatable, important in terms of public health considerations, and/or capable of providing a definitive diagnosis that will consequently limit other testing.

White Blood Cell (WBC) Count Elevations in the WBC count are often associated with infection, though many viral infections are associated with leukopenia. It is important to assess the WBC differential, given that different classes of microbes are associated with various leukocyte types. For example, bacteria are associated with an increase in polymorphonuclear neutrophils, often with elevated levels of earlier developmental forms such as bands; viruses are associated with an increase in lymphocytes; and certain parasites are associated with an increase in eosinophils. Table 119-2 lists the major infectious causes of eosinophilia.

Inflammatory Markers The erythrocyte sedimentation rate (ESR) and the C-reactive protein (CRP) level are indirect and direct measures of the acute-phase response, respectively, that can be used to assess a patient’s general level of inflammation. Moreover, these markers can be followed serially over time to monitor disease progress/resolution. It is noteworthy that the ESR changes relatively slowly, and its measurement more often than weekly usually is not useful; in contrast, CRP concentrations change rapidly, and daily measurements can be useful in the appropriate context. Although these markers are sensitive indicators of inflammation, neither is very specific. An extremely elevated ESR (>100 mm/h) has a 90% predictive value for a serious underlying disease (Table 119-3). Work is ongoing to identify other potentially useful inflammatory markers (e.g., procalcitonin, serum amyloid A protein); however, their clinical utility requires further validation.

Analysis of Cerebrospinal Fluid (CSF) Assessment of CSF is critical for patients with suspected meningitis or encephalitis. An opening pressure should always be recorded, and fluid should routinely be sent for cell counts, Gram’s stain and culture, and determination of glucose and protein levels. A CSF Gram’s stain typically requires >10⁵ bacteria/mL for reliable positivity; its specificity approaches 100%. Table 119-4 lists the typical CSF profiles for various infections. In general, CSF with lymphocytic pleocytosis and a low glucose concentration suggests either infection (e.g., with *Listeria*, *M. tuberculosis*, or a fungus) or a noninfectious disorder (e.g., neoplastic meningitis, sarcoidosis). Bacterial antigen tests of CSF (e.g., latex agglutination tests for *Haemophilus influenzae* type b, group B *Streptococcus*, *S. pneumoniae*, and *Neisseria meningitidis*) are not recommended for screening, given that these tests are no more sensitive than Gram’s stain; however, these assays can be helpful in presumptively identifying organisms seen on Gram’s stain. In contrast, other antigen tests (e.g., for *Cryptococcus*) and some CSF serologic testing (e.g., for *Treponema pallidum*, *Coccidioides*) are highly sensitive and are useful for select patients. In addition, polymerase chain reaction (PCR) analysis of CSF is increasingly being used for the diagnosis of bacterial (e.g., *N. meningitidis*, *S. pneumoniae*, mycobacteria) and viral (e.g., herpes simplex virus, enterovirus) infections; while these molecular tests permit rapid diagnosis with a high degree of sensitivity and specificity, they often do not allow determination of antimicrobial resistance profiles.

Cultures The mainstays of infectious disease diagnosis include the culture of infected tissue (e.g., surgical specimens) or fluid (e.g., blood, urine, sputum, pus from a wound). Samples can be sent for culture of bacteria (aerobic or anaerobic), fungi, or viruses. Ideally, specimens are collected before the administration of antimicrobial therapy; in instances where this order of events is not clinically feasible, microscopic examination of the specimen (e.g., Gram-stained or potassium hydroxide [KOH]-treated preparations) is particularly important. Culture of the organism(s) allows identification of the etiologic agent(s), determination of the antimicrobial

TABLE 119-2 Major Infectious Causes of Eosinophilia^a

ORGAN INVOLVED	ORGANISM	EXPOSURE	GEOGRAPHIC DISTRIBUTION	DEGREE OF EOSINOPHILIA ^b
Central nervous system	<i>Angiostrongylus</i>	Raw seafood	Asia	Mild
	<i>Gnathostoma</i>	Raw poultry and seafood	Asia	Moderate to extreme
Eye	<i>Loa loa</i>	Insect bite	Africa	Moderate (expatriates), mild (patients living in endemic areas)
	<i>Onchocerca</i>	Insect bite	Africa	Mild (expatriates), moderate (patients living in endemic areas)
Lung	<i>Chlamydia trachomatis</i>	Sexual transmission	Worldwide	Mild
	<i>Strongyloides</i>	Soil	Tropical	Moderate (acute), mild (chronic)
	<i>Toxocara canis/Toxocara cati</i> ^c	Dogs, soil	Worldwide	Moderate to extreme
	<i>Paragonimus</i>	Crabs and crayfish	Asia	Moderate (acute), mild (chronic)
	<i>Coccidioides immitis</i>	Soil	Southwestern United States	Mild (acute), extreme (disseminated)
	<i>Brugia malayi</i>	Insect bite	Asia	Mild to moderate
	<i>Pneumocystis jirovecii</i>	Air	Worldwide	Mild
Liver	<i>Schistosoma japonicum</i>	Freshwater swimming	Asia	Moderate (acute), mild (chronic)
	<i>Schistosoma mansoni</i>	Freshwater swimming	Africa, Middle East, Latin America	Moderate (acute), mild (chronic)
	<i>Fasciola</i>	Watercress	Worldwide	Moderate
	<i>Clonorchis</i>	Raw seafood	Asia	Mild to moderate
	<i>Opisthorchis</i>	Raw seafood	Asia	Mild to moderate
Intestines	<i>Ascaris</i> ^d	Raw fruits and vegetables, contaminated water	Worldwide	Mild to extreme
	Hookworm	Soil	Worldwide	Mild to moderate
	<i>Trichuris</i>	Raw fruits and vegetables, contaminated water	Tropical	Mild
	<i>Cystoisospora belli</i>	Contaminated water and food	Worldwide	Mild
	<i>Dientamoeba fragilis</i>	Unclear; spread via fecal–oral route	Worldwide	Mild
	<i>Capillaria</i>	Raw seafood	Asia	Extreme
	<i>Heterophyes</i>	Raw seafood	Asia, Middle East	Mild
	<i>Anisakis</i>	Raw seafood	Worldwide	Mild
	<i>Baylisascaris procyonis</i> ^e	Soil	North America	Moderate to extreme
	<i>Hymenolepis nana</i>	Contaminated water, soil	Worldwide	Mild
Bladder	<i>Schistosoma haematobium</i>	Freshwater swimming	Africa, Middle East	Moderate (acute), mild (chronic)
Muscle	<i>Trichinella</i>	Pork	Worldwide	Moderate to extreme
Lymphatics	<i>Wuchereria bancrofti</i> ^d	Insect bite	Tropical	Moderate to extreme ^f
	<i>Bartonella henselae</i>	Cats	Worldwide	Mild
Other	Recovery from bacterial or viral infections	—	—	Mild
	HIV	Contaminated bodily fluid	Worldwide	Mild
	<i>Cryptococcus neoformans</i>	Soil	Worldwide	Moderate to extreme (disseminated)

^aThere are numerous noninfectious causes of eosinophilia, such as atopic disease, DRESS (drug reaction with eosinophilia and systemic symptoms) syndrome, and pernicious anemia, which can cause mild eosinophilia; drug hypersensitivity and serum sickness, which can cause mild to moderate eosinophilia; collagen vascular disease, which can cause moderate eosinophilia; and malignancy, Churg–Strauss syndrome, and hyper-IgE syndromes, which can cause moderate to extreme eosinophilia.

^bMild: 500–1500 cells/μL; moderate: 1500–5000 cells/μL; extreme: >5000 cells/μL. ^cCan also affect the liver and the eyes. ^dCan also affect the lungs. ^eCan also affect the eyes and the central nervous system. ^fLevels are typically higher with pulmonary infections.

susceptibility profile, and—when there is concern about an outbreak—isolate typing. While cultures are extremely useful in the evaluation of patients, determining whether culture results are clinically meaningful or represent contamination (e.g., a non-*aureus*, non-*lugdunensis* staphylococcal species growing in a blood culture) can sometimes be challenging and requires an understanding of the patient's immune status, exposure history, and microbiota. In some cases, serial cultures to demonstrate clearance of the organism may be helpful.

Pathogen-Specific Testing Numerous pathogen-specific tests (e.g., serology, antigen testing, PCR testing) are commercially available, and many hospitals now offer some of these tests in-house to facilitate rapid turnaround that ultimately enhances patient care. The reader is directed to relevant chapters on the pathogens

of interest for specific details. Some of these tests (e.g., universal PCRs) identify organisms that currently are not easily cultivable and have unclear relationships to disease, thereby complicating diagnosis. As these tests become more commonplace and the work of the Human Microbiome Project progresses, the relevance of some of these previously unrecognized bacteria to human health will likely become more apparent.

Radiology Imaging provides an important adjunct to the physical examination, allowing evaluation for lymphadenopathy in regions that are not externally accessible (e.g., mediastinum, intraabdominal sites), assessment of internal organs for evidence of infection, and facilitation of image-guided percutaneous sampling of deep spaces. The choice of imaging modality (e.g., CT, MRI, ultrasound, nuclear medicine, use of contrast) is best made in consultation with

TABLE 119-3 Causes of an Extremely Elevated Erythrocyte Sedimentation Rate (>100 mm/h)

ETIOLOGIC CATEGORY (% OF CASES)	SPECIFIC CAUSES
Infectious diseases (35–40)	Subacute bacterial endocarditis Abscesses Osteomyelitis Tuberculosis Urinary tract infection
Inflammatory diseases (15–20)	Giant cell arteritis Rheumatoid arthritis Systemic lupus erythematosus
Malignancies (15–20)	Multiple myeloma Leukemias Lymphomas Carcinomas
Other (20–35)	Drug hypersensitivity reactions (drug fever) Ischemic tissue injury/trauma Renal diseases

a radiologist to ensure that the results will address the physician's specific concerns.

TREATMENT

Physicians often must balance the need for empirical antibiotic treatment with the patient's clinical condition. When clinically feasible, it is best to obtain relevant samples (e.g., blood, CSF, tissue, purulent exudate) for culture prior to the administration of antibiotics, as antibiotic treatment often makes subsequent diagnosis more difficult. Although a general maxim for antibiotic treatment is to use a regimen with as narrow a spectrum as possible (Chap. 144), empirical regimens are necessarily somewhat broad, given that a specific diagnosis has not yet been made. Table 119-5 lists empirical antibiotic treatment regimens for commonly encountered infectious presentations. These regimens should be narrowed as appropriate once a specific diagnosis is made. In addition to antibiotics, there is sometimes a role for adjunctive therapies, such as intravenous immunoglobulin G (IVIG) pooled from healthy adults or hyperimmune

globulin prepared from the blood of individuals with high titers of specific antibodies to select pathogens (e.g., cytomegalovirus, hepatitis B virus, rabies virus, vaccinia virus, *Clostridium tetani*, varicella-zoster virus, *Clostridium botulinum* toxin). Although the data suggesting efficacy are limited, IVIG is sometimes used for patients with suspected staphylococcal or streptococcal toxic shock syndrome.

INFECTION CONTROL

When evaluating a patient with a suspected infectious disease, the physician must consider what infection control methods are necessary to prevent transmission of any possible infection to other people. In 2007, the U.S. Centers for Disease Control and Prevention published guidelines for isolation precautions that are available for download at www.cdc.gov/hicpac/2007IP/2007isolationPrecautions.html. Persons exposed to certain pathogens (e.g., *N. meningitidis*, HIV, *Bacillus anthracis*) should receive postexposure prophylaxis to prevent disease acquisition. (See relevant chapters for details on specific pathogens.)

WHEN TO OBTAIN AN INFECTIOUS DISEASE CONSULT

At times, primary physicians need assistance with patient management from a diagnostic and/or therapeutic perspective. Multiple studies have demonstrated that an infectious disease consult is associated with improved outcomes, shorter length of hospital stay, and decreased costs for patients with various diseases. For example, in a prospective cohort study of patients with *S. aureus* bacteremia, infectious disease consultation was independently associated with a 56% reduction in 28-day mortality. In addition, infectious disease specialists provide other services (e.g., infection control, antimicrobial stewardship, management of outpatient antibiotic therapy, occupational exposure programs) that have been shown to benefit patients. Whenever such assistance would be advantageous to a patient with a possible infection, the primary physician should opt for an infectious disease consult. Specific situations that might prompt a consult include (1) difficult-to-diagnose patients with presumed infections, (2) patients who are not responding to treatment as expected, (3) patients with a complicated medical history (e.g., organ transplant recipients, patients immunosuppressed due to autoimmune or inflammatory conditions), and (4) patients with "exotic" diseases (i.e., diseases that are not typically seen within the region).

TABLE 119-4 Typical Cerebrospinal Fluid Profiles for Meningitis and Encephalitis^a

	NORMAL	BACTERIAL MENINGITIS	VIRAL MENINGITIS	FUNGAL MENINGITIS ^b	PARASITIC MENINGITIS	TUBERCULOUS MENINGITIS	ENCEPHALITIS
WBC count (per μ L)	<5	>1000	25–500	40–600	150–2000	25–100	50–500
Differential of WBC	60–70% lymphocytes, $\leq$ 30% monocytes/macrophages	$\uparrow\uparrow$ PMNs ($\geq$ 80%)	Predominantly lymphocytes ^c	Lymphocytes or PMNs, depending on specific organism	$\uparrow\uparrow$ Eosinophils ($\geq$ 50%) ^d	Predominantly lymphocytes ^c	Predominantly lymphocytes ^c
Gram's stain	Negative	Positive (in >60% of cases)	Negative	Rarely positive	Negative	Occasionally positive ^e	Negative
Glucose (mg/dL)	40–85	<40	Normal	$\downarrow$ to normal	Normal	<50 in 75% of cases	Normal
Protein (mg/dL)	15–45	>100	20–80	150–300	50–200	100–200	50–100
Opening pressure (mmH ₂ O)	50–180	>300	100–350	160–340	Normal	150–280	Normal to $\uparrow$
Common causes	—	<i>Streptococcus pneumoniae</i> , <i>Neisseria meningitidis</i>	Enteroviruses	<i>Candida</i> , <i>Cryptococcus</i> , and <i>Aspergillus</i> spp.	<i>Angiostrongylus cantonensis</i> , <i>Gnathostoma spinigerum</i> , <i>Baylisascaris procyonis</i>	<i>Mycobacterium tuberculosis</i>	Herpesviruses, enteroviruses, influenza virus, rabies virus

^aNumbers indicate typical results, but actual results may vary. ^bCerebrospinal fluid characteristics depend greatly on the specific organism. ^cNeutrophils may predominate early in the disease course. ^dPatients typically have striking eosinophilia as well. ^eSensitivity can be increased by examination of a smear of protein coagulum (pellicle) and the use of acid-fast stains.

Abbreviations: PMNs, polymorphonuclear neutrophils; WBC, white blood cell.

TABLE 119-5 Initial Empirical Antibiotic Therapy for Common Infectious Disease Presentations^a

CLINICAL SYNDROME	COMMON ETIOLOGIES	ANTIBIOTIC(S)	COMMENTS	SEE CHAPTER(S)
Septic shock	<i>Staphylococcus aureus</i> , <i>Streptococcus pneumoniae</i> , enteric gram-negative bacilli	Vancomycin, 15 mg/kg q12h ^b plus A broad-spectrum antipseudomonal β-lactam (piperacillin-tazobactam, 4.5 g q6h; imipenem, 1 g q8h; meropenem, 1 g q8h; or cefepime, 1–2 g q8–12h)	If a pseudomonal species is likely, a second antipseudomonal agent should be added.	304
Meningitis	<i>S. pneumoniae</i> , <i>Neisseria meningitidis</i>	Vancomycin, 15 mg/kg q12h ^b plus Ceftriaxone, 2 g q12h	Dexamethasone (0.15 mg/kg IV q6h for 2–4 d) should be added for patients with suspected or proven pneumococcal meningitis, with the first dose administered 10–20 min before the first dose of antibiotics.	138 and pathogen- specific chapters
CNS abscess	<i>Streptococcus</i> spp., <i>Staphylococcus</i> spp., anaerobes, gram-negative bacilli	Vancomycin, 15 mg/kg q12h ^b plus Ceftriaxone, 2 g q12h plus Metronidazole, 500 mg q8h	—	138
Acute endocarditis (native valve)	<i>S. aureus</i> , <i>Streptococcus</i> spp., coagulase-negative staphylococci	Vancomycin, 15 mg/kg q12h ^b plus Cefepime, 2 g q8h	—	128
Pneumonia Community-acquired, outpatient	<i>S. pneumoniae</i> , <i>Mycoplasma pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Chlamydia pneumoniae</i>	No comorbidities ^b : Azithromycin, 500 mg PO × 1, then 250 mg PO qd × 4 days With comorbidities ^b : Levofloxacin, 750 mg PO qd	If MRSA is a consideration, add vancomycin (15 mg/kg q8–12h ^b) or linezolid (600 mg q12h); daptomycin should not be used in patients with pneumonia.	126 and pathogen- specific chapters
Inpatient, non-ICU	Above plus <i>Legionella</i> spp.	A respiratory fluoroquinolone (moxifloxacin, 400 mg IV/PO qd; gemifloxacin, 320 mg PO qd; or levofloxacin, 750 mg IV/PO qd) or A β-lactam (cefotaxime, ceftriaxone, or ampicillin- sulbactam) plus azithromycin		
Inpatient, ICU	Above plus <i>S. aureus</i>	A β-lactam plus Azithromycin or a respiratory fluoroquinolone		
Hospital-acquired pneumonia ^d	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>S. aureus</i> , gram-negative bacilli (e.g., <i>Pseudomonas aeruginosa</i> , <i>Klebsiella pneumoniae</i> , <i>Acinetobacter</i> spp.)	An antipseudomonal β-lactam (cefepime, 2 g q8h; ceftazidime, 2 g q8h; imipenem, 500 mg q6h; meropenem, 1 g q8h; or piperacillin-tazobactam, 4.5 g q6h) plus An antipseudomonal fluoroquinolone (levofloxacin, 700 mg qd, or ciprofloxacin, 400 mg q8h) or an aminoglycoside (amikacin, 15–20 mg/kg q24h ^e ; gentamicin, 5–7 mg/kg q24h ^e ; or tobramycin, 5–7 mg/kg q24h ^e)	If MRSA is a consideration, add vancomycin (15 mg/kg q8–12h ^b) or linezolid (600 mg q12h); daptomycin should not be used in patients with pneumonia.	
Complicated intraabdominal infection	Anaerobes (<i>Bacteroides</i> spp., <i>Clostridium</i> spp.), gram- negative bacilli (<i>Escherichia coli</i>), <i>Streptococcus</i> spp.	Cefoxitin, 2 g q6h or A combination of metronidazole (500 mg q8–12h) plus one of the following : cefazolin (1–2 g q8h), cefuroxime (1.5 g q8h), ceftriaxone (1–2 g q12–24h), cefotaxime (1–2 g q6–8h), ciprofloxacin (400 mg q12h), levofloxacin (750 mg qd) A carbapenem (imipenem, 500 mg q6h; meropenem, 1 g q8h; doripenem, 500 mg q8h) or Piperacillin-tazobactam, 3.375 g q6h ^f or A combination of metronidazole (500 mg q8h) plus an antipseudomonal cephalosporin (cefepime, 2 g q8h; ceftazidime, 2 g q8h)	If MRSA is a consideration, add vancomycin (15 mg/kg q12h ^b)	132, 177 , and pathogen-specific chapters
Mild to moderate severity				
High-risk patient or high degree of severity	Same as above			

(Continued)

TABLE 119-5 Initial Empirical Antibiotic Therapy for Common Infectious Disease Presentations^a (Continued)

CLINICAL SYNDROME	COMMON ETIOLOGIES	ANTIBIOTIC(S)	COMMENTS	SEE CHAPTER(S)
Skin and soft tissue infection	<i>S. aureus</i> , <i>Streptococcus pyogenes</i>	Dicloxacillin, 250–500 mg PO qid or Cephalexin, 250–500 mg PO qid or Clindamycin, 300–450 mg PO tid or Nafcillin/oxacillin, 1–2 g q4h	If MRSA is a consideration, clindamycin, vancomycin (15 mg/kg q12h ^b), linezolid (600 mg IV/PO q12h), or TMP-SMX (1–2 double-strength tablets PO bid ^d) can be used.	129 and pathogen-specific chapters

^aThis table refers to immunocompetent adults with normal renal and hepatic function. All doses listed are for parenteral administration unless indicated otherwise. Local antimicrobial susceptibility profiles may influence the choice of antibiotic. Therapy should be tailored once a specific etiologic agent and its susceptibilities are identified. ^bTrough levels for vancomycin should be 15–20 µg/mL. ^cTrough levels for amikacin should be <4 µg/mL. ^dIn patients with late onset (i.e., after ≥5 days of hospitalization) or risk factors for multidrug-resistant organisms. ^eTrough levels for gentamicin and tobramycin should be <1 µg/mL. ^fIf *P. aeruginosa* is a concern, the dosage may be increased to 3.375 g IV q4h or 4.5 g IV q6h. ^gData on the efficacy of TMP-SMX in skin and soft tissue infections are limited. ^hComorbidities include chronic heart, lung, liver, or renal disease; diabetes mellitus; alcoholism; malignancy; or asplenia.

Abbreviations: CNS, central nervous system; ICU, intensive care unit; MRSA, methicillin-resistant *S. aureus*; TMP-SMX, trimethoprim-sulfamethoxazole.

PERSPECTIVE

The study of infectious diseases is really a study of host–microbial interactions and represents evolution by both the host and the microbe—an endless struggle in which microbes have generally been more creative and adaptive. Given that nearly one-fifth of deaths worldwide are still related to infectious diseases, it is clear that the war against infectious diseases has not been won. For example, a cure for HIV infection is still lacking, there have been only marginal improvements in the methods for detection and treatment of tuberculosis after more than a half century of research, new infectious disease outbreaks (e.g., viral hemorrhagic fevers, Zika, SARS-CoV-2) continue to emerge, and the threat of microbial bioterrorism remains high. The subsequent chapters in Part 5 detail—on both a syndrome and a microbe-by-microbe basis—the current state of medical knowledge about infectious diseases. At their core, all of these chapters carry a similar message: Despite numerous advances in the diagnosis, treatment, and prevention of infectious diseases, much work and research are required before anyone can confidently claim we have achieved “the virtual elimination of infectious disease.” In reality, this goal will never be attained, given the rapid adaptability of microbes.

FURTHER READING

BARTLETT JG: Why infectious diseases. Clin Infect Dis 59(Suppl 2):S85, 2014.

BHATRAJU PK et al: Covid-19 in critically ill patients in the Seattle region: Case series. N Engl J Med 382:2012, 2020.

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VERGHESE A et al: Inadequacies of physical examination as a cause of medical errors and adverse events: A collection of vignettes. Am J Med 128:1322, 2015.

specific and nonspecific, to pathogens. Among the microbial factors that promote disease are those that alter human cells, those that inhibit host immune responses, and those that respond to the microbes that constitute the microbiota and their metabolic products. The process of infection can be divided into several stages: the encounter with and entry of the microbe into the human body (*colonization*), the attachment of the microbe in its favored niche and microbial avoidance of host defenses (*infection*), the combined deployment of microbial factors that damage human tissue and host inflammatory responses to the presence of the microbe (*disease*), and the release of the pathogen into the environment, where it can infect others (*transmission*). It is notable that for most microbial pathogens, the host inflammatory response contributes substantially to symptoms and to tissue damage. Moreover, the human microbiota (the collection of microbes that reside in and on the human body) modulates, directly or indirectly, every stage of infection (Chap. 471). This chapter describes the best-understood molecular and cellular mechanisms that contribute to human disease caused by bacterial pathogens.

ENTRY INTO THE HUMAN HOST

Infectious diseases occur when either a live pathogen enters the human host or a toxic pathogen product is ingested by the host, with the former being much more common than the latter.

Most bacterial infections result from entry of a pathogen into the body. Entry can occur through a break in the skin into the underlying soft tissues or at the surface of a mucous membrane of the respiratory, gastrointestinal, or genitourinary tract; the skin may also be directly infected. Entry at these sites may result in infection of the bloodstream, which may in turn lead to infection of other organ systems.

Entry into the respiratory tract occurs via respiratory droplet nuclei (airborne particles 1–5 µm in diameter) or via fomites introduced on a hand that is contaminated by contact with a contaminated inert surface (e.g., a doorknob or a faucet). Infectious droplet nuclei are generated when an individual with a communicable respiratory infection (e.g., tuberculosis, Legionnaires’ disease, psittacosis, influenza, measles, chickenpox, aspergillosis, COVID-19 infection) sneezes, coughs, talks, plays a musical instrument by mouth, or sings. A cough may generate 3000 particles, whereas a sneeze may generate up to 40,000 particles; large particles may evaporate down to the 0.5- to 12-µm range, and particles that are 1.5 µm and hygroscopic increase in size as they pass through the moist nasal passages and lower respiratory tree.

Entry into the gastrointestinal tract occurs via ingestion of contaminated food or water or via person-to-person contact, typically with transfer from a contaminated hand into the mouth. Pathogens for which the infectious inoculum is large (e.g., 10⁸ organisms for epidemic spread of *Vibrio cholerae*, 10⁵ organisms for *Salmonella enterica* serovar Typhimurium) are generally acquired via contaminated food or water, whereas pathogens for which the infectious inoculum is small (e.g., 10¹ to 10² organisms for *Shigella* spp.) are more commonly acquired by person-to-person spread.

120 Molecular Mechanisms of Microbial Pathogenesis

Thomas E. Wood, Marcia B. Goldberg

Infectious diseases of humans involve intricate interactions among the infecting microbe, human tissue, and the host microbiome. The co-evolution of humans and microbes has led to the presence of numerous and varied specific microbial factors that promote the disease process and a corresponding wide range of human cellular responses, both

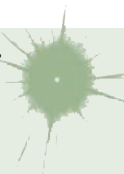


TABLE 119-5 Initial Empirical Antibiotic Therapy for Common Infectious Disease Presentations^a (Continued)

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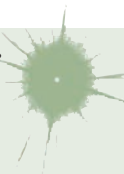
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Entry into the genitourinary tract generally takes place either via colonization of the urethral meatus or vaginal introitus with fecal organisms followed by ascension of the organisms into the bladder or kidneys or via instrumentation. Pyelonephritis can also result from seeding from the bloodstream.

ESTABLISHMENT OF INFECTION

■ NICHE

Live Pathogens Within the human host, many bacterial pathogens display tissue tropism; the sites of infection within the human body are pathogen-specific and restricted, such that even adjacent tissues may be uninvolved. For example, group A streptococci cause pharyngitis and soft tissue infection, but rarely pneumonia. Cholera is an infection of the small intestine, but not the stomach or the colon, where *Shigella* spp. cause disease only in the rectosigmoid. Therefore, to establish infection, the pathogen must access its niche and then remain within that niche (Table 120-1). In the respiratory tree, the sites at which pathogens initially settle can be determined by mode of spread. Droplet nuclei reach the bronchial tree or alveoli, whereas, after contaminated hands touch the face, fomites reach the pharynx or nasal passages. Pathogens move through the gastrointestinal tract via normal intestinal motility.

Tissue association and invasion are dictated by the interaction of the bacterium with host factors, commonly glycan-decorated receptors and/or the associated extracellular matrix. The environmental conditions of the niche trigger the expression of virulence factors

required for the establishment of infection; for example, bile salts in the gut stimulate the expression of *V. cholerae* adhesins and toxins and the germination of *Clostridioides difficile* spores. Bacteria commonly manipulate their particular niche environment in ways that facilitate infection. For example, the gastroduodenal pathogen *Helicobacter pylori* produces urease, which converts urea into ammonia, thereby increasing the pH of the acidic stomach environment and creating a more hospitable environment for its survival. This conversion also concomitantly alters the physiology of the gastric epithelium.

Preformed Toxins A small number of diseases are caused by ingestion in food of preformed bacterial toxins. The most common among these are *Staphylococcus aureus* enterotoxins, which can be present in prepared foods such as dairy, meat, eggs, salads, and produce, and the *Bacillus cereus* emetic enterotoxin, which is most often found in rice or other starchy food that has been improperly refrigerated. Less common but also important is botulinum toxin produced by *Clostridium botulinum*. Preformed toxins cause disease in the small intestine (nausea and vomiting) and may cause systemic symptoms.

Although the pathogens may be killed when food is cooked, their toxins are heat stable. *S. aureus* enterotoxins can be strongly emetic. *B. cereus* produces the heat-stable peptide toxin cereulide, which acts as a potassium ionophore and induces emesis. In botulism, a flaccid paralysis caused by botulinum toxin, uptake of the toxin blocks neurotransmitter release in motor neurons, inhibiting the central nervous system and resulting in potentially fatal respiratory failure.

■ ATTACHMENT

Attachment of bacteria to host tissue surfaces is a prerequisite for the pathogen's establishment of an infection and is mediated by specific receptor–ligand interactions. In this context, the tissue specificity of the host cell surface receptor repertoire is a critical factor in delimiting a pathogen's niche(s). These physical associations additionally facilitate the pathogen's avoidance of host clearance mechanisms (see “Avoidance of Innate Immune Responses,” below) and may contribute to formation of biofilms by the pathogen (see “Biofilms,” below). Because adhesion to cellular receptors often triggers cellular signal transduction and innate immune signaling, therapeutic blocking of this interaction may in some circumstances exacerbate infection.

Adhesins Bacterial pathogens have evolved a wide range of strategies by which to attach to the diverse host cell structures they encounter. For many bacterial pathogens, ligands or adhesins for specific host receptors are known. Adhesins comprise a wide variety of surface structures, including single proteins, carbohydrates, glycoproteins, lipids, lipoproteins, and multiprotein filamentous complexes that extend several micrometers from the bacterial surface, each anchored in the outer-surface cell envelope. Most bacteria produce multiple adhesins with varying specificity, enabling the pathogen to interact with multiple receptors, including those on several distinct cell and tissue types encountered during the process of infection. These interactions are often partially redundant, are serologically variable, and contribute additively or synergistically with other binding interactions.

Common classes of adhesins are pili (also known as fimbriae), flagella, and autotransporter proteins (Table 120-2). Pili are hairlike extensions consisting of a polymer of the major pilin subunit capped with minor pilins that provide the adherence function of the structure. Pili are classified by type and are produced by many gram-negative bacteria and a smaller number of gram-positive bacteria. To date, efforts to prevent infection with pilus-based vaccines have been unsuccessful.

Types of pili include type I, type P, and type IV. Type I pili frequently function at mucosal surfaces. For example, they mediate the close association of uropathogenic *Escherichia coli* (UPEC) with bladder epithelial cells and the ability of this pathogen to persist, causing relapsing urinary tract infections. UPEC also produces type P pili and afimbrial adhesins. These adhesins bind sugar moieties on host surface glycoproteins, with varied specificity depending on the adhesin. The minor pilin lectins at the tip of the pili generally bind D-mannose glycans, albeit with strain specificity. For example, the type I pilus adhesin FimH of intestinal *E. coli* strains often preferentially binds

TABLE 120-1 Bacterial Pathogens, Diseases, and Niches

MOST COMMON TROPISM	BACTERIUM	DISEASE
Skin, respiratory tract, small intestine	<i>Bacillus anthracis</i>	Anthrax
Respiratory tract	<i>Bordetella pertussis</i>	Pertussis
Systemic	<i>Borrelia burgdorferi</i>	Lyme disease
Systemic	<i>Brucella abortus</i>	Brucellosis
Systemic	<i>Burkholderia pseudomallei</i>	Melioidosis
Eyes, venereal	<i>Chlamydia trachomatis</i>	Various chlamydioses, including trachoma
Colon	<i>Clostridioides difficile</i>	Colitis
Pharynx	<i>Corynebacterium diphtheriae</i>	Diphtheria
Systemic	<i>Coxiella burnetii</i>	Q fever
Colon	Enterohemorrhagic <i>Escherichia coli</i>	
Stomach	<i>Helicobacter pylori</i>	Gastritis, gastric ulcers
Respiratory tract	<i>Legionella pneumophila</i>	Legionnaires' disease
Systemic, central nervous system	<i>Listeria monocytogenes</i>	Listeriosis
Respiratory tract	<i>Mycobacterium tuberculosis</i>	Tuberculosis
Urogenital tract	<i>Neisseria gonorrhoeae</i>	Gonorrhea
Respiratory tract	<i>Pseudomonas aeruginosa</i>	
Systemic	<i>Salmonella enterica</i> serovar Typhi	Typhoid fever
Gastrointestinal tract	<i>Salmonella enterica</i> serovar Typhimurium	
Colon, rectum	<i>Shigella</i> spp.	Dysentery, shigellosis
Multiple sites	<i>Staphylococcus aureus</i>	
Soft tissue	Group A <i>Streptococcus</i>	
Small intestine	<i>Vibrio cholerae</i>	Cholera
Systemic	<i>Yersinia pestis</i>	Plague

TABLE 120-2 Classes of Bacterial Adhesion Proteins and Their Host Receptors		
ADHESIN	EXAMPLE	RECEPTOR
Type I pili	Fim protein, uropathogenic <i>Escherichia coli</i>	Terminal mannose of uroplakin <i>N</i> -glycan in urinary epithelial cells
Type P pili	Pap protein, uropathogenic <i>E. coli</i>	Galactose disaccharides
Type IV pili	Tfp protein, <i>Neisseria gonorrhoeae</i>	CD64, CR3, I domain-containing integrins
MSCRAMM	SdrC protein, <i>Staphylococcus aureus</i>	β-Neurexin
Opa	Opa protein, <i>Neisseria meningitidis</i>	CEACAMs ^a
Flagellum	FliC protein, <i>Pseudomonas aeruginosa</i>	Asialo-GM1 ganglioside
Autotransporter	Invasin, <i>Yersinia pseudotuberculosis</i>	β1-Integrins
Autotransporter	Ag85, <i>Mycobacterium tuberculosis</i>	Fibronectin

^aCarcinoembryonic antigen-related cell adhesion molecules.

oligomannose, whereas the FimH of UPEC strains commonly binds monomannose. Thus, the same pilus structures in different bacterial strains can dictate adherence to distinct tissues. Type IV pili (Tfp) are widespread among gram-negative bacteria, and similar structures exist among gram-positive bacteria. These are evolutionarily related to the type II secretion system (T2SS) and, in addition to mediating adherence, allow the uptake of DNA into bacteria and the motility of bacteria on surfaces. The Tfp of *Neisseria* spp. and *V. cholerae* mediate aggregation of individual bacteria into microcolonies, which promotes colonization.

Flagella are polymeric helical filaments that propel bacteria through liquid environments by rotating about their long axis. Because flagella confer the ability to swim toward a target surface, often following a chemotactic gradient where chemical sensing influences the direction of motility, they are vital virulence factors of many pathogenic bacteria. Flagella can be localized to one end of the bacterial cell (*polar*)

or distributed around the bacterial surface (*peritrichous*). Flagella are evolutionarily related to type III secretion systems (T3SSs) (see “Replicative Niche” and “Survival in the Vacuole,” below) and have been shown in some instances to be responsible for the secretion of bacterial toxins. Flagella may also act as adhesins, binding to mucins of mucosal surfaces in the case of the gastrointestinal pathogens enteropathogenic *E. coli* (EPEC) and enterohemorrhagic *E. coli* (EHEC).

Autotransporter proteins comprise a subdivision of type V secretion systems (T5SSs), which are prevalent among gram-negative bacteria (Fig. 120-1). Extended adhesive projections anchored in the bacterial outer membrane, autotransporter proteins are pivotal virulence determinants for several human pathogens, including the filamentous hemagglutinin of *Bordetella pertussis* (the etiologic agent of whooping cough) and the IcsA adhesin and intracellular motility determinant of the *Shigella* spp. responsible for dysenteric diseases. In addition, the autotransporter proteins intimin and invasins are required for the intimate association of adhering and effacing pathogens, such as EPEC, and invasion by *Yersinia* spp., respectively. Besides these main classes of adhesins, other bacterial surface proteins are also involved in adhesion, including the Opa family of membrane proteins of *Neisseria* spp. and the gram-positive cell wall-anchored microbial surface component recognizing adhesive matrix molecules (MSCRAMMs) of *S. aureus* and various enterococci.

Receptors Carbohydrates (glycans) on the surface of and secreted by human cells play major roles in the adherence of bacterial pathogens. The surfaces of human cells are coated with glycoproteins and glycolipids, whereas the extracellular matrix (ECM) scaffold of tissues is mainly composed of secreted proteoglycans. Most mucosal surfaces are covered with a layer of mucus, which consists primarily of mucins, a family of proteins that are heavily glycosylated. Among the human enzymes involved in glycan decoration of secreted proteins is *FUT2*, for which ~20% of the human population harbors two nonfunctional alleles; the importance of human glycans in infection is highlighted by the observation that individuals who lack functional *FUT2* display increased susceptibility to certain bacterial infections, decreased susceptibility to certain viral infections, and altered susceptibility to chronic noninfectious inflammatory diseases. These data are confounded by a relative increase in secretion of sialylated glycans in individuals lacking functional *FUT2*. Glycans participate in many adhesive

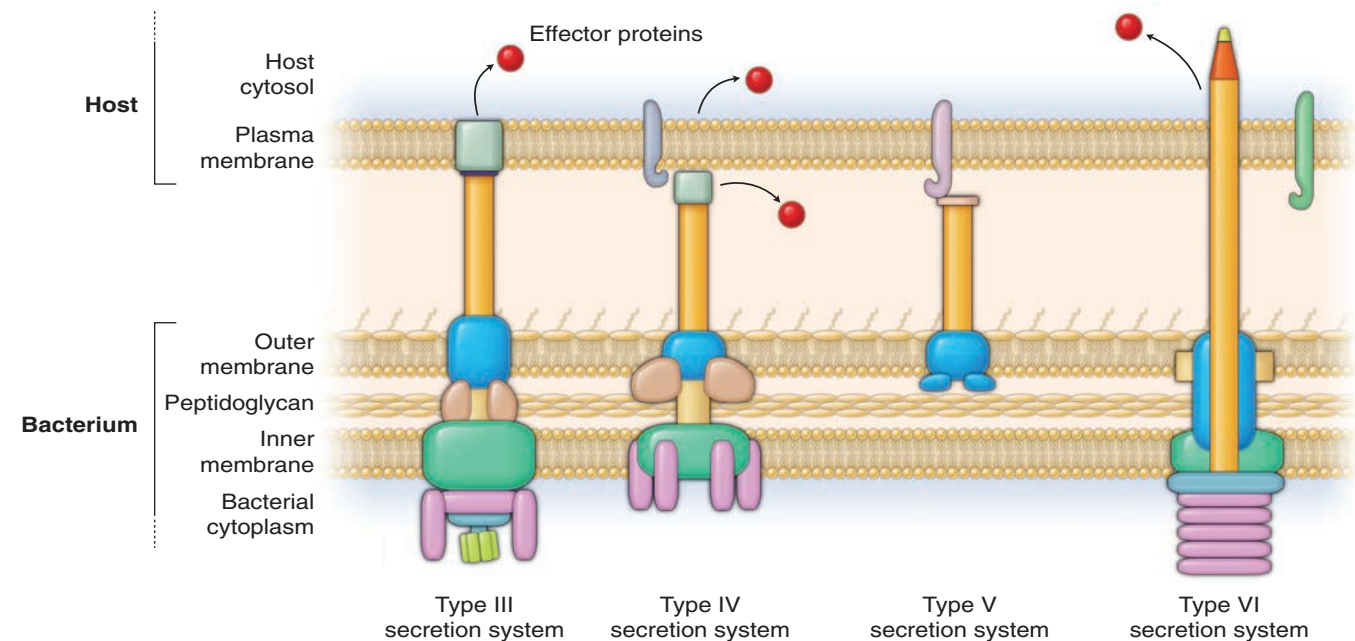


FIGURE 120-1 Major bacterial secretion systems involved in pathogenesis. Schematic of the types III, IV, V, and VI secretion systems (T3SS, T4SS, T5SS, and T6SS, respectively) of gram-negative bacteria. The T3SS, T4SS, and T6SS deliver bacterial effector proteins into host cells, whereas the T5SS participates in adhesion to the surface of cells. The architecture of the extracellular portion of the T4SS and how it translocates effector proteins remain poorly defined. Colored shapes with hooks in the host plasma membrane represent host membrane proteins that, it is thought, may participate in these processes.

interactions and serve as receptors for certain bacterial toxins. Certain bacteria enzymatically alter host glycans in a manner that enables improved access to the epithelial surface, as in the case of *V. cholerae* and the oral pathogen *Tannerella forsythia*.

The GM1 ganglioside, a glycolipid, is the receptor for cholera toxin (CTX), *Pseudomonas aeruginosa* flagella, and the *Clostridium perfringens* toxin, whereas terminal sialic acid residues on GM1 play an important role in tropism of the gastric pathogen *H. pylori* through interactions with the bacterial membrane proteins BabA and SabA. Heparan sulfate proteoglycans and other glycosaminoglycan-conjugated proteins are commonly associated with the basolateral membranes of epithelial layers and act as ligands for the chlamydial OmcB protein and the *E. coli* cytotoxic necrotizing factor toxin; these interactions promote initial bacterial adherence in proximity to the plasma membrane and lead to engagement of additional surface molecules.

In mammals, the ECM and integrin proteins are ubiquitous. The ECM consists of laminin, vimentin, and type IV collagen, which interact via fibronectin with integrin receptors in the plasma membrane. Because of direct interaction with plasma membrane-embedded integrins, ECM alterations can result in signal transduction that directly influences immune cell behavior. Moreover, many bacterial pathogens engage integrins and the functionally related carcinoembryonic antigen-related cell adhesion molecules (CEACAMs) as host cell receptors and as triggers for their internalization into human cells (see “Mechanisms of Microbial Entry into Cells,” below). The receptor for adherent-invasive *E. coli*, a pathogenic type of *E. coli*, is CEACAM6, whose levels are increased on epithelial cells in inflammatory bowel disease; adherent-invasive *E. coli* adhere at increased levels in Crohn’s disease, and sites of excessive bacterial adherence display increased inflammation.

Among the many other host cell surface proteins that serve as receptors for bacterial adherence is the cystic fibrosis transmembrane conductance regulator (CFTR), a chloride channel involved in the maintenance of adequate hydration of mucosal surfaces. Mutation of the *CFTR* gene gives rise to the hereditary disease cystic fibrosis. These patients are hypersusceptible to respiratory infection because the cilia are unable to clear viscous mucus from the bronchial epithelial surface. The remarkably high frequency of cystic fibrosis (>2.5%) in Caucasian populations has been attributed to relative resistance to *Salmonella enterica* serovar Typhi infection, since *S. Typhi* adheres to CFTR via type IV pili. It has also been reported that epithelial cells utilize CFTR during internalization-mediated clearance of the extracellular pathogen *P. aeruginosa*, potentially contributing to the decreased ability of cystic fibrosis patients to clear infections by this pathogen.

■ REPLICATIVE NICHE

Once bacterial pathogens find their niche and associate with target host cells, they must replicate to persist. Some pathogens remain predominantly associated with the surface of epithelial cells (e.g., *Bordetella*, *Pseudomonas*, *Vibrio*, and *Clostridium* spp.), whereas others predominantly enter into cells (e.g., *Salmonella*, *Shigella*, *Francisella*, and *Listeria* spp.). Phagocytic cells actively engulf pathogens for destruction; thus extracellular bacteria must avoid or inhibit this process to remain associated with the cell surface. Many pathogens deliver cytoskeleton-disrupting proteins that perturb the phagocytic process, whereas others create structures that impede phagocytosis; examples include biofilm formation by *P. aeruginosa* and cell chaining by streptococci. In contrast, nonphagocytic cells, such as epithelial cells, may also internalize bacteria in processes that can be triggered by either the host cell (as a clearance mechanism) or the pathogen (for tissue invasion). A subpopulation of any infecting pathogen will be phagocytosed by macrophages, neutrophils, and/or dendritic cells. Many pathogens have evolved mechanisms to survive within phagocytes; for example, during colonization of the nasopharynx, *Streptococcus pneumoniae* survives within vacuoles in dendritic cells and alveolar or splenic macrophages.

Extracellular Pathogens Extracellular pathogens replicate at the surface of human cells. Many secrete enzymes that may liberate nutrients from extracellular factors and fashion a hospitable niche. For

example, *V. cholerae*, an extracellular pathogen of the small intestine, secretes the mucinase vibriolysin that degrades the mucus barrier overlying intestinal epithelial cells, giving the bacterium access to the cell surface. *H. pylori*, an extracellular pathogen of the stomach, secretes a urease that converts urea into carbon dioxide and ammonia, thereby buffering the acidic pH of the stomach. *Yersinia* spp., which associates with the surface of leukocytes, delivers into these naturally phagocytic cells secreted bacterial proteins that inhibit phagocytosis. Once phagocytosed, *V. cholerae* can inhibit further phagocytosis of extracellular bacteria.

EPEC and EHEC efface the microvilli of the brush border of intestinal epithelial cells and adhere tightly to the surface of these cells. These bacteria induce formation of actin-rich pedestals at the plasma membrane, which may help prevent bacterial internalization by the epithelial cells. Epithelial polarity is important for tissue function, influencing ion flux, barrier integrity, and protein sorting. Many pathogens subvert epithelial polarity to redistribute membrane complexes in a manner that facilitates tissue penetration and dissemination and avoids internalization and subsequent cell-autonomous immune mechanisms (see “Avoidance of Innate Immune Responses,” below). For certain pathogens that do not efface microvilli (e.g., *Listeria monocytogenes* and *Shigella* spp.), the basolateral compartment of the plasma membrane—rather than the mucus-covered microvillus apical surface—is the preferential site of epithelial cell invasion. Tight junctions mediate intimate associations between neighboring epithelial cells, thereby maintaining tissue barrier function; the targeted dissolution of tight junctions by pathogens facilitates bacterial penetration into the tissue.

BIOFILMS Some extracellular pathogens, including *S. aureus* and *P. aeruginosa*, establish chronic infections through the production of extracellular polymeric matrices called *biofilms*, which encase the bacteria. Biofilms commonly develop where tissue integrity has been compromised, such as in burn wounds. The biofilm matrix is composed of extracellular polysaccharides and DNA, to which the bacteria adhere. The biofilm mass protects the bacterial residents from phagocytosis while impairing the diffusion and efficacy of antibodies and administered antibiotics. Therefore, the bacteria avoid elimination and persist; meanwhile, ongoing recognition of virulence factors by the immune system may lead to massive inflammation and local tissue damage, exacerbating the infection.

Mechanisms of Microbial Entry Into Cells Essentially all bacterial pathogens that are predominantly intracellular during infection possess mechanisms for inducing internalization into human cells. Even pathogens that survive for prolonged periods in normally phagocytic cells such as macrophages (*Salmonella* spp., *L. monocytogenes*) possess mechanisms for inducing internalization into these cells.

TRIGGER MECHANISM Bacterial systems that deliver proteins into human cells are critical to many aspects of pathogenesis (Fig. 120-1). One of these systems is the type III secretion system, a specialized system that delivers proteins from the bacterial cytoplasm directly into the cytosol of eukaryotic cells. Evolutionarily conserved T3SSs are found in many gram-negative bacterial pathogens. In all cases, delivery of the proteins that are injected into the cell (*effector* proteins) has dramatic effects on the cell; for many human pathogens, these effects include the induction of bacterial uptake into the cell and the alteration of other cellular processes in ways that promote infection.

The T3SS forms an apparatus on the bacterial surface that resembles a needle and syringe, with a hollow tube down its long axis (Fig. 120-1). The base of the organelle spans the two bacterial membranes, forming a conduit from the bacterial cytoplasm, where proteins are synthesized. Anchored to the organelle base and protruding from the bacterial surface is a long needle-like structure. Upon contact of the needle tip with the plasma membrane of the eukaryotic cell, two proteins secreted through the apparatus form a pore in the plasma membrane. The tip of the needle docks onto the extracellular face of the pore, thereby forming a continuous conduit between the bacterial cytoplasm and the host cell cytosol. It is through this conduit that bacterial effector proteins are then delivered into the cell.

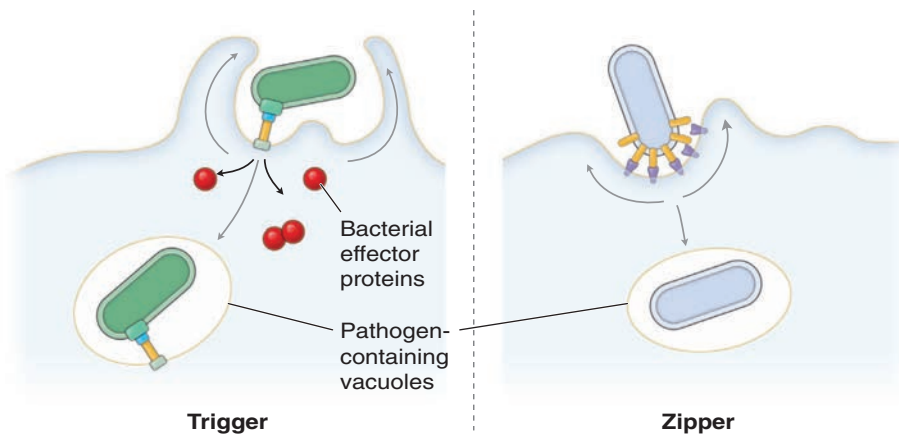


FIGURE 120-2 Common mechanisms of bacterial invasion. The mechanisms for internalization of bacteria into nonphagocytic cells are typically classified as *trigger* or *zipper* mechanisms. As examples of the trigger mechanism, *Shigella* and *Salmonella* spp. use their T3SSs to deliver into host cells effector proteins that manipulate the cytoskeleton in ways leading to the formation of cytoskeleton-supported membrane ruffles. These membrane ruffles extend and surround the pathogen, with consequent endocytosis of the bacterium. As examples of the zipper mechanism, bacterial membrane proteins of *Yersinia* and *Listeria* spp. induce clustering of host receptors. Clustering and subsequent intracellular signaling result in the uptake of the bacterium in a tightly opposed vacuole.

For those pathogens that use a T3SS to induce entry into human cells (*Salmonella*, *Shigella*, and *Chlamydia* spp.), among the first effector proteins secreted into the cell are several that activate polymerization of cellular actin immediately beneath the point where the bacterium is docked on the plasma membrane. Actin polymerization pushes outwardly against the plasma membrane, generating large ruffles of the membrane that engulf the bacterium and take it up into a membrane-bound vacuole (Fig. 120-2). *Shigella* spp. then escape the vacuole and reside in the host cell cytosol, whereas *Salmonella* and *Chlamydia* spp. predominantly reside within the vacuole.

ZIPPER MECHANISM Several invasive pathogens that do not utilize a T3SS for internalization enter cells using a zipper-like mechanism. In these instances, bacterial surface molecules engage with and stimulate the clustering of host cell receptors, which form close associations with the bacterium in a processive fashion until the plasma membrane is tightly opposed to and surrounding the bacterium (Fig. 120-2). For example, the *Yersinia pseudotuberculosis* invasion molecules YadA and invasin bind to $\beta 1$ -integrins; this binding stimulates phosphoinositide 3-kinase and protein kinase B (Akt) signaling pathways, leading to zipper-mediated entry of *Y. pseudotuberculosis* into otherwise nonphagocytic cells. In another example, *L. monocytogenes* internalins InlA and InlB engage E-cadherin and the Met receptor, activating β -catenin signaling and zipper-mediated entry of *L. monocytogenes* into otherwise nonphagocytic cells. Subpopulations of bacteria classically considered extracellular may also promote their own uptake in a similar fashion, with the lectin LecA of *P. aeruginosa* and fibronectin-binding protein of *S. aureus* performing similar functions.

SURVIVAL IN THE VACUOLE

After uptake into cells, the majority of invasive bacterial pathogens remain in a vacuole. Uptake vacuoles normally enter the endosomal pathway, acquiring host proteins that promote vesicle maturation, acidification, and degradation of contents. Many bacterial pathogens contained in these endosomes subvert this intracellular trafficking in ways that prevent endosomal maturation and block lysosomal degradation of the pathogen. A distinct subset of bacteria actively damage vacuolar membrane integrity, disrupting the vacuole and thereby escaping into the cytosol, where they replicate. For bacteria that remain in the vacuole, various bacterial effector proteins are delivered by secretion systems either into the vacuolar space or across the vacuolar membrane into the cytosol (Fig. 120-1), where these proteins manipulate host processes to the benefit of the pathogen.

TYPE III SECRETION SYSTEMS

T3SSs (Fig. 120-1) are versatile virulence systems, helping bacteria such as *P. aeruginosa* remain extracellular, but promoting both the uptake

and the intracellular survival of the intracellular pathogens *S. enterica* serovar Typhimurium, *Chlamydia* spp., and *Shigella* spp. *S. enterica* serovar Typhimurium uses distinct T3SSs for the invasion and subsequent maintenance of the *Salmonella*-containing vacuole (SCV), whereas *Chlamydia* and *Shigella* each encode only one such system. The effector proteins delivered are the major determinants of the lifestyles of these pathogens; *Shigella* quickly escapes the vacuole in a T3SS-dependent manner, whereas *Chlamydia* remains in the vacuolar compartment (the inclusion) and uses effector proteins to hijack cellular trafficking and perturb innate immune responses. The SCV and the *Chlamydia* inclusion exhibit distinct traits: the SCV is associated with thin membranous tubules that aid nutrient acquisition, whereas the inclusion is localized at the microtubule-organizing center, which is thought to facilitate recruitment of vesicles.

TYPE IV SECRETION SYSTEMS

For other pathogens, type IV secretion systems (T4SSs, Figure 120-1)—conceptually similar, evolutionarily distinct effector protein delivery systems—are key to the ability to survive in cellular phagocytic vacuoles. These T4SSs are similar to the T3SSs described above in that they form a multiprotein apparatus that contains a continuous channel between the bacterial cytoplasm and the human cell cytosol. However, T4SSs are functionally more diverse than T3SSs, in that (1) they are present among both gram-negative and gram-positive bacteria as well as some archaea, (2) a large subset of T4SSs transport DNA in a process called conjugation, and (3) some T4SSs deliver proteins, typically toxins, into bacterial cells rather than eukaryotic cells. Not surprisingly, T4SSs also display great structural diversity.

For the human pathogens *Legionella pneumophila* (the etiologic agent of Legionnaires' disease) and *Coxiella burnetii* (the etiologic agent of Q fever), bacterial effector proteins delivered across the vacuolar membrane into the cell cytosol by a T4SS alter maturation of the vacuole in a manner that makes it hospitable for bacterial survival; the resulting vacuole is known as a *Legionella*-containing or *Coxiella*-containing vacuole (LCV or CCV, respectively). The delivered effector proteins block fusion of lysosomes with the vacuole, thereby preventing bacterial degradation by lysosomal enzymes; manipulate host vesicular trafficking pathways; and remodel intracellular membranes to alter the lipid and protein content of the vacuolar membrane. For example, the *L. pneumophila* phospholipase VipD, a T4SS substrate, reduces the levels of phosphatidylinositol 3-phosphate on the LCV, thereby preventing recruitment of the host protein Rab5 GTPase, which is involved in endosomal maturation. The mature LCV membrane contains many features that resemble cellular endoplasmic reticulum. In the case of *C. burnetii*, the vacuole displays more lysosome-like characteristics, including a relatively low pH, which induces the activation of T4SS effector delivery. The delivered bacterial effectors stimulate efficient vesicle recruitment from endosomal and autophagosomal (see "Autophagy," below) networks, massively increasing the membrane of the CCV until it occupies the majority of the cytosol. For both *L. pneumophila* and *C. burnetii*, the mature bacterium-containing vacuoles are hospitable to bacterial replication and long-term survival.

OTHER SECRETION SYSTEMS

Francisella tularensis, the etiologic agent of the zoonotic infection tularemia, is a gram-negative facultative intracellular bacterium that displays a tropism for macrophages. *F. tularensis* resides in a phagosome, the maturation of which it delays by remodeling the lipid content of the vacuolar membrane through the action of its type VI secretion system (T6SS; Fig. 120-1). *F. tularensis* then avoids destruction by lysosomal

enzymes by escaping into the cytosol, where the bacteria replicate. Another pathogen that escapes into the cytosol of macrophages is the respiratory pathogen *Mycobacterium tuberculosis*. Here, the ESX-1 type VII secretion system (T7SS) is required for lysis of the phagosomal membrane. For both of these pathogens, few effectors of their cognizant secretion system have been characterized and the process of phagosomal membrane lysis remains poorly understood.

SURVIVAL IN THE CYTOSOL

All invasive bacteria initially enter cells into a vacuole (nonphagocytic cells) or a phagosome (phagocytic cells), yet, for some pathogens, residence in the vacuole or phagosome is transient. *L. monocytogenes*, *Shigella* spp., spotted fever group *Rickettsia* spp. (the etiologic agents of Rocky Mountain spotted fever, rickettsialpox, and other spotted fever diseases), and *Burkholderia pseudomallei* (the etiologic agent of melioidosis) lyse the vacuole rapidly after invasion. Infections caused by these agents are characterized by their cytosolic lifestyles, actin-based motility, and cell-to-cell spread. The gram-positive bacterium *L. monocytogenes* produces a pore-forming toxin, listeriolysin O, and phospholipases that mediate escape into the cytosol, whereas the mechanism of vacuolar escape of *Shigella* spp. is still uncertain except for a requirement for its T3SS. Once in the cytosol, bacteria readily access nutrients such as glucose and amino acids but must also evade cell-autonomous immune mechanisms, including autophagy (see “Autophagy,” below). *L. monocytogenes*, *Shigella* spp., *B. pseudomallei*, and spotted fever group *Rickettsia* spp. display actin-based motility, where expression of a cell-surface protein (ActA in *L. monocytogenes*, the autotransporter IcsA in *Shigella* spp., the autotransporter BimA in *B. pseudomallei*, and RickA and Sca2 in spotted fever group *Rickettsia* spp.) recruits host factors that polymerize actin. Actin polymerization occurs at the end of the bacterium and propels the microbe through the cell cytosol. This subversion of the host cytoskeleton aids bacterial survival in the cytosol and promotes bacterial dissemination through epithelial tissues. In the case of *Shigella*, *Listeria*, and *Rickettsia*, actin-based motility pushes the bacteria into the host cell membranes to form protrusions that are subsequently taken up by the neighboring cell, resolving into a double-membrane vacuole containing the bacterium. This vacuole is lysed once more, releasing the bacteria into the cytosol, wherein they begin another round of replication. The modes of actin-based motility exhibit many similarities despite being utilized by evolutionarily diverse bacteria, indicating the importance of this virulence mechanism for cytosolic pathogens. *Burkholderia* spp., on the other hand, induce formation of multinucleated giant cells. Here, the bacteria trigger the infected host cell to fuse its membranes with adjacent cells in a T6SS-dependent manner, forming a contiguous cytoplasm.

NUTRIENT ACQUISITION

Regardless of the site of bacterial infection, microbes must acquire nutrients to replicate. For both extracellular and vacuolar pathogens, a global defensive strategy used by the host is the maintenance of nutrient-poor conditions. Also impacting extracellular bacteria is the microbiota, which contributes to nutritional immunity through competition for resources (Chap. 471). Bacterial pathogens possess an arsenal of nutrient acquisition mechanisms to overcome these challenges. For example, upon oxidative stress, to maximize nutrient uptake, several pathogens secrete metal ion-binding proteins through T4SSs (Fig. 120-1) and upregulate expression of cell surface receptors. Many bacteria manipulate the host to subvert the processes restricting nutrient availability, with intracellular bacteria commonly delivering factors that inhibit host mRNA translation and thus increase the available pool of amino acids. Pathogens within vacuoles have evolved methods of redirecting host nutrient transport to these vacuoles.

CELLULAR TRAFFICKING

Because the vacuolar environment is poor in nutrients, bacteria that survive in vacuoles have evolved intricate strategies for the manipulation of host processes to meet their metabolic needs. Once internalized, vacuolar pathogens modulate endosomal trafficking to

prevent degradation by lysosomal compartments. Whereas cytoskeletal rearrangements are regulated by small GTPases of the Rho family, the transport of vesicles through the cell cytosol is controlled by Rab and Arf GTPases. The co-evolution of vacuole-residing bacteria with their eukaryotic hosts has promoted the acquisition of eukaryotic-like domains in many secreted bacterial effector proteins. Many of these effector proteins mimic host cellular trafficking proteins such as guanine nucleotide exchange factors (GEFs), which regulate GTPase activities, and soluble *N*-ethylmaleimide-sensitive factor attachment protein receptors (SNAREs), which control vesicle fusion with vacuolar and other membranes. These bacterial effectors cause the accumulation of host protein and lipid markers on the vacuolar membranes, thereby causing the vacuoles to resemble host compartments. Consequently, host vesicles are redirected from the endoplasmic reticulum, Golgi apparatus, and secretory pathways to the pathogen-containing vacuole, thus delivering nutrients to the vacuole. *Chlamydia* spp., which are obligate intracellular bacteria, acquire host-derived lipids in the bacterial membranes. Dissection of the modes of action of bacterial effectors has revealed novel biochemical modifications of host trafficking regulators. For example, the *L. pneumophila* T4SS secretes two effector proteins that inactivate the cellular GTPase Rab1: AnkX inactivates Rab1 by transferring to it the phospholipid phosphatidylcholine, and SidM modifies Rab1 by adding an adenosine monophosphate molecule to it. This bacterium delivers more than 300 effector proteins through its T4SS; some of these proteins (*meta-effectors*) modulate the activity of other effectors rather than host factors. The coordinated activity of the numerous effector proteins provides precise spatiotemporal control of host processes.

AVOIDANCE OF INNATE IMMUNE RESPONSES

For the host to respond to and clear an infection, it must first be capable of sensing it. The route of infection and properties of the pathogen contribute to determining the nature of the immune response. The human immune system is generally considered to consist of two arms: the innate immune system, initiated by germline-encoded sensors, and the adaptive immune system, where lymphocytes are clonally selected against specific antigens. Initial interactions of pathogens are with specific host cell types, influenced by bacterial tropism; the resulting release of danger signals and cytokines from the innate immune response shapes the later adaptive immune response.

Epithelia (e.g., in the intestinal and respiratory tracts) are commonly the first tissues that a pathogen encounters. The mucosal surfaces of these epithelia are covered in a gel-like mucus layer that contains secretory immunoglobulins, antimicrobial peptides, and commensal bacteria, which offer the first line of defense that infecting pathogens must overcome (Chap. 471). In the gastrointestinal tract, this barrier can be overcome through hijacking of microfold cells (M cells) in the follicle-associated epithelia. M cells sample antigens, including intact pathogenic bacteria, from the gut lumen and deliver them to the underlying gut-associated lymphoid tissue in a process known as *transcytosis*. Within the gut-associated lymphoid tissue, professional phagocytic immune cells, such as dendritic cells, macrophages, and neutrophils, engulf transcytosed material for destruction. Lysosomal degradation of pathogenic bacteria in macrophages and dendritic cells can then lead to antigen presentation, where the hydrolytic products are processed for display by the major histocompatibility complexes (MHCs). These antigen-presenting cells migrate through the lymphatic system to lymph nodes, where the adaptive immune response is stimulated through activation of B-cell and T-cell clones.

COMPLEMENT

Once bacteria have crossed epithelial barriers into the lamina propria, they encounter components of humoral immunity. The complement system is a complex of plasma and tissue-resident proteins that can undergo activation from their pre-protein or zymogen forms to either label pathogens for phagocytosis (*opsonization*) or lyse them directly (Fig. 120-3). Complement activation commences with detection of the pathogen through the binding of circulating antibodies, which are

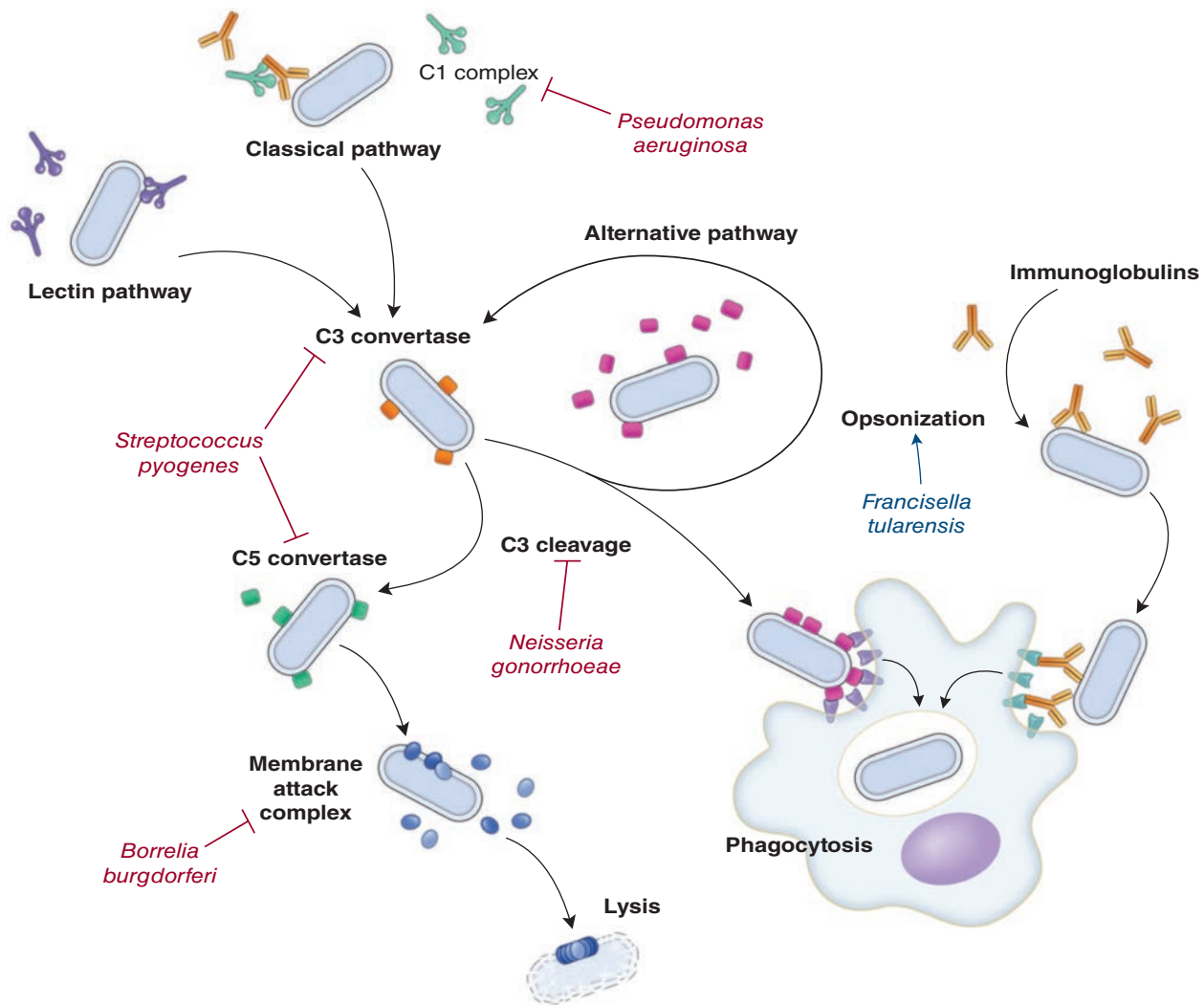


FIGURE 120-3 Overview of the complement system in bacterial infection. The *classical* and *lectin* pathways of complement activation are initiated by the binding of the C1 complex to antibodies bound to bacteria or to lectins binding carbohydrate moieties on the bacterial surface, respectively. A cascade of proteolytic cleavage generates the C3 convertase complex on the bacterial surface, which, upon activation, leads to C5 convertase formation. In the *alternative* pathway, spontaneous C3 cleavage forms alternative C3 products that lead to bacterial opsonization and to amplification of C3 convertase activation. The three complement pathways converge at the point of C3 convertase activation. In a manner similar to that seen for bacteria opsonized with circulating immunoglobulin, opsonized bacteria bind to surface receptors of phagocytes, triggering their engulfment for destruction and stimulation of the adaptive immune system. The C5 convertase, generated by conventional C3 convertase activation, recruits further complement proteins to form the membrane attack complex that directly lyses bacteria. Many bacterial pathogens inhibit specific points in the complement cascade, a selection of which are shown with blunt-ended arrows; in contrast, others, such as *Francisella* spp., promote their own opsonization and subsequent phagocytosis, enabling them to reach their intracellular replicative niche.

recognized by the C1 complex (*classical pathway*), or of lectins (*lectin pathway*) to surface carbohydrates. These pathways converge at the deposition of the C3 convertase complex on the bacterium. The *alternative pathway* is the third route of complement activation, whereby an alternative C3 convertase complex is formed either spontaneously or by the action of plasma factors, promoting amplification of the cascade. Activation of further complement components results in chemokine production, bacterial opsonization, or assembly of the membrane attack complex pore on the bacterial surface, which lyses the pathogen. Bacterial virulence factors have been described that inhibit each step of the complement pathway, highlighting the importance of this antimicrobial system to host immunity. The various strategies of bacterial interference include prevention of initial recognition by assembly of a polysaccharide capsule, modification of bacterial lipopolysaccharide (LPS), degradation of complement components, and masking of surfaces with host proteins. Streptococci employ numerous mechanisms for complement evasion and are thus a formidable foe. The streptococcal proteases ScpA and SpeB degrade many complement proteins, whereas the bacterial surface-exposed M proteins bind host proteins, including the complement inhibitor C4BP, thereby preventing C3b

deposition. The thick peptidoglycan layer of gram-positive bacteria provides modest resistance to membrane attack complex-mediated lysis. Furthermore, streptococci produce a hyaluronic capsule that shields the bacterial surface from complement recognition. In comparison, *Borrelia burgdorferi*, the etiologic agent of Lyme disease, encodes a CD59-like protein that inhibits the completion of membrane attack complex assembly. Whereas many pathogens inhibit opsonization, thereby avoiding phagocytosis, other bacteria, including *Francisella* and *Yersinia* spp., promote this process, leading to uptake into the phagosomal compartment, wherein they replicate and prevent lysosomal fusion.

■ LYSOSOMES

Lysosomes are vesicular organelles of the endosomal system that contain dozens of hydrolytic and antimicrobial enzymes in a low pH environment. One such enzyme is lysozyme, which hydrolyzes the polymeric glycan chains in the bacterial cell wall peptidoglycan, resulting in bacterial lysis. In addition, release of peptidoglycan fragments stimulates other components of the innate immune system. Lysozyme is highly effective against gram-positive bacteria; however,

in gram-negative bacteria, because the peptidoglycan layer lies between the outer and inner membranes, it is inaccessible to lysozyme. Gram-positive bacteria can decrease their susceptibility to lysozyme by peptidoglycan modification, such as decreased acetylation of the sugars in the glycan strand.

■ ANTIMICROBIAL PEPTIDES

Antimicrobial peptides, such as the positively charged defensins and cathelicidins, are concentrated in degradative compartments, where they bind the anionic surface of bacteria and perturb the integrity of the LPS of the gram-negative outer membrane and the wall teichoic acids in the gram-positive cell wall. Many bacterial pathogens modify these surfaces to reduce the net negative charge, thereby decreasing the binding of negatively charged antimicrobial peptides. In addition, *S. aureus* and the respiratory pathogen *Burkholderia cenocepacia* secrete proteases that cleave antimicrobial peptides; this action facilitates bacterial survival at mucosal surfaces and within phagosomes.

■ OXIDATIVE BURST

Pathogens are exposed to reactive oxygen species and reactive nitrogen species produced by cells responding to infection. A well-characterized example is the neutrophil oxidative burst, where the membrane complex nicotinamide adenine dinucleotide phosphate oxidase (NADPH oxidase) produces superoxide radicals through the transfer of electrons to molecular oxygen. NADPH oxidase activation at the site of infection generates reactive oxygen species in the phagosome of neutrophils and/or in the extracellular space. Bacterial detoxification enzymes, including superoxide dismutases, peroxiredoxins, and catalases, enable bacteria to survive this oxidative burst. Many bacteria can also block the activity of NADPH oxidase; several pathogens prevent assembly of the NADPH oxidase complex by T3SS effector protein-mediated post-translational modification of small GTPases. The importance of NADPH oxidase in immune defense is highlighted clinically by the marked susceptibility to bacterial and fungal infections in patients with chronic granulomatous disease, which results from nonfunctional alleles for essential subunits of this complex.

■ NEUTROPHIL EXTRACELLULAR TRAPS

The oxidative burst also stimulates the release of the pro-inflammatory tumor necrosis factor α , azurophilic granules, and neutrophil extracellular traps (NETs). NETs are an extruded mesh of decondensed chromatin that are thought to ensnare pathogens, thereby restricting their spread and destroying them. Streptococci encode deoxyribonucleases (DNases) that enable pathogens to evade NETs. *S. aureus* nuclease degrades NETs; the nucleotide products of this degradation are then converted into deoxyadenosine, which functions as a potent macrophage toxin. This process showcases how bacterial pathogens may utilize antimicrobial responses to their own benefit.

■ PATTERN RECOGNITION RECEPTORS AND THEIR EVASION

The first line of detection of pathogens by the innate immune system is via recognition of molecular patterns that are indicative of the presence of pathogens and that distinguish self from nonself. These are the pathogen- and damage-associated molecular patterns (PAMPs and DAMPs, respectively). Recognition of these patterns enables cells to establish that an infection is occurring and to determine the nature of the pathogen, as dictated by the specific PAMP detected. Receptors for the molecular patterns include Toll-like receptors (TLRs) and pattern recognition receptors (PRRs). TLRs, transmembrane proteins at the cell surface or in the endosomal compartment, surveil the extracellular milieu, whereas other PRRs monitor threats within the cell cytosol. On phagocytic cells, TLRs may employ scavenger receptors as co-stimulatory surface receptors. TLRs recognize conserved molecules that include LPS, peptidoglycan, flagella, and nucleic acids (Table 120-3). Stimulation of these receptors by ligand binding triggers assembly of supramolecular organizing centers (SMOCs), large protein complexes that are signaling platforms for all-or-nothing signal transduction via

TABLE 120-3 Pattern Recognition Receptors of the Innate Immune System and Their Ligands

PATTERN RECOGNITION RECEPTOR	LIGAND OR MODE OF ACTIVATION
TLR2 (with TLR1 or TLR6)	Lipoproteins
TLR4	LPS
TLR5	Flagellin
TLR9	CpG DNA
NLRP1	Enzymatic cleavage
NLRP3	Ionic flux, mitochondrial damage
NLRP6	Lipoteichoic acid
NAIPs/NLRC4	Flagellin/T3SS
STING	Cyclic dinucleotides
NOD1 and NOD2	Peptidoglycan

ubiquitination and phosphorylation of target proteins. Ultimately, signaling results in translocation of transcription factors, including nuclear factor- κ B (NF- κ B), AP-1, and interferon-regulatory factors (IRFs), into the nucleus, where they stimulate expression of cytokines, chemokines, and other immunity-related genes (Fig. 120-4). Among the signaling pathways activated upon recognition of certain intracellular bacterial pathogens is the cyclic guanosine monophosphate-adenosine monophosphate synthase (cGAS) stimulator of interferon genes (STING) pathway, which was previously thought to be relevant only to recognition of intracellular viruses. In response to recognition of microbial nucleic acid, cGAS-STING activates the translocation of interferon regulatory factor 3 (IRF3) and NF- κ B into the nucleus, whereupon they activate a set of interferon-regulated pro-inflammatory genes. Cytokine release alerts bystander cells and recruits immune cells from the circulation, creating a heightened immune state in the local area. The nature of the stimulus influences the type of immune response. Cytosolic ligands tend to stimulate stronger pro-inflammatory responses than extracellular ligands, a difference reflecting the higher threat perceived. Moreover, detection of cytosolic peptidoglycan, which is derived from bacteria, elicits a response distinct from that to dsRNA, which is derived from viruses.

Pathogens have evolved a plethora of strategies to avoid detection by the innate immune system. A common strategy is to mask PAMPs. For example, LPS on the surface of gram-negative bacteria contains a lipid A component that usually harbors six acyl chains. LPS that is hexa-acylated is recognized by TLR4 at the cell surface. The etiologic agent of plague, *Yersinia pestis*, produces hexa-acylated LPS in the flea vector, but, upon the organism's transmission to human hosts, the temperature shift to 37°C induces acyltransferase expression that results in production instead of tetra-acylated LPS, which does not activate the TLR4 receptor. Tetra-acylated LPS is also synthesized by the etiologic agent of tularemia, *F. tularensis*; the periodontal pathogen *Porphyromonas gingivalis*; and the gastric pathogen *H. pylori*. The flagellum of *H. pylori* is also essential for virulence, enabling the bacterium to reach the gastric epithelium after ingestion. Flagellin, the monomeric protein subunit of the flagellum, is an extracellular ligand for TLR5, but *H. pylori* bypasses flagellin recognition since its flagellin epitope is divergent from the TLR5 recognition motif. DNA in intracellular membrane compartments of the endosomal system is recognized by TLR9, which binds to CpG-containing unmethylated DNA. Because CpG motifs are usually methylated in mammalian DNA, the methylation state provides a method of discriminating self from nonself. Group A streptococci secrete the Sda1 DNase, which degrades CpG DNA, therefore reducing the immunogenicity of lysed bacteria.

Inhibition of NF- κ B Signaling Upon binding to bacterial peptidoglycan, the cytosolic PRRs NOD1 and NOD2 activate NF- κ B signaling, thereby priming cell immune responses. Polymorphisms in the *NOD1* and *NOD2* genes confer susceptibility to inflammatory bowel disease, an action that highlights the role of cytosolic sensing of bacterial cell wall components in immune homeostasis. The co-evolution

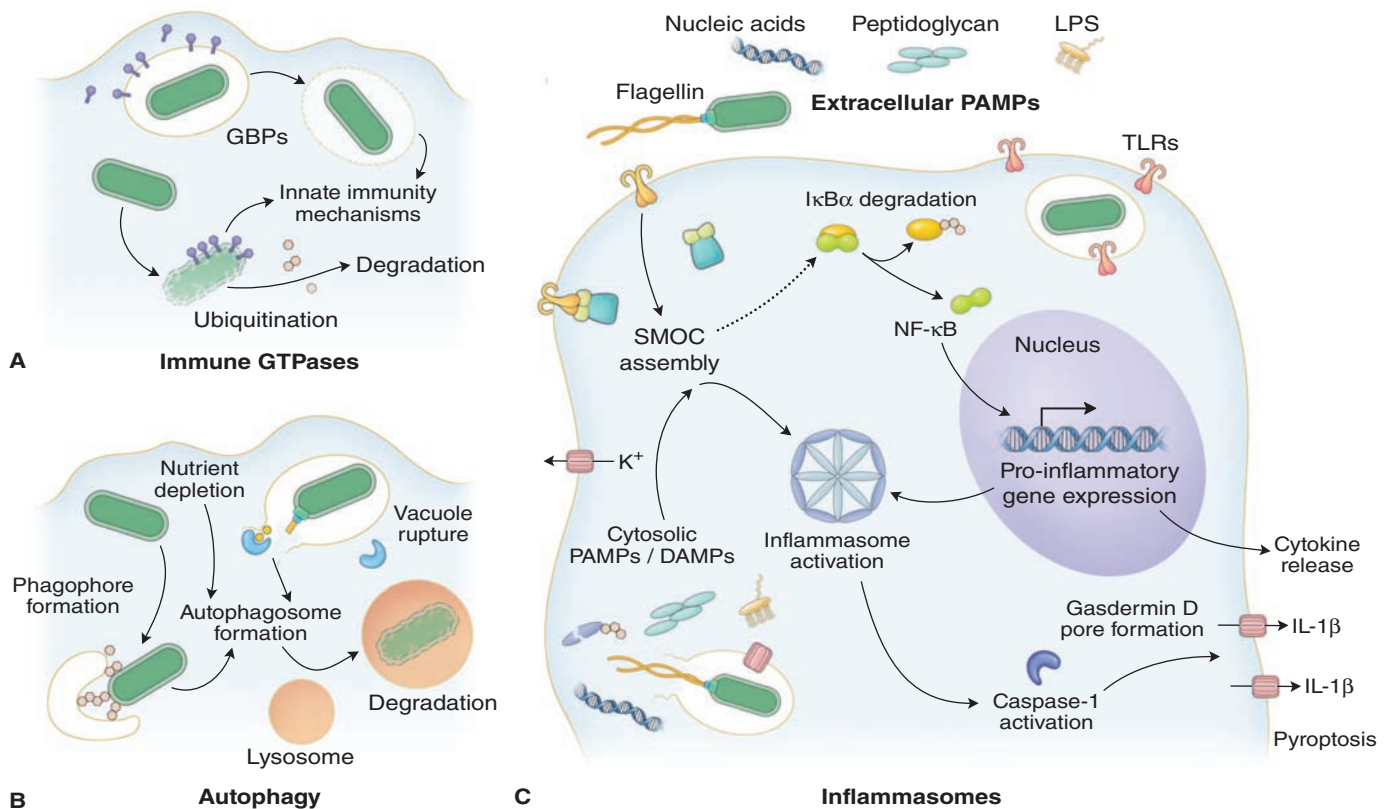


FIGURE 120-4 Overview of innate immune recognition and response to bacterial pathogens. **A.** Immune GTPases. Proteins of the GBP family associate with the bacterial surface, LPS shed from the bacterial surface, or pathogen-containing vacuoles, disrupting the integrity of the pathogen and leading to lysis. Labeling of intracellular pathogens with immune GTPases may instead stimulate ubiquitination of the bacterium and its destruction by cell-autonomous immune mechanisms such as autophagy. **B.** Autophagy. Recognition of bacterial invasion leads to host-mediated sequestration of the pathogen in an autophagosome and destruction of the pathogen upon fusion of the autophagosome with lysosomes. Bacteria may be directly labeled by ubiquitination, damaged vacuoles are recognized by the binding of galectins to exposed glycans, and/or the cellular autophagy may be induced in response to sensing starvation. **C.** Pattern recognition receptors and innate immune signaling. Conserved PAMPs, DAMPs, and danger signals are recognized by all cells, triggering innate immune signaling pathways and pro-inflammatory gene expression. TLRs surveil the cell surface and endosomal interior for PAMPs that include LPS, nucleic acids, and flagellin. Cytosolic PRRs, including NLRs and ALRs, recognize similar ligands as well as distinct stimuli, including dysregulation of ion homeostasis. TLR activation promotes SMOC assembly and NF-κB signaling. Under resting conditions, NF-κB is sequestered in the cytosol by IκBα, but TLR activation leads to IκBα phosphorylation, ubiquitination, and degradation, with the consequent release of NF-κB. Resultant translocation of NF-κB into the nucleus leads to pro-inflammatory gene expression and the production and release of cytokines. Stimulation of cytosolic PRRs leads to assembly of SMOCs known as inflammasomes, resulting in activation of caspases. A well-characterized example is the activation of caspase-1 by canonical and noncanonical inflammasomes, subsequent activation of the membrane pore protein GSDMD, maturation and release of pro-inflammatory cytokines IL-1β and IL-18, and pyroptotic death of the cell.

of bacterial pathogens with the host has led to the acquisition in many bacterial virulence factors of eukaryotic-like domains that enable the pathogen to hijack host signal transduction. *Ubiquitination*, a post-translational modification of eukaryotic proteins that regulates many cellular processes, is a common target among known bacterial mechanisms of host signaling inhibition. In ubiquitination, the small ubiquitin protein is covalently linked to a target protein, and additional ubiquitin molecules are added on; the consequence is branched or linear chains that signal the fate of the target. Ubiquitination-mediated target fates are degradation by the proteasome (most common), targeting to particular vesicular compartments, or signaling. The enzymes responsible for labeling targets with ubiquitin are E3 ubiquitin ligases, which transfer ubiquitin from an E2 ubiquitin-conjugating enzyme to the target protein.

A plethora of bacterial effector proteins specifically target components of the NF-κB pathway in ways that maintain repression of NF-κB-dependent pro-inflammatory genes. In doing so, these pathogens counteract activation of innate immune responses triggered by their conserved PAMPs. The translocation of NF-κB into the nucleus requires the successive phosphorylation, ubiquitination, and degradation of IκBα by the IκB kinase (IKK) complex, which in resting cells functions to inhibit NF-κB; the dissociation of the IKK complex from NF-κB enables the latter to translocate into the nucleus, where it functions as a transcriptional factor (Fig. 120-4C). The T3SS of *Shigella* spp. delivers into cells several E3 ubiquitin ligases that target

components of the NF-κB pathway for degradation. The intracellular pathogens *Chlamydia trachomatis* and *B. pseudomallei* secrete bacterial deubiquitinases through T3SS and T2SS, respectively, which cleave the ubiquitin moiety from IκBα, thereby preventing its degradation, maintaining inhibited NF-κB in the cytosol, and repressing pro-inflammatory gene expression.

Other enzymatic activities of bacterial effectors can also inhibit NF-κB activation by ubiquitin-dependent processes. *L. pneumophila* delivers, via its T4SS, the MavC transglutaminase, which performs noncanonical ubiquitination of an E2 enzyme, preventing activation of the IKK complex; the *Shigella* spp. T3SS cryptic kinase OspG binds and inhibits cellular E2 ubiquitin-conjugating enzymes involved in NF-κB activation; and NleE and OspZ from EPEC and *Shigella* spp., respectively, are cysteine methyltransferases that inhibit IKK activation.

Another cellular pathway of pro-inflammatory signaling stimulated by PRRs is a phosphorylation cascade involving mitogen-activated protein kinases (MAPKs) that leads to activation of the transcription factor AP-1. The potent *Bacillus anthracis* exotoxin lethal factor exhibits metalloprotease activity that cleaves the N-terminal region of most MAPKs, thereby ablating the kinase cascade. The homologous T3SS effectors YopJ, AvrA, and VopA of *Yersinia* spp., *S. enterica* serovar Typhimurium, and *Vibrio parahaemolyticus*, respectively, acetylate key serine residues in MAPKs, which blocks their phosphorylation. Finally, OspF of *Shigella* spp. and the ortholog SpvC from *S. enterica* serovar Typhimurium are phosphothreonine lyases that irreversibly cleave

phosphorylated threonine residues from MAPKs, thereby preventing their activities. *S. enterica* serovar Typhimurium also targets NF- κ B directly with several T3SS metalloproteases. The broad range of effector protein mechanisms and the large number of distinct host targets underscore the need for bacterial pathogens to subvert innate immune signaling in order to establish a successful infection (Fig. 120-4).

Immune GTPases For many infectious processes, the initial interaction of the pathogen with an immune cell primes an enhanced subsequent immune response by augmenting innate immune mechanisms—for example, through upregulation of immunity-related gene expression and/or recruitment of neutrophils to the site of infection. A second stimulus then activates a more robust innate immune response that is directly geared toward clearing the pathogen or, if the threat is too great, leads to suicide of the infected cell. One set of molecules that primes cellular responses is the immune GTPases, of which the best characterized is the guanylate binding protein (GBP) family. Production of these cytosolic or membrane-associated sensors is induced by interferon signaling. GBPs target pathogen-containing vacuoles, bacterial membranes, or released fragments of bacterial membranes, such as LPS. Analogous to scavenger receptors, GBPs promote the release of PAMPs and DAMPs from these membrane structures and present the PAMP or DAMP to PRRs. For example, some GBPs extract LPS molecules from the surface of cytosolic gram-negative bacterial pathogens, and this extraction leads to activation of the noncanonical inflammasome (see below). Recruitment of GBPs to *Shigella* spp. prevents bacterial actin-based motility and promotes labeling of bacteria with ubiquitin, which targets them for degradation by the proteasome. In this case, the shigellae are able to interfere with this cellular response by secreting an E3 ubiquitin ligase effector that targets the GBPs for degradation.

Inflammasomes Whereas the extracellular environment is surveilled for signs of infection by TLRs, the cytosol is monitored by other PRRs, including members of the NOD- and AIM2-like receptor (NLR and ALR, respectively) families, which form SMOCs known as *inflammasomes*. Upon binding to ligand (for example, cytosolic wall teichoic acids and dsDNA), these PRRs oligomerize, often with the ASC scaffold protein. Oligomerization is associated with recruitment of pro-caspases, zymogens that are activated within the complexes by auto-cleavage. The cleaved forms of pro-inflammatory caspases are themselves proteases that cleave zymogen substrates. For example, caspase-1 cleaves the substrate gasdermin D (GSDMD), enabling the free amino-terminal fragment of GSDMD to assemble into a pore in the plasma membrane. Caspase-1 also cleaves and activates pro-IL-1 β and pro-IL-18, releasing the mature cytokines IL-1 β and IL-18, which are secreted from the cell through the GSDMD pore, and enabling cytokine signaling (Fig. 120-4). The formation of GSDMD pores also leads to lytic death of the cell, a process designated *pyroptosis* (inflammatory cell death). Pyroptosis is triggered by either of two types of inflammasomes, the canonical caspase-1 inflammasome and the non-canonical inflammasome.

The PRRs of the best characterized canonical inflammasomes are NLRP1, NLRP3, and NAIP-NLRC4. The NLRP3 inflammasome appears to be activated upon sensing of homeostatic dysregulation, including ion flux and mitochondrial damage. Thus, it may be activated by pore-forming toxins such as the phenol-soluble modulins of *S. aureus* and α -hemolysin of UPEC. NAIP-NLRC4 senses the needle protein of the T3SS and the related flagellin protein. Many pathogens have evolved mechanisms to overcome activation of the NAIP-NLRC4 inflammasome, including downregulation of flagella expression by *L. pneumophila* upon entry into cells as well as activation of pro-survival Akt signaling by the *S. enterica* serovar Typhimurium T3SS effector SopB, which reduces *nlr4* expression during intracellular infection. In addition, caspases of the noncanonical inflammasome are directly inhibited by the binding of the T3SS effector proteins OspC3 and NleF from *Shigella* spp. and EHEC, respectively. Overall, the inhibition of inflammasomes by bacterial pathogens enables pathogens to avoid elimination by the innate immune system.

Inflammasomes have been described as organelles that assess the level of threat posed by pathogens, since they not only recognize PAMPs and DAMPs but also act as sentinels for dysregulation of host processes. Both the NOD1 and pyrin inflammasomes sense cytoskeletal perturbations by monitoring modification of small GTPases. Pyrin senses a range of covalent modifications of Rho GTPases, including glucosylation by *C. difficile* TcdB, adenylation by *C. botulinum* C3 toxin, and deamidation by *B. cenocepacia* T6SS effector TecA. Meanwhile, recent work suggests that the NLRP1 inflammasome may act as bait for cleavage by microbial proteases or ubiquitin-targeted degradation, processes that stimulate its activation and subsequent pyroptosis. Once more, the importance of inflammasome signaling is underscored by a plethora of polymorphisms that exist in inflammasome-associated genes and that may give rise to autoinflammatory disorders or may confer high susceptibility to microbial infection.

Inhibition of Nonpyroptotic Cell Death An outcome of inflammasome activation in macrophages, and occasionally in non-phagocytic cells, is pyroptosis. This inflammatory form of cell death both destroys the pathogen residing within the cell and alerts bystander cells of the threat. Several other types of regulated (*programmed*) cell death occur, of which inflammatory necroptosis and noninflammatory apoptosis are best characterized. In *necroptosis*, activation of any of a variety of innate immune signaling pathways (interferon- γ , TLRs) leads to kinase-dependent assembly of a plasma membrane pore complex that induces cell lysis and release of DAMPs into the extracellular space, events that in turn induce an intense inflammatory response. Necroptosis has been described in bacterial pneumonia and sepsis.

In *apoptosis*, cells condense and form membranous blebs without exposing or releasing cytosolic contents into the surrounding milieu and thus without inducing inflammation in adjacent tissues. The apoptotic blebs and dead cells are disposed of by macrophages in a process known as *efferocytosis*. Apoptosis is not instigated solely upon infection; it plays a key role in development and also occurs normally in many epithelial tissues, such as the intestine, where, during normal cell turnover, epithelial cells migrate to the tips of intestinal villi, from which they extrude and detach.

Apoptosis can be initiated by intrinsic or extrinsic cellular stimuli, resulting in activation of apoptotic caspases that demolish cellular contents. The intrinsic pathway occurs through dissolution of the mitochondrial membrane potential, which commonly occurs during infection because of cellular stress or the action of bacterial toxins. The extrinsic pathway can be stimulated by interactions of cell-surface death receptors with pro-inflammatory ligands like TNF- α .

The gastrointestinal pathogens *S. enterica* serovar Typhimurium, *Yersinia enterocolitica*, and pathogenic *E. coli* inhibit signal transduction from cell-surface death receptors, thereby blocking cell death. The cell's sensing of inhibition of extrinsic apoptotic signaling leads to stimulation of alternative inflammatory cell death pathways; this scenario highlights the importance of functional redundancy of critical host immune responses. These pathogens can block signaling by modifying arginine residues of death receptor SMOCs with sugar moieties or by direct cleavage of SMOC components (Fig. 120-4).

The intracellular vacuole-residing *Chlamydia* spp. activate survival pathways through phosphoinositide 3-kinase (PI3K) and Wnt/ β -catenin signaling, while simultaneously inhibiting apoptosis. T4SS effectors of vacuole-residing *L. pneumophila* and *C. burnetii* inhibit apoptosis through the sequestration of pro-apoptotic host factors. Cyclomodulins, a group of bacterial effector proteins that arrest the cell cycle, are delivered into cells by the extracellular attaching and effacing pathogens EPEC and EHEC. The EPEC and EHEC cyclomodulin Cif deamidates ubiquitin and related proteins, inactivating the host E3 ligases involved in cell cycle progression and thereby reducing epithelial turnover. The recognition by the cell of this disruption of important host cell processes triggers cell death pathways. Thus, co-evolution of bacterial pathogens with their hosts results in a tug-of-war.

Autophagy Once detection of a pathogen has been established and cytokine secretion has been triggered, nonimmune cells are able

to deploy cell-autonomous immunity mechanisms to neutralize the threat. In addition to assembly of the NADPH oxidase complex and production of immune GTPases, the responses include upregulation of autophagy. Autophagy disposes of damaged organelles and recycles long-lived protein complexes through generation of double-membrane compartments (*autophagosomes*) around the cargo and delivery of the engulfed cargo to the lysosome, wherein the cargo is degraded to amino acids and other constituents. Autophagy can isolate and dispose of cytosolic pathogens. Recognition of the pathogen can be direct or can occur through the sensing of DAMPs. Here, host galectins, polysaccharide-binding molecules, participate in the recognition of pathogens and induce ubiquitination of the bacterial surface and recruitment of the host autophagy protein LC3, which triggers engulfment in autophagosomes.

The survival of intracellular bacterial pathogens is dependent on their ability to inhibit the autophagy machinery. Common mechanisms utilized by bacteria to interfere with autophagy include evasion of ubiquitination by inhibition of host E3 ligases, blocking of ubiquitination sites through decoration of surface-exposed virulence factors with host proteins, or alteration of host pathways that regulate autophagy. The cytosolic pathogen *L. monocytogenes* induces actin-based motility and cell-to-cell spread through recruitment of the host factor Arp2/3, and Arp2/3 recruitment helps mask *Listeria* surface proteins from ubiquitination. An important metabolic trigger of autophagy is intracellular depletion of amino acids, which occurs upon replication of metabolically active intracellular pathogens. The master regulator of cell metabolism is mammalian target of rapamycin (mTOR); mTOR is inactivated by intracellular starvation conditions, and its inactivation derepresses autophagy. Invasive pathogens, including *M. tuberculosis*, *S. enterica* serovar Typhimurium, and *Shigella* spp., have evolved strategies to re-activate mTOR signaling during infection, thereby evading autophagy. *S. enterica* serovar Typhimurium also secretes two effectors, SseF and SseG, that block the activation of a kinase required for autophagosome generation. Because of the ubiquity of autophagy and other cell-autonomous immunity pathways in the body, bacterial pathogens have evolved mechanisms to overcome these key host defenses regardless of cellular tropism.

Dysregulated or excessive immune responses are detrimental to the host, as in sepsis or severe COVID-19 infection, causing morbidity and sometimes death. Too weak an immune response may lead to pathogen-induced morbidity and mortality. A cellular mechanism that promotes appropriate immune responses is innate immune training, which enables cells to be primed at the transcriptional level through epigenetic modifications. Here, epigenetic alterations in chromatin structure cause stable changes in gene expression profiles. Epigenetic regulation consists of post-translational modification of histones, with alteration of their phosphorylation, acetylation, and/or methylation states—changes that influence chromatin structure and access of transcription factors to regulatory elements in the DNA. Mounting evidence indicates that the human microbiota promotes this innate immune training, modifying the methylation state of promoters of genes involved in metabolism and immunity and allowing the host to respond to infection in a well-tuned manner. Not surprisingly, pathogens also regulate host responses at the epigenetic level. *B. anthracis* and *L. pneumophila* deliver into cells histone methyltransferases that control inflammation and ribosome activity. *L. monocytogenes* listeriolysin O, the *P. aeruginosa* T3SS pore, and other plasma membrane pore-forming complexes exert epigenetic modulation through alterations in ion homeostasis, thereby manipulating the gene expression profiles of infected host cells.

■ INHIBITION OF ADAPTIVE IMMUNE RESPONSES

Early interactions of pathogens with host cells trigger the innate immune response, releasing cytokines that recruit additional antigen-presenting immune cells to the site of infection. To establish chronic infection, bacterial pathogens need not only to suppress the innate immune system but also to avoid elimination by the adaptive immune response. The adaptive immune system comprises clonally expanded lineages of B and T lymphocytes that have been activated by

antigen-presenting cells, the best characterized of which are dendritic cells and macrophages. B cells are required for humoral immunity; genetic diseases that affect B-cell function often manifest as the inability to produce adequate antibody titers to clear extracellular bacterial infections. T cells generally mediate cell-mediated immunity, helping the host clear infected cells. The activation of T cells occurs by the presentation of processed antigen on MHC molecules of antigen-presenting cells. The activation of T cells is controlled by the specificity of MHC molecules and of their receptors. The large variety of receptors possessed by T cells enables collective recognition of a wide variety of antigens. Mutations that ablate the development of the adaptive immune system result in severe combined immunodeficiency, rendering the individual extremely vulnerable to infection.

M. tuberculosis is notorious for its ability to establish latent infections, avoiding elimination by the immune system for decades. Once phagocytosed, the pathogen induces secretion of immunosuppressive cytokines, including IL-6, IL-10, and transforming growth factor β . The result is inhibition of interferon- γ -dependent gene expression, with downregulation of MHC class II and other immune-stimulatory molecules, which inhibits induction of helper CD4⁺ T lymphocytes. Similarly, *B. pertussis* induces dendritic cells to produce IL-10, which skews T-cell maturation into regulatory T cells, dampening the immune response. The *S. enterica* serovar Typhimurium T3SS effector protein SteD depletes MHC class II molecules from the surface of dendritic cells by the activation of the E3 ubiquitin ligase MARCH8. MARCH8 ubiquitinates MHC class II, interfering with its trafficking to the cell surface and thereby decreasing interaction with and subsequent activation of T lymphocytes.

In addition to preventing antigen-presenting cells from stimulating lymphocytes, pathogens may also directly alter B-cell and T-cell activity. The *Yersinia* T3SS effector YopH, a protein tyrosine phosphatase, dephosphorylates B-cell and T-cell receptors and thus prevents both signal transduction upon stimulation by antigen-presenting cells and lymphocyte activation. Staphylococcal toxic shock syndrome is induced by the production of superantigens by *S. aureus*. These extremely inflammatory exotoxins are potent activators of T cells, stimulating exuberant and at times fatal cytokine production. Rather than binding the specificity-determining variable regions of MHC molecules and the T-cell receptor, superantigens bind invariable regions and are therefore able to nonspecifically activate vast numbers of T cells, with a consequent cytokine storm.

■ BACTERIAL CYTOTOXINS

Many bacterial toxins are *cytotoxins*—toxins that trigger host cell death. The best-characterized group of bacterial cytotoxins are the AB toxins, which are defined by the presence of an active (A) subunit, which mediates the enzymatic activity of the toxin, and a binding (B) subunit, which mediates binding to the cellular receptor. This family of toxins includes Shiga toxins of *Shigella* spp. and pathogenic *E. coli*, ϵ toxin of *C. perfringens*, diphtheria toxin of *Corynebacterium diphtheriae*, pertussis toxin of *B. pertussis*, and CTX of *V. cholerae*. In general, upon binding to cell surface receptors, the toxin is taken up into endosomes, whereupon the A-subunit is translocated across the endosome membrane and released into the cytosol, where its toxic enzymatic activity is stimulated. CTX and pertussis toxin exhibit ADP-ribosyl transferase activity, targeting G-protein regulators of adenylate cyclases in a manner that increases cellular cyclic AMP concentrations and thereby perturbing ion homeostasis and apoptosis. Increased intracellular cyclic AMP induces chloride secretion via CFTR and inhibits sodium chloride absorption; the results are massive fluid secretion into the lumen of the small intestine and the diarrheal symptoms of cholera. Many other bacterial toxins that are not AB family toxins, including *C. perfringens* ϵ toxin and *P. aeruginosa* exotoxin A, display ADP-ribosyl transferase activity that targets elongation factor-2, thereby inhibiting host translation. In contrast, the *S. enterica* serovar Typhimurium protein SpvB and *C. difficile* binary toxins target actin, inhibiting normal cellular cytoskeletal rearrangements. Despite the range of enzymatic activities displayed by the bacterial toxins described above, the effects on the host cell generally fall into the few broad categories of cytoskeletal

manipulation, inhibition of innate immune response signaling, and hijacking of cellular trafficking.

Pore-forming toxins are prevalent virulence factors of extracellular pathogens. They include the cholesterol-dependent cytolysins intermedilysin of *S. intermedius*, θ toxin of *C. perfringens*, anthrolysin O of *B. anthracis*, and listeriolysin O of *L. monocytogenes*. *S. aureus* produces several cytolysins, including the Panton-Valentine leucocidin; α -, β -, γ -, and δ -hemolysins; and phenol-soluble modulins. Following their release by bacteria, leukocidins and α - and γ -hemolysins form oligomeric complexes in host plasma membranes, with consequent lysis of the host cell. These pore-forming toxins have different cell-type specificities that are thought to be driven by binding of pore components with specific cell surface receptors. Phenol-soluble modulins are small peptides that, as a result of their amphipathic nature, appear to directly insert into cell membranes, such that no receptor is thought to be necessary. Lytic toxins protect *S. aureus* from phagocytosis, yet recent work has uncovered a host defense mechanism that counters cytolysins through the release of vesicles (termed *exosomes*), which act as decoys by removing cytolysins from the local environment.

■ TISSUE DAMAGE AND PATHOGEN DISSEMINATION

Much of the pathology associated with bacterial infection results from pro-inflammatory immune responses. Infected cells may continually signal in ways that alert the immune system, even though, as described above, many bacteria avoid elimination by cell-autonomous and immune cell-mediated mechanisms. In the intestine, tight junctions mediate intimate associations between epithelial cells that maintain tissue barrier function, linking the cytoskeletal networks of adjacent cells through intimate association of protein complexes across the cell membranes. Many intestinal pathogens perturb the integrity of the gut epithelium either by manipulating cell polarity or by disrupting intercellular junctions. *C. perfringens*, *V. cholerae*, pathogenic *E. coli*, *Shigella* spp., and *S. enterica* serovar Typhi all produce toxins in the gastrointestinal tract that disrupt tight junctions, consequently disrupting the barrier function of the tissue and facilitating access of the pathogen to deeper tissue. The RtxA multifunctional repeats-in-toxin (MARTX) toxin of *V. cholerae* causes cell rounding and barrier failure through actin-crosslinking activity, yet avoids eliciting substantial immune responses by simultaneously inactivating phospholipases and Rho GTPases. The serine protease autotransporters (SPATEs) of *Shigella* spp. and some pathogenic *E. coli* are typically secreted into the gut lumen or mucus layer, whereupon they cleave components of epithelial junctions and mucins in ways that facilitate tissue penetration. Tissue damage permits access to the underlying mucosal layers, the lymphatics, and the bloodstream; for some pathogens, this access enables seeding of other organs.

Transmission to New Hosts The host represents a replicative niche for bacterial pathogens, in which they multiply and are transmitted to new hosts. The mode of transmission is typically aligned with the mode of entry. For example, for respiratory pathogens, coughing induced by tissue damage in the lung aerosolizes the pathogen, enabling inhalation by and colonization of a new host. Similarly, gastrointestinal pathogens elicit diarrhea and are transmitted via the direct fecal–oral route or via contamination of crops or food with waste from an infected individual. The understanding of the spread of infectious diseases permits the institution of basic hygiene procedures that greatly diminish transmission rates—for example, hand washing, decontamination of communal surfaces, and adoption of social distancing measures.

Bacteriophages and Pathogen Lifestyle The reservoir of *V. cholerae* is aquatic environments. Disease is acquired through the ingestion of contaminated seawater or seafood. In regions where cholera is endemic, the disease displays seasonal peaks. This seasonality is associated with blooms of bacteria-targeting viruses (*bacteriophages*), which infect *V. cholerae* organisms, replicate, and lyse and kill the bacterial host. Bacterial lysis releases viral particles into the aquatic

environment, whereupon they can infect other *V. cholerae*. Consequently, the bacteriophages regulate the abundance of pathogens, with ramifications for the epidemiology of the disease. Whether bacteriophages contribute to cyclical control of other pathogens is currently unknown.

■ COMPETITION WITH COMMENSAL MICROBES

In recent years, it has become clear that the host microbiota constitutes an ecosystem that contributes to host defense against infecting pathogens. This community of commensal microorganisms (bacteria, fungi, viruses) is essential for the metabolic homeostasis and immune development of the host. Furthermore, the presence of commensal organisms on external surfaces (e.g., the skin), in the nasopharynx, and in the intestinal tract provides colonization resistance against pathogens. The gut microbiota is among the densest ecosystems known. Consequently, pathogens are greatly outnumbered by flora, and, within each niche, pathogens must compete with commensals for nutrients. The importance of the microbiota is underscored by *C. difficile* infection, wherein antibiotic treatment, which eradicates components of the gut microbiota, enables *C. difficile* spores to germinate, with consequent toxin production and disease. It is notable that an effective treatment for *C. difficile* colitis is repopulation of the microbiota through fecal transplantation.

Many commensal organisms possess antibacterial systems, including bacteriocins and T6SSs. One activity of T6SSs is the delivery of toxins directly into neighboring bacteria, resulting in lysis or growth inhibition of the target. T6SSs are present in both pathogenic and commensal bacteria, and T6SS-mediated killing contributes both to the establishment of a stable gut microbial community and to colonization by invading pathogens, including *Shigella sonnei* and *V. cholerae*. In cystic fibrosis patients, upregulation of a T6SS of *P. aeruginosa* occurs; this alteration suggests a role for this system in the persistence of chronic *P. aeruginosa* infection in the cystic fibrosis lung, which is a polymicrobial environment.

■ SUMMARY

Bacterial pathogens display myriad mechanisms of colonization, adhesion, invasion, dissemination, and manipulation of host pathways. Infectious diseases are the result when pathogens successfully establish themselves within the host, and symptoms are usually the result of the ensuing fights between the pathogens and the immune system. The sheer diversity of virulence determinants highlights the success of the host in combating infection. Further elucidation of how bacteria cause infection will better our understanding of human biology and pathogens and will provide new opportunities for successful therapeutic intervention.

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and Asia, transforming the recognition and management of incident tuberculosis and multidrug resistance where they are most prevalent. Since rifampin resistance frequently accompanies resistance to other antibiotics, this test can suggest the presence of multidrug-resistant *M. tuberculosis* within hours instead of weeks, without the infrastructure required for culture.

High-resolution genomic tracking of the spread of epidemics—from cholera to Ebola to Zika to COVID-19—has yielded insights into which public health measures may prove most effective in controlling local epidemics. Many genomic tracking efforts have involved close collaborations with local scientists and public health officials, and considerable investment in sequencing infrastructure in sub-Saharan Africa has made on-location epidemic tracking in the event of another such outbreak feasible. Such investment can not only enable real-time outbreak recognition and tracking but also provide the infrastructure needed to capitalize on the many other benefits of high-throughput sequencing as they are developed. The early returns of such investments are exemplified by the rapid reporting of genome sequences for SARS-CoV-2, with 55,000 viral genome sequences reported within the first 6 months of the pandemic. Overall, sequencing efforts have become cheaper and have moved closer to point-of-care with each passing year. As these technologies synergize with efforts to globalize information-technology resources, global implementation of genomic methods promises to spread state-of-the-art methods for diagnosis, treatment, and epidemic tracking of infections to areas that need these capabilities the most.

GENOMICS AND THE COVID-19 PANDEMIC

The COVID-19 pandemic, which began in 2019 and spread worldwide in 2020, resulted in hundreds of millions of documented infections and millions of deaths and serves as a prime example of the pandemic potential of infectious pathogens. It also demonstrated the central role that genomic tools now play in response to infectious outbreaks, ranging from enabling diagnostics and vaccines to tracking evolution, virulence, and transmissibility of the pathogen. The rapid discovery of SARS-CoV-2 and sequencing of its genome was complete within weeks of the recognition of the clinical syndrome. The rapid public sharing of this genome sequence led directly to two key interventions: diagnostic assay development via RT-qPCR and vaccine design. Crucially, vaccine development was informed by homology of the SARS-CoV-2 sequence to SARS and MERS coronaviruses. The dominant antigen of those viruses, the surface protein Spike, was well characterized, enabling the design of the first SARS-CoV-2 vaccines to begin the day after the genome sequence was shared. The progress of the most rapidly developed and validated vaccine in human history was unquestionably accelerated by genomic technology. WGS has also played a large role in outbreak tracking and confirmation of case clusters in institutional settings such as hospitals or congregate living facilities, in helping to distinguish reinfections from recrudescence or prolonged viral shedding, in monitoring spread through societies, and in tracking pathogen evolution, including the emergence of new variants of concern with altered transmissibility, severity, and/or partial evasion of the immune response generated to prior versions of the virus, vaccines, or monoclonal antibody therapeutics. Finally, cutting edge genomic methods including single-cell transcriptional profiling and genome-wide association studies are contributing to our understanding of the wide variability in outcomes of SARS-CoV-2 infection, ranging from asymptomatic carriage to death. Overall, just as the global response to the COVID-19 pandemic underscores the indispensable role that genomics methods have come to play in the clinical and public health management of infectious diseases, the devastating impact of this pandemic reveals the urgent need for further development and implementation of tools for disease surveillance and response.

SUMMARY

By illuminating the genetic information that encodes the most fundamental processes of life, genomic technologies are transforming many aspects of medicine. In infectious diseases, methods such as next-generation sequencing and genome-scale expression analysis

offer information of unprecedented depth about individual microbes as well as microbial communities. This information is expanding our understanding of the interactions of microorganisms with each other, their human hosts, and the environment. Despite technological and financial barriers that continue to slow the widespread adoption of large-scale pathogen sequencing in clinical and public health settings, genomic methodologies have utterly transformed the research landscape in infectious disease and are beginning to make meaningful inroads into clinical settings. As even vaster amounts of data are generated, innovations in data storage, development of bioinformatics tools to manipulate the data, standardization of methods, and training of end-users in both the research and clinical realms will be required. The cost-effectiveness and applicability of whole genome sequencing, particularly in the clinic, remain to be studied, and studies of the impact of genome sequencing on patient outcomes will be needed to clarify the contexts in which these new methodologies can make the greatest contributions to patient well-being. The ongoing efforts to overcome limitations through collaboration, teaching, and reduction of financial obstacles should be applauded and expanded. With advances in genomic technologies and computational analysis, our ability to detect, characterize, treat, monitor, prevent, and control infections has advanced rapidly in recent years and will continue to do so, with the hope of heralding a new era where the clinician is better armed to combat infection and promote human health.

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Approach to the Acutely Ill Infected Febrile Patient

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The physician treating the acutely ill febrile patient must be able to recognize infections that require emergent attention. If such infections are not adequately evaluated and treated at initial presentation, the opportunity to alter an adverse outcome may be lost. In this chapter, the clinical presentations of and approach to patients with infectious disease emergencies are discussed. These infectious processes and their treatments are discussed in detail in other chapters.

APPROACH TO THE PATIENT

Acute Febrile Illness

Before the history is elicited and a physical examination is performed, an immediate assessment of the patient's general appearance can yield valuable information. The perceptive physician's subjective sense that a patient is septic or toxic often proves accurate. Visible agitation or anxiety in a febrile patient can be a harbinger of critical illness.

HISTORY

Presenting symptoms are frequently nonspecific. Detailed questions should be asked about the onset and duration of symptoms and about changes in severity or rate of progression over time. Host factors, such as extremes of age, and comorbid conditions may increase the risk of infection with certain organisms or of a more fulminant course than is usually seen. Lack of splenic function, alcoholism with significant liver disease, IV drug use, HIV infection, diabetes, malignancy, morbid obesity, organ transplantation, and chemotherapy all predispose to specific infections and frequently to increased severity. The patient should be questioned about factors that might help identify a nidus for invasive infection, such as recent upper respiratory tract infections, influenza, or varicella; prior trauma; disruption of cutaneous barriers due to lacerations, burns, surgery, body piercing, or decubiti; and the presence of foreign bodies or prosthetic devices. Travel, presence during a natural disaster such as a hurricane or tsunami, contact with pets or other animals, or activities that might result in tick or mosquito exposure can lead to diagnoses that would not otherwise be considered. Recent dietary intake, medication use, social or occupational contact with ill individuals, vaccination history, recent sexual contacts, and menstrual history may be relevant. Pregnancy might increase the risk and severity of some illnesses, such as influenza or COVID-19, or increase the risk of significant morbidity for the fetus, as in *Listeria* or Zika virus infection. A detailed review of systems should include any neurologic signs or sensorium alterations, rashes or skin lesions, and focal pain or tenderness.

PHYSICAL EXAMINATION

A complete physical examination should be performed, with special attention to several areas that are sometimes given short shrift in routine examinations such as assessment of the patient's general appearance and a detailed skin, soft tissue and neurologic evaluation.

The patient may appear either anxious and agitated or lethargic and apathetic. Fever is usually present, although elderly patients and compromised hosts (e.g., patients who are uremic or cirrhotic and those who are taking glucocorticoids or nonsteroidal anti-inflammatory drugs) may be afebrile despite serious underlying infection. Critically ill patients may be hypothermic, with a high risk of organ failure and mortality. Mortality at 30 days has been shown to decrease with increasing temperature at presentation. Measurement of blood pressure, heart rate, and respiratory rate and oxygen saturation helps determine the degree of hemodynamic and metabolic compromise. The patient's airway must be evaluated to rule out the risk of obstruction from an invasive oropharyngeal infection.

The etiologic diagnosis may become evident in the context of a thorough skin examination (**Chap. 19**). Petechial rashes are typically seen with meningococcemia or Rocky Mountain spotted fever (RMSF; **see Fig. A1-16**); erythroderma is associated with toxic shock syndrome (TSS). On soft tissue and muscle examination, areas of erythema or duskeness, edema, and tenderness may indicate underlying necrotizing fasciitis, myositis, or myonecrosis. The neurologic examination must include a careful assessment of mental status for signs of early encephalopathy. Evidence of nuchal rigidity or focal neurologic findings should be sought.

DIAGNOSTIC WORKUP

After a quick clinical assessment, diagnostic material should be obtained rapidly and antibiotic and supportive treatment begun. Blood (for cultures; baseline complete blood count with differential;

measurement of serum electrolytes, blood urea nitrogen, serum creatinine, and serum glucose; liver function tests and serum lactate; C-reactive protein, lactate dehydrogenase, and D-dimer) can be obtained at the time an IV line is placed and before antibiotics are administered. Three sets of blood cultures should be performed for patients with possible acute endocarditis. Blood smears from patients at risk for severe parasitic disease, such as malaria or babesiosis (**Chaps. 224, 225, and A2**), must be examined for the diagnosis and quantitation of parasitemia. Blood smears may also be diagnostic in ehrlichiosis and anaplasmosis. Testing of a nasopharyngeal sample for possible COVID-19 may be indicated.

Patients with possible meningitis should have cerebrospinal fluid (CSF) drawn before the initiation of antibiotic therapy. Focal findings, depressed mental status, or papilledema should be evaluated by brain imaging prior to lumbar puncture, which, in this setting, could initiate herniation. *Antibiotics should be administered before imaging but after blood for cultures has been drawn.* If CSF cultures are negative, blood cultures will provide the diagnosis in 50–70% of cases. Molecular diagnostic techniques (e.g., broad-range 16S rRNA gene polymerase chain reaction testing for bacterial meningitis pathogens) are of increasing importance in the rapid diagnosis of life-threatening infections.

Focal abscesses necessitate immediate CT or MRI as part of an evaluation for surgical intervention. Other diagnostic procedures, such as wound cultures, should not delay the initiation of treatment for more than minutes. Once emergent evaluation, diagnostic procedures, and (if appropriate) surgical consultation (see below) have been completed, other laboratory tests can be conducted. Appropriate radiography, CT and/or MRI imaging, urinalysis, measurement of the erythrocyte sedimentation rate, C-reactive protein, and/or procalcitonin, and transthoracic or transesophageal echocardiography all may prove important.

TREATMENT

The Acutely Ill Patient

In the acutely ill patient, empirical antibiotic therapy for presumed bacterial or fungal infection is critical and should be administered without undue delay in addition to fluid resuscitation and vasopressor support as needed. Increased prevalence of antibiotic resistance in community-acquired bacteria must be considered when antibiotics are selected. **Table 122-1** lists first-line empirical regimens for infections considered in this chapter. In addition to the rapid initiation of antibiotic therapy, several of these infections require urgent surgical attention. Neurosurgical evaluation for subdural empyema, otolaryngologic surgery for possible mucormycosis, and cardiothoracic surgery for critically ill patients with acute endocarditis are as important as antibiotic therapy. For infections such as necrotizing fasciitis and clostridial myonecrosis, rapid surgical intervention supersedes other diagnostic or therapeutic maneuvers.

Adjunctive treatments may reduce morbidity and mortality rates and include dexamethasone for bacterial meningitis or IV immunoglobulin for TSS. Adjunctive therapies should usually be initiated within the first hours of treatment; however, dexamethasone for bacterial meningitis must be given before or at the time of the first dose of antibiotic. Glucocorticoids may also be harmful—e.g., when given in the setting of cerebral malaria or viral hepatitis.

SPECIFIC PRESENTATIONS

The infections considered below according to common clinical presentation can have rapidly catastrophic outcomes, and their immediate recognition and treatment can be life-saving. Recommended empirical therapeutic regimens are presented in **Table 122-1**.

■ SEPSIS WITHOUT AN OBVIOUS FOCUS OF PRIMARY INFECTION

Patients initially have a brief prodrome of nonspecific symptoms and signs that progresses quickly to hemodynamic instability with

TABLE 122-1 Empirical Treatment for Common Infectious Disease Emergencies ^a				
CLINICAL SYNDROME	POSSIBLE ETIOLOGIES	TREATMENT	COMMENTS	SEE CHAP(S).
Sepsis without a Clear Focus				
Septic shock	<i>Pseudomonas</i> spp., gram-negative enteric bacilli, <i>Staphylococcus</i> spp., <i>Streptococcus</i> spp.	Vancomycin (15 mg/kg q12h) ^b plus either Piperacillin/tazobactam (4.5 g) q8h via extended infusion (EI) ^c or cefepime (2 g) q8h via EI	Empirical therapy should be tailored to local resistance patterns. Carbapenem or aminoglycoside antibiotics should be considered for empirical therapy when rates of multidrug-resistant gram-negative organisms are high or for patients with risk factors for resistant organisms. Adjust treatment when culture data become available.	147, 148, 161, 164, 304
Overwhelming post-splenectomy sepsis	<i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Neisseria meningitidis</i>	Ceftriaxone (2 g q12h) plus vancomycin (15 mg/kg q12h) ^b	If a β -lactam-sensitive strain is identified, vancomycin can be discontinued and a narrower-spectrum agent, such as penicillin, considered based on susceptibility testing.	304
Babesiosis	<i>Babesia microti</i> (U.S.), <i>B. divergens</i> (Europe)	Atovaquone (750 mg q12h) plus azithromycin (500 mg q24h)	Clindamycin (600 mg q8h) plus quinine (650 mg q8h) can be used in severe disease not responding to atovaquone and azithromycin. Treatment with doxycycline (100 mg bid) for potential co-infection with <i>Borrelia burgdorferi</i> or <i>Anaplasma</i> spp. may be prudent.	222, 225
Sepsis with Skin Findings				
Meningococcemia	<i>N. meningitidis</i>	Ceftriaxone (2 g q12h) or penicillin (4 mU q4h)	Ceftriaxone eradicates nasopharyngeal carriage of the organism. Close contacts require chemoprophylaxis with rifampin (600 mg q12h for 2 days) or ciprofloxacin (a single dose, 500 mg).	155
Rocky Mountain spotted fever (RMSF)	<i>Rickettsia rickettsii</i>	Doxycycline (100 mg bid)	If both meningococcemia and RMSF are being considered, use ceftriaxone (2 g q12h) plus doxycycline (100 mg bid). If RMSF is diagnosed, doxycycline is the proven superior agent.	187
Purpura fulminans	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>N. meningitidis</i>	Ceftriaxone (2 g q12h) plus vancomycin (15 mg/kg q12h) ^b	If a β -lactam-sensitive strain is identified, vancomycin can be discontinued.	146, 155, 157, 304
Erythroderma: toxic shock syndrome	Group A <i>Streptococcus</i> , <i>Staphylococcus aureus</i>	Vancomycin (15 mg/kg q12h) ^b plus clindamycin (600 mg q8h)	If a penicillin- or oxacillin-sensitive strain is isolated, these agents are superior to vancomycin (penicillin, 2 mU q4h; or oxacillin, 2 g IV q4h). The site of toxigenic bacteria should be debrided; IV immunoglobulin can be used in severe cases. ^d	147, 148
Sepsis with Soft Tissue Findings				
Necrotizing fasciitis	Group A <i>Streptococcus</i> , mixed aerobic/anaerobic flora, CA-MRSA ^e	Vancomycin (15 mg/kg q12h) ^b plus piperacillin/tazobactam (4.5 g q8h via EI) ^c plus clindamycin (600 mg q8h)	Urgent surgical evaluation is critical. Empirical therapy should be tailored to local resistance patterns. For mixed aerobic/anaerobic infections, clindamycin can be discontinued. Adjust treatment when culture data become available.	129, 147, 148
Clostridial myonecrosis	<i>Clostridium perfringens</i>	Penicillin (2 mU q4h) plus clindamycin (600 mg q8h)	Urgent surgical evaluation is critical.	154
Neurologic Infections				
Bacterial meningitis	<i>S. pneumoniae</i> , <i>N. meningitidis</i>	Ceftriaxone (2 g q12h) plus vancomycin (15 mg/kg q12h) ^b	If a β -lactam-sensitive strain is identified, vancomycin can be discontinued. If the patient is >50 years old or has comorbid disease, add ampicillin (2 g q4h) for <i>Listeria</i> coverage. Dexamethasone (10 mg q6h for 4 days) started before, or at the time of, the first dose of antibiotic improves outcome in adults with meningitis (especially pneumococcal).	138
Brain abscess, suppurative intracranial infections	<i>Streptococcus</i> spp., <i>Staphylococcus</i> spp., anaerobes, gram-negative bacilli	Vancomycin (15 mg/kg q12h) ^b plus metronidazole (500 mg q8h) plus ceftriaxone (2 g q12h)	Urgent surgical evaluation is critical. If a penicillin- or oxacillin-sensitive strain is isolated, these agents are superior to vancomycin (penicillin, 4 mU q4h; or oxacillin, 2 g q4h).	138
Cerebral malaria	<i>Plasmodium falciparum</i>	Artesunate (2.4 mg/kg IV at 0, 12, and 24 h; then once daily) ^f	Avoid glucocorticoids. Until IV artesunate is available, treatment can be initiated with oral artemether-lumefantrine. Atovaquone-proguanil, quinine, and mefloquine are other options.	222, 224
Spinal epidural abscess	<i>Staphylococcus</i> spp., gram-negative bacilli	Vancomycin (15 mg/kg q12h) ^b plus either Piperacillin/tazobactam (4.5 g q8h via EI) ^c or cefepime (2 g q8h via EI) ^c	Surgical evaluation is essential. If a penicillin- or oxacillin-sensitive strain is isolated, these agents are superior to vancomycin (penicillin, 4 mU q4h; or oxacillin, 2 g q4h).	442
Focal Infections				
Acute bacterial endocarditis	<i>S. aureus</i> , β -hemolytic streptococci, HACEK group, ^g <i>Neisseria</i> spp., <i>S. pneumoniae</i>	Ceftriaxone (2 g q12h) or cefepime (2 g q8h via EI) ^c plus vancomycin (15 mg/kg q12h) ^b	Adjust treatment when culture data become available. Surgical evaluation is essential.	128

^aThese empirical regimens include coverage for gram-positive pathogens that are resistant to β -lactam antibiotics. Local resistance patterns should be considered and may alter the need for empirical vancomycin or for expanded coverage for antibiotic-resistant gram-negative pathogens. ^bA vancomycin loading dose of 20–25 mg/kg can be considered in critically ill patients. Dosing must be adjusted based on pharmacokinetic/pharmacodynamic monitoring. ^cEI=extended infusion. β -Lactam antibiotics may exhibit unpredictable pharmacodynamics in sepsis. Prolonged or continuous infusions are often used. ^dThe optimal dose of IV immunoglobulin has not been determined, but the median dose in observational studies is 2 g/kg (total dose administered for 1–5 days). ^eCommunity-acquired methicillin-resistant *S. aureus*. ^fIn the United States, artesunate must be obtained through the Centers for Disease Control and Prevention. ^g*Haemophilus* spp., *Aggregatibacter* spp., *Cardiobacterium hominis*, *Eikenella corrodens*, and *Kingella kingae*.

976 hypotension, tachycardia, tachypnea, respiratory distress, and altered mental status. Disseminated intravascular coagulation (DIC) with clinical evidence of a hemorrhagic diathesis is a poor prognostic sign.

Septic Shock (See also Chap. 304) Patients with bacteremia leading to septic shock may have a primary site of infection (e.g., pneumonia, pyelonephritis, or cholangitis) that is not evident initially. Elderly patients who may have atypical presentations and often have comorbid conditions, hosts compromised by malignancy and neutropenia, and patients who have recently undergone a surgical procedure or hospitalization are at increased risk for an adverse outcome. Gram-negative bacteremia with organisms such as *Pseudomonas aeruginosa* or *Escherichia coli* and gram-positive infection with organisms such as *Staphylococcus aureus* (including methicillin-resistant *S. aureus* [MRSA]) or group A streptococci can present as intractable hypotension and multiorgan failure. Treatment can usually be initiated empirically on the basis of the presentation, host factors (Chap. 304), and local patterns of bacterial resistance. Outcomes are worse when antimicrobial treatment is delayed or when the responsible pathogen ultimately proves not to be susceptible to the initial regimen. The increasing prevalence of multidrug-resistant organisms makes this especially relevant. Broad-spectrum antimicrobial agents are therefore recommended and should be instituted rapidly, preferably within the first hours after presentation. Pharmacodynamics are altered in sepsis due to increased volume of distribution and renal clearance, so it is important to adequately dose antimicrobials. Risk factors for fungal infection should be assessed, as the incidence of fungal septic shock is increasing. Nonbacterial causes of shock, such as dengue virus infection, should be considered in endemic areas as mortality can increase if undiagnosed. Biomarkers such as C-reactive protein and procalcitonin have not proved reliable diagnostically or prognostically but, when measured over time, can facilitate appropriate de-escalation of therapy with improved outcomes. Glucocorticoids are often considered for patients with severe sepsis who do not respond to fluid resuscitation and vasopressor therapy, but conclusive evidence for efficacy in this setting is lacking.

Overwhelming Infection in Asplenic Patients (See also Chap. 304) Patients without splenic function are at risk for overwhelming bacterial sepsis. Asplenic adult patients succumb to sepsis at 58 times the rate of the general population. Most infections occur within the first 1 or 2 years, but the increased risk persists throughout life. The median interval between splenectomy and sepsis is 4–6 years, with a range of 1–19 years. In asplenia, encapsulated bacteria cause the majority of infections. Adults, who are more likely to have antibody to these organisms, are at lower risk than children. *Streptococcus pneumoniae* is the most common isolate, causing 40–70% of cases. The risk of infection with *Haemophilus influenzae* or *Neisseria meningitidis* also is greater in patients without splenic function, but reported cases are declining. Severe clinical manifestations of infections due to other organisms, such as *E. coli*, *S. aureus*, *Bordetella holmesii*, *Capnocytophaga*, *Babesia*, and *Plasmodium* species, have been described.

Babesiosis (See also Chap. 225) A history of recent travel to endemic areas raises the possibility of infection with *Babesia*. Between 1 and 4 weeks after a tick bite, the patient experiences chills, fatigue, anorexia, myalgia, arthralgia, shortness of breath, nausea, and headache; ecchymosis and/or petechiae are occasionally seen. The tick that most commonly transmits *Babesia*, *Ixodes scapularis*, also transmits *Borrelia burgdorferi* (the agent of Lyme disease) and *Anaplasma*; coinfection can occur, and may result in more severe disease. Infection with the European species *Babesia divergens* is more frequently fulminant than that due to the U.S. species *Babesia microti*. *B. divergens* causes a febrile syndrome with hemolysis, jaundice, hemoglobinemia, and renal failure and is associated with a mortality rate of >40%. Severe babesiosis is especially common in asplenic hosts but does occur in hosts with normal splenic function, particularly those >60 years of age and those with underlying immunosuppressive conditions such as HIV infection or malignancy. Complications include renal failure, acute respiratory failure, DIC, and splenic rupture.

Other Sepsis Syndromes Tularemia (Chap. 170) has been reported in every U.S. state except Hawaii. This disease is associated with wild rabbit, tick, horse-fly, and tabanid fly contact. It can be transmitted by arthropod bite, handling of infected animal carcasses, consumption of contaminated food and water, or inhalation. The typhoidal form can be associated with gram-negative septic shock and a mortality rate of >30%, especially in patients with underlying comorbid or immunosuppressive conditions. Plague occurs infrequently in the United States (Chap. 171), primarily after contact with ground squirrels, prairie dogs, or chipmunks, but is endemic in other parts of the world; >90% of all cases occur in Africa with Madagascar especially affected. The septic form is particularly rare and is associated with shock, multiorgan failure, and a 30% mortality rate. Pneumonic plague is rapidly progressive and fatal without treatment. These infections should be considered in the appropriate epidemiologic setting. The Centers for Disease Control and Prevention (CDC) lists *Francisella tularensis* and *Yersinia pestis* (the agents of tularemia and plague, respectively) along with *Bacillus anthracis* (the agent of anthrax) as important organisms that might be used for bioterrorism (Chap. S3).

■ SEPSIS WITH SKIN MANIFESTATIONS (SEE ALSO CHAP. 19)

Maculopapular rashes may reflect early meningococcal or rickettsial disease but are usually associated with nonemergent infections. Exanthems are usually viral. Primary HIV infection commonly presents with a rash that is typically maculopapular and involves the upper part of the body but can spread to the palms and soles. The patient is usually febrile and can have lymphadenopathy, severe headache, dysphagia, diarrhea, myalgias, and arthralgias. Recognition of this syndrome provides an opportunity to prevent transmission and to institute early treatment.

Petechial rashes caused by viruses are seldom associated with hypotension or a toxic appearance, although there can be exceptions (e.g., severe measles or arboviral infection). Petechial rashes limited to the distribution of the superior vena cava are rarely associated with severe disease. In other settings, petechial rashes require more urgent attention.

Meningococcemia (See also Chap. 155) Almost three-quarters of patients with *N. meningitidis* bacteremia have a rash. Meningococcemia most often affects young children (i.e., those 6 months to 5 years old). In sub-Saharan Africa, the high prevalence of serogroup A meningococcal disease has been a threat to public health for more than a century. Thousands of deaths occur annually in this area, which is known as the “meningitis belt,” and large epidemic waves occur approximately every 8–12 years. Serogroups W135, X, and C also are important emerging pathogens in Africa. For example, Nigeria experienced a large serogroup C outbreak in 2016–2017 in the setting of aggressive vaccination programs against serogroup A. Outside Africa, outbreaks for the past 50 years reported in the United States and Europe are caused primarily by serogroup C (approximately 60%) followed by serogroup B (29%). In the United States, sporadic cases and outbreaks occur in day-care centers, schools (grade school through college, particularly among college freshmen living in residential halls), and army barracks. Household contacts of index cases are at 400–800 times greater risk of disease than the general population. Patients may have fever, headache, nausea, vomiting, myalgias, changes in mental status, and meningismus. However, the rapidly progressive form of disease is not usually associated with meningitis. The rash is initially pink, blanching, and maculopapular, appearing on the trunk and extremities, but then becomes hemorrhagic, forming petechiae. Petechiae are first seen at the ankles, wrists, axillae, mucosal surfaces, and palpebral and bulbar conjunctiva, with subsequent spread on the lower extremities and to the trunk. A cluster of petechiae may be seen at pressure points—e.g., where a blood pressure cuff has been inflated. In rapidly progressive meningococcemia (10–20% of cases), the petechial rash quickly becomes purpuric (see Fig. A1-41), and patients develop DIC, multiorgan failure, and shock; 50–60% of these patients die, and survivors often require extensive debridement or

amputation of gangrenous extremities. Hypotension with petechiae for <12 h is associated with significant mortality. Cyanosis, coma, oliguria, metabolic acidosis, and elevated partial thromboplastin time also are associated with a fatal outcome. Antibiotics given in the office by the primary care provider before hospital evaluation and admission may improve prognosis; this observation suggests that early initiation of treatment may be life-saving. Meningococcal conjugate vaccines are protective against serogroups A, C, Y and W135 and are recommended for children 11–12 years of age with a booster dose at 16 years of age, and for other high-risk patients. Vaccines active against serogroup B are recommended for high-risk individuals ≥10 years of age and may be appropriate for teens and young adults (16 through 23 years of age).

Rocky Mountain Spotted Fever and Other Rickettsial Diseases (See also Chap. 187) RMSF is a tickborne disease that occurs throughout North and South America. It is caused primarily by *Rickettsia rickettsii* but can be caused by other rickettsiae (e.g., *R. parkeri*, *R. akari*). Up to 40% of patients do not report a history of a tick bite, but a history of travel or outdoor activity (e.g., camping in tick-infested areas) can often be ascertained. For the first 3 days, headache, fever, malaise, myalgias, nausea, vomiting, and anorexia are documented. By day 3, half of patients have skin findings. Blanching macules develop initially on the wrists and ankles and then spread over the legs and trunk. The lesions become hemorrhagic and are frequently petechial. The rash spreads to palms and soles later in the course. The centripetal spread is a classic feature of RMSF but occurs in a minority of patients. Moreover, 10–15% of patients with RMSF never develop a rash. The patient can be hypotensive and develop noncardiogenic pulmonary edema, confusion, lethargy, and encephalitis progressing to coma. The CSF contains 10–100 cells/μL, usually with a predominance of mononuclear cells. The CSF glucose level is often normal; the protein concentration may be slightly elevated. Renal and hepatic injury as well as bleeding secondary to vascular damage are noted. Delayed recognition and treatment are associated with a greater risk of death; mortality rates are 20–30% if untreated. Native Americans, Alaskan natives, Pacific Islanders, children 5–9 years of age, adults >70 years old, and persons with underlying immunosuppression are at increased risk of death as well.

Other rickettsial diseases cause significant morbidity and mortality worldwide. *Mediterranean spotted fever* caused by *Rickettsia conorii* is found in Africa, southwestern and south-central Asia, and southern Europe. Patients have fever, flu-like symptoms, and an inoculation eschar at the site of the tick bite. A maculopapular rash develops within 1–7 days, involving the palms and soles but sparing the face. Elderly patients or those with diabetes, alcoholism, uremia, or congestive heart failure are at risk for severe disease characterized by neurologic involvement, respiratory distress, and gangrene of the digits or purpura fulminans. Mortality rates associated with this severe form of disease approach 50%. *Epidemic typhus*, caused by *Rickettsia prowazekii*, is transmitted in louse-infested environments and emerges in conditions of extreme poverty, war, and natural disaster. Patients experience a sudden onset of high fevers, severe headache, cough, myalgias, and abdominal pain. A maculopapular rash develops (primarily on the trunk) in more than half of patients and can progress to petechiae and purpura. Serious signs include delirium, coma, seizures, noncardiogenic pulmonary edema, skin necrosis, and peripheral gangrene. Mortality rates approached 60% in the preantibiotic era and continue to exceed 10–15% in contemporary outbreaks. *Scrub typhus*, caused by *Orientia tsutsugamushi* (a separate genus in the family Rickettsiaceae), is transmitted by larval mites or chiggers and is one of the most common infections in southeastern Asia and the western Pacific. The organism is found in areas of heavy scrub vegetation (e.g., along riverbanks). Patients may have an inoculation eschar and may develop a maculopapular rash, lymphadenopathy, and dyspnea. Severe cases progress to pneumonia, meningoencephalitis, myocarditis, DIC, and renal failure. Mortality rates range from 1% to 70% and vary by location, increasing age, myocarditis, delirium, pneumonitis, or signs of hemorrhage.

If recognized in a timely fashion, rickettsial disease is very responsive to treatment. Doxycycline (100 mg twice daily for 3–14 days) is

the treatment of choice for both adults and children. Mortality rates are higher when tetracycline-based treatment is not given.

Purpura Fulminans (See also Chaps. 155 and 304) Purpura fulminans is the cutaneous manifestation of DIC and presents as large ecchymotic areas and hemorrhagic bullae. Progression of petechiae to purpura, ecchymoses, and gangrene is associated with congestive heart failure, septic shock, acute renal failure, acidosis, hypoxia, hypotension, and death. Purpura fulminans has been associated primarily with *N. meningitidis* but, in splenectomized patients, may be associated with *S. pneumoniae*, *H. influenzae*, and *S. aureus*.

Ecthyma Gangrenosum Septic shock caused by *P. aeruginosa* or, less often, *Aeromonas hydrophila* or other gram-negative organisms, can be associated with ecthyma gangrenosum (see Figs. 164-1 and A1-34): hemorrhagic vesicles surrounded by a rim of erythema with central necrosis and ulceration. Ecthyma gangrenosum is most common among patients with neutropenia, extensive burns, and hypogammaglobulinemia.

Other Infections Associated with Rash *Vibrio vulnificus* and other noncholera *Vibrio* bacteremic infections (Chap. 168) can cause focal skin lesions and overwhelming sepsis in hosts with chronic liver disease, heavy alcohol consumption, iron storage disorders, diabetes, renal insufficiency, hematologic disease, or malignancy or other immunocompromising conditions. Over 95% of the cases are in the subtropical Pacific and Atlantic Oceans coastal regions in the Northern Hemisphere. After ingestion of contaminated raw shellfish (typically oysters from the Gulf Coast in U.S. cases), there is a sudden onset of malaise, chills, fever, and hypotension. The patient develops bullous or hemorrhagic skin lesions, usually on the lower extremities, and 75% of patients have leg pain. The mortality rate can be as high as 35–60%, particularly when the patient presents with hypotension. Outcomes are improved when patients are treated with fluoroquinolones with or without cephalosporins or with tetracycline-containing regimens. Other infections, caused by agents such as *Aeromonas*, *Klebsiella*, and *E. coli*, can cause hemorrhagic bullae and death due to overwhelming sepsis in cirrhotic patients. *Capnocytophaga canimorsus* can cause septic shock in asplenic or cirrhotic patients. Infection typically follows a dog bite. Serovars A–C appear more virulent, constituting 92% of human infections but only 7.6% of canine isolates. Patients present with fever, chills, myalgia, vomiting, diarrhea, dyspnea, confusion, and headache. Findings can include an exanthem or erythema multiforme (see Figs. 56-9 and A1-24), cyanotic mottling or peripheral cyanosis, petechiae, and ecchymosis. About 30% of patients with this fulminant form die of overwhelming sepsis and DIC, and survivors may require amputation because of gangrene.

Erythroderma TSS (Chaps. 147 and 148) is usually associated with erythroderma. The patient presents with fever, malaise, myalgias, nausea, vomiting, diarrhea, and confusion. There is a sunburn-type rash that may be subtle and patchy but is usually diffuse and is found on the face, trunk, and extremities. Erythroderma, which desquamates after 1–2 weeks, is more common in *Staphylococcus*-associated than in *Streptococcus*-associated TSS. Hypotension develops rapidly—often within hours—after the onset of symptoms. Multiorgan failure occurs. Early renal failure may precede hypotension and distinguishes this syndrome from other septic shock syndromes. There may be no indication of a primary focal infection, although possible cutaneous or mucosal portals of entry for the organism can be ascertained when a careful history is taken. Colonization rather than overt infection of the vagina or a postoperative wound, for example, is typical with staphylococcal TSS, and the mucosal areas appear hyperemic but not infected. Streptococcal TSS is more often associated with skin or soft tissue infection (including necrotizing fasciitis), and patients are more likely to be bacteremic. TSS caused by *Clostridium sordellii* is associated with childbirth or with skin injection of black-tar heroin. The diagnosis of TSS is defined by the clinical criteria of fever, rash, hypotension, and multiorgan involvement. (Of note, fever is typically absent when TSS is caused by *C. sordellii*.) The mortality rate is 5% for

978 menstruation-associated TSS, 10–15% for nonmenstrual TSS, 30–70% for streptococcal TSS, and up to 90% for obstetric *C. sordellii* TSS. Clindamycin improves outcomes when included in the treatment regimen. Some studies have shown that use of IV immunoglobulin is associated with improved survival as well.

Viral Hemorrhagic Fevers Viral hemorrhagic fevers (Chaps. 209 and 210) are zoonotic illnesses caused by viruses that reside in either animal reservoirs or arthropod vectors. These diseases occur worldwide and are restricted to areas where the host species live. They are caused by four major groups of viruses: Arenaviridae (e.g., Lassa fever in Africa), Bunyaviridae (e.g., Rift Valley fever in Africa; hantavirus hemorrhagic fever with renal syndrome in Asia; and Crimean-Congo hemorrhagic fever, which has an extensive geographic distribution), Filoviridae (e.g., Ebola and Marburg virus infections in Africa), and Flaviviridae (e.g., yellow fever in Africa and South America and dengue in Asia, Africa, and the Americas). Lassa fever and Ebola and Marburg virus infections are also transmitted from person to person. The vectors for most viral fevers are found in rural areas; dengue and yellow fever are important exceptions. After a prodrome of fever, myalgias, and malaise, patients develop evidence of vascular damage, petechiae, and local hemorrhage. Shock, multifocal hemorrhaging, and neurologic signs (e.g., seizures or coma) predict a poor prognosis. Dengue (Chap. 209) is the most common arboviral disease worldwide. More than half a million cases of dengue hemorrhagic fever occur each year, with at least 12,000 deaths. Patients have a triad of symptoms: hemorrhagic manifestations, evidence of plasma leakage, and platelet counts of <100,000/μL. Mortality rates are 10–20%. If dengue shock syndrome develops, mortality rates can reach 40%. Ebola infection has been associated with outbreaks with high mortality rates. The 2014 outbreak in West Africa had a mortality rate of >50%. Symptoms can appear 2–21 days after exposure, but most patients become ill within 9 days. The patient first presents with fatigue, fever, headache, and muscle pains, and the illness can progress to multiorgan failure and hemorrhaging. Careful volume-replacement therapy to maintain blood pressure and intravascular volume is key to survival in these infections. Ribavirin also may be useful against Arenaviridae and Bunyaviridae.

Other viral illnesses with rash, such as measles, can be associated with significant mortality rates. Measles continues to be responsible for more than 100,000 deaths per year, worldwide, and to cause outbreaks in populations with low vaccination rates.

■ SEPSIS WITH A SOFT TISSUE/MUSCLE PRIMARY FOCUS See also Chap. 129.

Necrotizing Fasciitis This infection is characterized by extensive necrosis of the subcutaneous tissue and fascia. It may arise at a site of minimal trauma or surgical incision and may also be associated with recent varicella, childbirth, or muscle strain. The most common causes of necrotizing fasciitis are group A streptococci alone (Chap. 148) and a mixed facultative and anaerobic flora (Chap. 129); the incidence of group A streptococcal necrotizing fasciitis has been increasing for the past quarter-century. Diabetes mellitus, IV drug use, chronic liver or renal disease, and malignancy are associated risk factors. Physical findings are initially minimal compared with the severity of pain and the degree of fever. The examination is often unremarkable except for soft tissue edema and erythema. The infected area is red, hot, shiny, swollen, and exquisitely tender. In untreated infection, the overlying skin develops blue-gray patches after 36 h, and cutaneous bullae and necrosis develop after 3–5 days. Necrotizing fasciitis due to a mixed flora, but not that due to group A streptococci, can be associated with gas production. Without treatment, pain decreases because of thrombosis of the small blood vessels and destruction of the peripheral nerves—an ominous sign. The mortality rate is 15–34% overall, >70% in association with TSS, and nearly 100% without surgical intervention. With surgery, outcomes are significantly better; e.g., mortality is reduced to <40% in the setting of TSS. Necrotizing fasciitis may also be due to *Clostridium perfringens* (Chap. 154); in this condition, the patient is extremely toxic and the mortality rate is high. Within 48 h,

rapid tissue invasion and systemic toxicity associated with hemolysis and death ensue. The distinction between this entity and clostridial myonecrosis is made by muscle biopsy. Necrotizing fasciitis caused by community-acquired MRSA also has been reported.

Clostridial Myonecrosis (See also Chap. 154) Myonecrosis is often associated with trauma or surgery but can develop spontaneously. The incubation period is usually 12–24 h long, and massive necrotizing gangrene develops within hours of onset. Systemic toxicity, shock, and death can occur within 12 h. The patient's pain and toxic appearance are out of proportion to physical findings. On examination, the patient is febrile, apathetic, tachycardic, and tachypneic and may express a feeling of impending doom. Hypotension and renal failure develop later, and hyperalertness is evident preterminally. The skin over the affected area is bronze-brown, mottled, and edematous. Bullous lesions with serosanguineous drainage and a mousy or sweet odor can develop. Crepitus can occur secondary to gas production in muscle tissue. The mortality rate is >65% for spontaneous myonecrosis, which is often associated with *Clostridium septicum* or *C. tertium* and underlying malignancy. The mortality rates associated with trunk and limb infection are 63% and 12%, respectively, and any delay in surgical treatment increases the risk of death.

■ NEUROLOGIC INFECTIONS WITH OR WITHOUT SEPTIC SHOCK

Bacterial Meningitis (See also Chap. 138) Bacterial meningitis is one of the most common infectious disease emergencies involving the central nervous system. Although hosts with cell-mediated immune deficiency (including transplant recipients, diabetic patients, elderly patients, and cancer patients receiving certain chemotherapeutic agents) are at particular risk for *Listeria monocytogenes* meningitis, most cases in adults are due to *S. pneumoniae* (30–60%) and *N. meningitidis* (10–35%). The classic presentation of fever, meningismus, and altered mental status is seen in only one-half to two-thirds of patients and less than one-half of adults. The elderly can present without fever or meningeal signs. Cerebral dysfunction is evidenced by confusion, delirium, and lethargy that can progress to coma. In some cases, the presentation is fulminant, with sepsis and brain edema; papilledema at presentation is unusual and suggests another diagnosis (e.g., an intracranial lesion). Focal signs, including cranial nerve palsies (IV, VI, VII), can be seen in 10–20% of cases; 50–70% of patients have bacteremia. A poor outcome is associated with coma, seizures, hypotension, a pneumococcal etiology, respiratory distress, a CSF glucose level of <0.6 mmol/L (<10 mg/dL), a CSF protein level of >2.5 g/L, a peripheral white blood cell count of <5000/μL, and a serum sodium level of <135 mmol/L. Rapid initiation of treatment is essential; the odds of an unfavorable outcome may increase by 30% for each hour that treatment is delayed. Dexamethasone is an adjunctive treatment for meningitis in adults, especially for infections caused by *S. pneumoniae*. It must be given before or with the first dose of antibiotics; otherwise, it is unlikely to improve outcomes.

Suppurative Intracranial Infections (See also Chap. 140) Suppurative intracranial infections present along with sepsis and hemodynamic instability. Rapid recognition of the toxic patient with central neurologic signs is crucial to improvement of the prognosis of these entities. Patients with diabetes or hematologic disease may be at increased risk for these infections. *Subdural empyema* arises from the paranasal sinus in 60–70% of cases. Microaerophilic streptococci and staphylococci are the predominant etiologic organisms. The patient is toxic, with fever, headache, and nuchal rigidity. Of all patients, 75% have focal signs and 6–20% die. Despite improved survival rates, 15–44% of patients are left with permanent neurologic deficits. *Septic cavernous sinus thrombosis* follows a facial or sphenoid sinus infection; 70% of cases are due to staphylococci (including MRSA), and the remainder are due primarily to aerobic or anaerobic streptococci. Fungi have been common in some series. A unilateral or retro-orbital headache progresses to a toxic appearance and fever within days. Three-quarters of patients have unilateral periorbital edema that

becomes bilateral and then progresses to ptosis, proptosis, ophthalmoplegia, and papilledema. The mortality rate is as high as 30% in older studies. Recent reports indicate improved survival as high as 90% with fewer sequelae. *Septic thrombosis of the superior sagittal sinus* spreads from the ethmoid or maxillary sinuses and is caused by *S. pneumoniae*, other streptococci, and staphylococci. The fulminant course is characterized by headache, nausea, vomiting, rapid progression to confusion and coma, nuchal rigidity, and brainstem signs. If the sinus is totally thrombosed, the mortality rate exceeds 80%. Broad-spectrum antibiotics and early surgical intervention at the primary site of infection may improve outcomes. Anticoagulation or steroids are of uncertain benefit.

Brain Abscess (See also Chap. 140) Brain abscess often occurs without systemic signs. Almost half of patients are afebrile, and presentations are more consistent with a space-occupying lesion in the brain; 70% of patients have headache and/or altered mental status, 50% have focal neurologic signs, and 25% have papilledema. Abscesses can present as single or multiple lesions resulting from contiguous foci or hematogenous infection, such as endocarditis, or after surgery or trauma. The infection progresses over several days from cerebritis to an abscess with a mature capsule. More than half of infections are polymicrobial, with an etiology consisting of aerobic bacteria (primarily streptococcal species) and anaerobes. Abscesses arising hematogenously are especially apt to rupture into the ventricular space, causing a sudden and severe deterioration in clinical status and a high mortality rate. Otherwise, mortality is low (<20%) but morbidity is high (30–55%). Patients presenting with stroke and a parameningeal infectious focus, such as sinusitis or otitis, may have a brain abscess, and physicians must maintain a high level of suspicion. Prognosis worsens in patients with a fulminant course, delayed diagnosis, abscess rupture into the ventricles, multiple abscesses, or abnormal neurologic status at presentation. In one study, mortality at 1 year was 19%.

Cerebral Malaria (See also Chap. 224) This entity should be urgently considered for inhabitants of, or recent travelers to, areas endemic for malaria. Patients present with a febrile illness and lethargy or other neurologic signs. Fulminant malaria is caused by *Plasmodium falciparum* and is associated with temperatures of >40°C (>104°F), hypotension, jaundice, acute respiratory distress syndrome, and bleeding. By definition, any patient with a change in mental status or repeated seizure in the setting of fulminant malaria has cerebral malaria. In adults, this nonspecific febrile illness progresses to coma over several days; occasionally, coma occurs within hours and death within 24 h. Nuchal rigidity and photophobia are rare. On physical examination, symmetric encephalopathy is typical, and upper motor neuron dysfunction with decorticate and decerebrate posturing can be seen in advanced disease. Unrecognized infection results in a 20–30% mortality rate. Children with neurologic deficit at hospital discharge, seizure recurrence during treatment, and/or ischemic neural injury on MRI are at particular risk for neurologic and mental health sequelae.

Intracranial and Spinal Epidural Abscesses (See also Chap. 442) Spinal and intracranial epidural abscesses (SEAs and ICEAs) can result in permanent neurologic deficits, sepsis, and death. At-risk patients include those with diabetes mellitus; IV drug use; chronic alcohol abuse; recent spinal trauma, surgery, or epidural anesthesia; and other comorbid conditions, such as HIV infection. Fungal epidural abscess and meningitis can follow epidural or paraspinal glucocorticoid injections. In the United States and Canada, where early treatment of otitis and sinusitis is typical, ICEA is rare but the number of cases of SEA is on the rise. In areas with limited access to health care, SEAs and ICEAs cause significant morbidity and mortality. ICEAs typically present as fever, mental status changes, and neck pain, while SEAs often present as fever, localized spinal tenderness, and back pain. ICEAs are typically polymicrobial, whereas SEAs are most often due to hematogenous seeding, with staphylococci the most common etiologic agent. Early diagnosis and treatment, which may include surgical drainage, minimize rates of mortality and permanent neurologic sequelae. Outcomes are worse for SEA due to MRSA, for

infection at a higher vertebral-body level, for impaired neurologic status on presentation, and for dorsal rather than ventral abscess location. Elderly patients and persons with renal failure, malignancy, and other comorbidities also have less favorable outcomes.

■ OTHER FOCAL SYNDROMES WITH A FULMINANT COURSE

Infection at virtually any primary focus (e.g., osteomyelitis, pneumonia, pyelonephritis, or cholangitis) can result in bacteremia and sepsis. Lemierre's syndrome—jugular septic thrombophlebitis caused by *Fusobacterium necrophorum*—is associated with metastatic infectious emboli (primarily to the lung but sometimes to the liver or other organs) and sepsis, with mortality rates of >15%. *Fusobacterium* bloodstream infections have been associated with occult gastrointestinal or genitourinary malignancy. TSS has been associated with focal infections such as septic arthritis, peritonitis, sinusitis, and wound infection. Rapid clinical deterioration and death can be associated with destruction of the primary site of infection, as is seen in endocarditis and in infections of the oropharynx (e.g., Ludwig's angina or epiglottitis, in which edema suddenly compromises the airway).

Rhinocerebral Mucormycosis (See also Chap. 218) Individuals with diabetes or immunocompromising conditions such as solid organ transplants or hematologic malignancies are at risk for invasive rhinocerebral mucormycosis. Patients present with low-grade fever, dull sinus pain, diplopia, decreased mental status, decreased ocular motion, chemosis, proptosis, dusky or necrotic nasal turbinates, and necrotic hard-palate lesions that respect the midline. Without rapid recognition, surgical intervention and antifungal therapy, the process continues on an inexorable invasive course, with mortality rates of 50–85% or greater. Uncontrolled diabetes and increasing age are negative prognostic factors.

Acute Bacterial Endocarditis (See also Chap. 128) This entity presents with a much more aggressive course than subacute endocarditis. Bacteria such as *S. aureus*, *S. pneumoniae*, *L. monocytogenes*, *Haemophilus* species, and streptococci of groups A, B, and G attack native valves. Native-valve endocarditis caused by *S. aureus* (including MRSA strains) is increasing. Mortality rates range from 10% to 40%. The host may have comorbid conditions such as underlying malignancy, diabetes mellitus, IV drug use, or alcoholism. The patient presents with fever, fatigue, and malaise <2 weeks after onset of infection. On physical examination, a changing murmur and congestive heart failure may be noted. Hemorrhagic macules on palms or soles (*Janeway lesions*) sometimes develop. Petechiae, Roth's spots, splinter hemorrhages, and splenomegaly are unusual. Rapid valvular destruction, particularly of the aortic valve, results in pulmonary edema and hypotension. Myocardial abscesses can form, eroding through the septum or into the conduction system and causing life-threatening arrhythmias or high-degree conduction block. Large friable vegetations can result in major arterial emboli, metastatic infection, or tissue infarction. Older patients with *S. aureus* endocarditis are especially likely to present with nonspecific symptoms—a circumstance that delays diagnosis and worsens prognosis. Rapid intervention is crucial for a successful outcome.

Inhalational Anthrax (See also Chap. S3) Inhalational anthrax, the most severe form of disease caused by *B. anthracis*, had not been reported in the United States for more than 25 years until the use of this organism as an agent of bioterrorism in 2001. Patients presented with malaise, fever, cough, nausea, drenching sweats, shortness of breath, and headache. Rhinorrhea was unusual. All patients had abnormal chest roentgenograms at presentation. Pulmonary infiltrates, mediastinal widening, and pleural effusions were the most common findings. Hemorrhagic meningitis was documented in 38% of these patients. Survival was more likely when antibiotics were given during the prodromal period and when multidrug regimens were used. In the absence of urgent intervention with antimicrobial agents and supportive care, inhalational anthrax progresses rapidly to hypotension, cyanosis, and death.

980 Viral Respiratory Tract Illness Viral respiratory tract illnesses can cause severe disease; several new syndromes have been described in the past decade. For patients who present with a respiratory illness and a relevant exposure and travel history, these viral illnesses must be considered and appropriate infection control measures instituted in addition to supportive care.

Avian and Swine Influenza (See also Chap. 200) Human cases of avian influenza have occurred primarily in Southeast Asia, particularly Vietnam (H5N1) and China (H7N9). Avian influenza should be considered in patients with severe respiratory tract illness, particularly if they have been exposed to poultry. Patients present with high fever, an influenza-like illness, and lower respiratory tract symptoms; this illness can progress rapidly to bilateral pneumonia, acute respiratory distress syndrome, multiorgan failure, and death. Younger age appears to be associated with a lower risk of complications. Early antiviral treatment with neuraminidase inhibitors should be initiated along with aggressive supportive measures. Unlike avian influenza, whose human-to-human transmission has so far been rare and has not been sustained, influenza caused by a novel swine-associated A/H1N1 virus has spread rapidly throughout the world; by 2012, 214 countries had diagnosed cases of influenza A/H1N1, with 18,449 deaths. Patients most at risk of severe disease are children <5 years of age, elderly persons, patients with underlying chronic conditions, and pregnant women. Obesity also has been identified as a risk factor for severe illness. Immunosuppression and co-infection with *S. aureus* at presentation are independent risk factors for increased mortality.

SARS, COVID-19, and MERS (See Chap. 199) Severe acute respiratory syndrome (SARS) caused by what is now labeled SARS-CoV-1 was identified in 2002 in China but has been diagnosed in several countries, primarily in Asia. Possible animal reservoirs include bats and civets. SARS-CoV-1 is characterized by efficient human transmission but relatively low mortality. It spreads from person to person via droplets; “super-spreader” airborne events have occurred. The potential SARS-CoV-1 pandemic was controlled through identification and isolation of infected patients. A 3- to 7-day prodrome characterized by fever, malaise, headache, and myalgia can progress to nonproductive cough, dyspnea, and respiratory failure. The risk of contagion is low during the prodrome. Older patients and those with diabetes mellitus, chronic hepatitis B, and other comorbidities can have less favorable outcomes.

COVID-19 caused by SARS-CoV-2 has resulted in a pandemic of historic proportion. Originally linked to cases in China, SARS-CoV-2 has spread internationally at a rapid pace. As of June 2021, cases approached 174 million worldwide with 3.7 million deaths; the United States had more than 33 million cases and ~592,000 deaths. Actual infection may exceed reported cases by 8–10 fold. The primary mode of transmission of COVID-19 is through direct person-to-person contact via respiratory droplets; transmission by the airborne route or by contact with contaminated surfaces has been documented but is much less common. Greatest risk of infection is associated with close and prolonged contact such as in household or congregate settings. Rates of secondary infection among household contacts are high, over 50% in recent studies.

Although ~80% of patients with symptomatic COVID-19 infection have mild disease such as cough, fever, gastrointestinal symptoms, and taste and smell alterations, ~5% of patients develop respiratory failure, shock, and multiorgan system failure. Patients typically have been symptomatic for 5–7 days before progression to severe pneumonia and hypoxemia. Shock, often in the setting of cardiac injury/myocarditis, arrhythmias or cardiomyopathy, and thromboembolic complications such as pulmonary emboli or stroke, can occur. Other manifestations include acute kidney or liver injury. Laboratory findings include lymphopenia, elevated lactate dehydrogenase, and often evidence of a cytokine-release type syndrome with elevated C-reactive protein, D-dimer, ferritin, and proinflammatory cytokines such as interleukin 6. Chest radiographs and CT imaging reveal diffuse ground-glass opacities. Consolidation, predominantly in the lower lobes and peripherally, is a frequent

finding. Mortality rates for patients requiring intensive care and mechanical ventilation have improved significantly over the course of the pandemic but remain ≥20%. Risk factors for severe disease and poor outcomes include age ≥65 years, morbid obesity, diabetes, cardiovascular or cerebrovascular disease, hypertension, chronic obstructive lung disease, and chronic kidney disease. Severe disease and mortality is also increased in males and pregnant women.

Treatment currently includes supportive care, antiviral therapy, anti-inflammatory therapy, and anticoagulation prophylaxis or treatment. Patients with respiratory distress are hypoxemic but maintain pulmonary mechanical functioning; thus, invasive mechanical ventilation should be delayed until other interventions have been exhausted. In a meta-analysis, dexamethasone or other glucocorticoids reduced 28-day mortality in patients with severe COVID-19 disease compared with standard care: 32% and 40%, respectively (odds ratio, 0.66; 95% confidence interval, 0.53–0.82). Secondary bacterial pneumonia is uncommon in COVID-19, and antibacterial agents should not be routinely prescribed. With the availability of highly effective vaccines, prevention is the most important strategy to control COVID-19.

Middle East respiratory syndrome (MERS) is caused by a novel betacoronavirus with a likely bat reservoir and was first recognized in 2012 in Saudi Arabia. Human cases have been associated with direct and indirect contact with dromedary camels. Unlike SARS and COVID-19, MERS exhibits inefficient human transmission but carries a high mortality rate. As of 2017, >2000 cases had been confirmed, most in the Arabian Peninsula, with 35% mortality. MERS ranges from asymptomatic infection to acute respiratory distress syndrome, multiorgan failure, and death. Elderly men with comorbidities appear to be at highest risk for poor outcomes. Despite little documented human-to-human transmission in the community, nosocomial infection must be prevented by adherence to strict infection control practices. MERS is currently a low-level public health threat and is likely to remain so unless the virus mutates and its transmissibility increases.

Hantavirus Pulmonary Syndrome (See also Chap. 209) Hantavirus pulmonary syndrome has been documented in the United States since 1993 (primarily the southwestern states, west of the Mississippi River), Canada, and South America. Most cases occur in rural areas and are associated with exposure to rodents. Patients present with a nonspecific viral prodrome of fever, malaise, myalgias, nausea, vomiting, and dizziness that may progress to pulmonary edema, respiratory failure, and death. Hantavirus pulmonary syndrome causes myocardial depression and increased pulmonary vascular permeability; therefore, careful fluid resuscitation and use of pressor agents are crucial. Aggressive cardiopulmonary support during the first few hours of illness can be life-saving in this high-mortality syndrome. The early onset of thrombocytopenia may help distinguish this syndrome from other febrile illnesses in an appropriate epidemiologic setting.

Clostridioides difficile Infection *C. difficile* infection (CDI) is a toxin-mediated diarrheal syndrome that is strongly associated with prior antibiotic use. Proton-pump inhibitors also have been identified as a potential risk factor for the disease. Although most cases of CDI have occurred in the health care setting, community-onset CDI is increasing. Overall, community-onset cases occur in younger patients than nosocomial cases. Patients with community-onset CDI are less likely to have a history of antibiotic or protein-pump inhibitor use. CDI is associated with significant morbidity and mortality, particularly among older patients. The CDC named *C. difficile* infection as one of the top three health threats associated with antibiotic use.

SUMMARY

Acutely ill febrile patients with the syndromes discussed in this chapter require close observation, aggressive supportive measures, and—in most cases—admission to intensive care units. The most important task of the physician is to distinguish these patients from other infected febrile patients whose illness will not progress to fulminant disease. The alert physician must recognize the acute infectious disease emergency and proceed with appropriate urgency.

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Immunization Principles and Vaccine Use

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Few medical interventions of the past century can rival the effect that immunization has had on longevity, economic savings, and quality of life. Seventeen diseases are now preventable through vaccines routinely administered to children and adults in the United States (Table 123-1), and most vaccine-preventable diseases of childhood are at historically low levels (Table 123-2). Health care providers deliver the vast majority of vaccines in the United States in the course of providing routine

CONDITION	TARGET POPULATION(S) FOR ROUTINE USE
Pertussis	Children, adolescents, adults
Diphtheria	Children, adolescents, adults
Tetanus	Children, adolescents, adults
Poliomyelitis	Children
Measles	Children
Mumps	Children
Rubella, congenital rubella syndrome	Children
Hepatitis B	Children and high-risk adults
<i>Haemophilus influenzae</i> type b infection	Children and high-risk adults
Hepatitis A	Children and high-risk adults
Influenza	Children, adolescents, adults
Varicella	Children
Pneumococcal disease	Children, older adults, and high-risk adults ^a
Serogroups A, C, W, Y meningococcal disease	Adolescents and high-risk children and adults
Serogroup B meningococcal disease	High-risk children and adults ^a
Rotavirus infection	Infants
Human papillomavirus infection, cervical and anogenital cancers	Adolescents and young adults ^a
Zoster	Older adults

^aOthers in certain age groups may be vaccinated based on shared clinical decision-making.

TABLE 123-2 Decline in Vaccine-Preventable Diseases in the United States Following Widespread Implementation of National Vaccine Recommendations

CONDITION	ANNUAL NO. OF PREVACCINE CASES (AVERAGE)	NO. OF CASES REPORTED IN 2019 ^a	REDUCTION (%) IN CASES AFTER WIDESPREAD VACCINATION
Smallpox	29,005	0	100
Diphtheria	21,053	2	>99
Measles	530,217	1287	>99
Mumps	162,344	3509	98
Pertussis	200,752	15,662	92
Polio (paralytic)	16,316	0	100
Rubella	47,745	3	>99
Congenital rubella syndrome	152	0	100
Tetanus	580	19	97
<i>Haemophilus influenzae</i> type b infection	20,000	14 ^b	>99
Hepatitis A	117,333	4000 ^c	97
Hepatitis B (acute)	66,232	20,900 ^c	68
Invasive pneumococcal infection: all ages	63,067	30,400 ^d	52
Rotavirus hospitalizations (<3 years old)	62,500	30,625 ^e	51
Varicella	4,085,120	102,128 ^f	98

^a2019 reported cases unless otherwise specified (provisional as of January 2020).

^bAn additional 12 type b infections are estimated to have occurred among 243 reports of *H. influenzae* infection caused by unknown types among children <5 years of age. ^cData from the CDC's Viral Hepatitis Surveillance, 2016. ^dUnpublished data from the CDC's Active Bacterial Core Surveillance, 2016. ^eUnpublished data from the CDC's New Vaccine Surveillance Network, 2017. ^fU.S. rotavirus disease now has biennial pattern. ^gUnpublished data from CDC's varicella program, 2017.

Source: Adapted from SW Roush et al: *JAMA* 298:2155, 2007 and *Morb Mortal Wkly Rep* 65:924, 2017.

health services and therefore play an integral role in the nation's public health system.

The COVID-19 pandemic has highlighted the importance of a strong immunization program: robust surveillance systems to detect emerging infectious disease threats; public-private partnerships to accelerate the development of novel vaccines; and systems in place to rapidly implement a vaccination program and monitor vaccine safety and effectiveness. As of July 2021, multiple COVID-19 vaccines have received authorization for emergency use in the United States and over two-thirds of adults have received at least one dose. Furthermore, the largest and most comprehensive vaccine safety program in U.S. history has demonstrated the safety of COVID-19 vaccines, and evaluations have demonstrated the vaccines to be effective against SARS-CoV-2 infections, including against severe disease that would otherwise result in hospitalization and death.

VACCINE IMPACT

Direct and Indirect Effects Immunizations against specific infectious diseases protect individuals against infection and thereby prevent symptomatic illnesses. In addition, specific vaccines may also blunt the severity of clinical illness (e.g., rotavirus vaccines and severe gastroenteritis) or reduce complications (e.g., zoster vaccines and postherpetic neuralgia). Some immunizations also reduce transmission of infectious disease agents from immunized people to others, thereby reducing the impact of infection spread. This indirect impact is known as *herd immunity*. The level of immunization in a population that is required to achieve indirect protection of unimmunized people varies substantially with the specific vaccine and disease.